

Real Analysis

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Preface

This is a reading note of measure theory and real analysis. The note is revised and edited using ChatGPT. The note presumes basic knowledge in mathematical analysis. Topics of the note include: measure and integration, differentiation in $\mathbb{R}$ and in general measure spaces, and L^p spaces. In particular, Chapter 1, 2, 3, and 5 are mainly based on the book *Real Analysis: Modern Techniques and Their Applications* by Gerald B. Folland, and Chapter 4 and 6 are based on the book *Measure, Integraion, and Real Analysis* by Sheldon Axler. The note also contains some additional materials from *Real Analysis* by H. L. Royden and P. M. Fitzpatrick in Chapter 3 and 6.

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Chapter 1 Preliminaries

1.1 Maximal principles and choice

Definition 1.1.1 (Partially ordered sets). A *partially ordered set* is a pair $(X, \preceq)$ for which $\preceq$ is reflexive, antisymmetric, and transitive. A subset $C \subseteq X$ is a *chain* if every two elements of C are comparable. An element $u \in X$ is an *upper bound* for C if $c \preceq u$ for every $c \in C$. An element $m \in X$ is *maximal* if $m \preceq x$ implies $x = m$.

Theorem 1.1.2 (Hausdorff maximal principle). Every partially ordered set contains a maximal chain.

Theorem 1.1.3 (Zorn's lemma). Let $(X, \preceq)$ be a partially ordered set. If every chain in X has an upper bound in X , then X contains a maximal element.

Proposition 1.1.4. The Hausdorff maximal principle and Zorn's lemma are equivalent.

Proof. Assume the Hausdorff maximal principle and let C be a maximal chain in X . By hypothesis, C has an upper bound $u \in X$. If $u \preceq y$, then y is comparable with every element of C , so $C \cup \{y\}$ is a chain. Maximality of C gives $y \in C$, and because u is an upper bound of C , we also have $y \preceq u$. Thus $y = u$, and u is maximal.

Conversely, let $\mathcal{C}$ be the set of all chains in X , ordered by inclusion. If $\mathcal{D} \subseteq \mathcal{C}$ is a chain of chains, then $\bigcup \mathcal{D}$ is again a chain in X and is an upper bound of $\mathcal{D}$. Zorn's lemma therefore yields a maximal element of $\mathcal{C}$, which is a maximal chain in X . $\square$

Theorem 1.1.5 (Well-ordering theorem). Every nonempty set can be well ordered.

Proof. Let X be nonempty. Consider all pairs $(A, \leq_A)$ such that $A \subseteq X$ and $\leq_A$ well orders A . Order these pairs by *initial-segment extension*: $(A, \leq_A) \preceq (B, \leq_B)$ when A is an initial segment of B and the two orders agree on A .

The union of a chain of such partial well-orderings is well defined and is again a well-ordering of the union of their underlying sets. Zorn's lemma gives a maximal pair $(M, \leq_M)$. If $M \neq X$, choose $x \in X \setminus M$ and extend the order by declaring every element of M to be smaller than x . This contradicts maximality.

Hence $M = X$. $\square$

Theorem 1.1.6 (Axiom of choice). For every family $(X_\alpha)_{\alpha \in A}$ of nonempty sets, the Cartesian product $\prod_{\alpha \in A} X_\alpha$ is nonempty.

Proof. Well order $\bigcup_{\alpha \in A} X_\alpha$. For each α , let $f(\alpha)$ be the least element of X_α . Then $f \in \prod_{\alpha \in A} X_\alpha$. $\square$

Corollary 1.1.7. If $(X_\alpha)_{\alpha \in A}$ is a pairwise disjoint family of nonempty sets, there is a set $C \subseteq \bigcup_{\alpha \in A} X_\alpha$ containing exactly one element of each X_α .

Proof. Choose $f(\alpha) \in X_\alpha$ and set $C = \{f(\alpha) : \alpha \in A\}$. Disjointness ensures that these chosen elements are distinct. $\square$

1.2 Well-ordered sets and transfinite arguments

Definition 1.2.1 (Initial segment). Let $(X, <)$ be well ordered. For $x \in X$, the initial segment determined by x is

$$I_x = \{y \in X : y < x\}.$$

A subset $J \subseteq X$ is an initial segment if $y \in J$ and $x < y$ imply $x \in J$.

Theorem 1.2.2 (Transfinite induction). Let $(X, <)$ be well ordered and let $A \subseteq X$. Suppose that for every $x \in X$,

$$I_x \subseteq A \implies x \in A.$$

Then $A = X$.

Proof. If $A \neq X$, let x_0 be the least element of $X \setminus A$. Every predecessor of x_0 belongs to A , so $I_{x_0} \subseteq A$. The hypothesis gives $x_0 \in A$, a contradiction. $\square$

Proposition 1.2.3 (Unions of initial segments). If $(X, <)$ is well ordered and $A \subseteq X$, then

$$J = \bigcup_{x \in A} I_x$$

is either X or an initial segment I_y for a unique $y \in X$.

Proof. The set J is downward closed. If $J \neq X$, let y be the least element of $X \setminus J$. Then every $z < y$ lies in J , while no $z \geq y$ lies in J ; otherwise downward closure would force $y \in J$. Hence $J = I_y$. $\square$

Proposition 1.2.4 (Comparison theorem). If X and Y are well ordered, exactly one of the following possibilities occurs:

- (a). X is order isomorphic to Y ;
- (b). X is order isomorphic to a proper initial segment of Y ;
- (c). Y is order isomorphic to a proper initial segment of X .

Proof. Consider all order isomorphisms between initial segments of X and initial segments of Y , ordered by extension. The union of a chain is again such an isomorphism, so Zorn's lemma gives a maximal one,

say $f : A \rightarrow B$. If both A and B were proper initial segments, their least omitted elements could be matched, extending f . Thus at least one of $A = X$ or $B = Y$ holds. The three alternatives follow, and two distinct alternatives cannot hold simultaneously because no well-ordered set is isomorphic to a proper initial segment of itself. $\square$

Theorem 1.2.5 (The first uncountable well-order). There exists an uncountable well-ordered set Ω such that every proper initial segment of Ω is countable. Any two such well-ordered sets are order isomorphic.

Proof. Well order an uncountable set X . If every proper initial segment of X is countable, take $\Omega = X$. Otherwise, let x be the least point for which I_x is uncountable and take $\Omega = I_x$. Every proper initial segment of Ω is then countable by minimality.

For uniqueness, compare two candidates Ω and Ω' . If one were isomorphic to a proper initial segment of the other, it would be countable, contradicting uncountability. Hence they are isomorphic. $\square$

Proposition 1.2.6. Every countable subset $A \subseteq \Omega$ has an upper bound in Ω .

Proof. The set $J = \bigcup_{x \in A} I_x$ is a countable union of countable sets and is therefore countable. Hence $J \neq \Omega$, so $J = I_y$ for some $y \in \Omega$. For each $x \in A$, the inclusion $I_x \subseteq I_y$ implies $x \leq y$. Thus y is an upper bound for A . $\square$

1.3 Unordered sums

Definition 1.3.1. For a function $f : X \rightarrow [0, \infty]$, define the unordered sum

$$\sum_{x \in X} f(x) = \sup \left\{ \sum_{x \in F} f(x) : F \subseteq X \text{ finite} \right\}.$$

Proposition 1.3.2. Let $A = \{x \in X : f(x) > 0\}$. If A is uncountable, then $\sum_{x \in X} f(x) = \infty$. If A is countable and $g : \mathbb{N} \rightarrow A$ is a bijection, then

$$\sum_{x \in X} f(x) = \sum_{n=1}^{\infty} f(g(n)).$$

Proof. Write

$$A = \bigcup_{n=1}^{\infty} \{x : f(x) \geq 1/n\}.$$

If A is uncountable, one of these level sets is infinite. Choosing finite subsets of arbitrarily large cardinality shows that the finite partial sums are unbounded.

If A is countable, every finite subset of A is contained in some initial segment $\{g(1), \dots, g(N)\}$, so each finite partial sum is bounded by a sufficiently long series partial sum. Conversely, every series partial sum is one of the finite partial sums appearing in the defining supremum. Taking suprema gives equality. $\square$

Chapter 2 Measures

2.1 Sigma-algebras and generated structures

Definition 2.1.1 (Algebras and sigma-algebras). Let X be a set. An *algebra* $\mathcal{A} \subseteq \mathcal{P}(X)$ contains $\emptyset$ and is closed under complements and finite unions. A *sigma-algebra* $\mathcal{M} \subseteq \mathcal{P}(X)$ is an algebra that is closed under countable unions. A pair $(X, \mathcal{M})$ is called a *measurable space*.

Remark. Because an algebra is closed under complements, it is a sigma-algebra if and only if it is closed under countable intersections. This is an immediate consequence of De Morgan's laws.

Example. The power set $\mathcal{P}(X)$ and the trivial sigma-algebra $\{\emptyset, X\}$ are sigma-algebras. If X is uncountable, the collection of all countable or co-countable subsets of X is also a sigma-algebra.

Definition 2.1.2 (Generated sigma-algebra). For $\mathcal{E} \subseteq \mathcal{P}(X)$, the sigma-algebra generated by $\mathcal{E}$ is

$$\sigma(\mathcal{E}) = \bigcap \{ \mathcal{M} : \mathcal{M} \text{ is a sigma-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{M} \}.$$

Lemma 2.1.3 (Generator principle). Let $\mathcal{E} \subseteq \mathcal{P}(X)$ and let $\mathcal{M}$ be a sigma-algebra on X . If $\mathcal{E} \subseteq \mathcal{M}$, then $\sigma(\mathcal{E}) \subseteq \mathcal{M}$.

Definition 2.1.4 (Borel sigma-algebra). If X is a topological space, the *Borel sigma-algebra* $\mathcal{B}_X$ is the sigma-algebra generated by the open subsets of X .

Proposition 2.1.5 (Generators of $\mathcal{B}_{\mathbb{R}}$). The Borel sigma-algebra on $\mathbb{R}$ is generated by any one of the following families:

- (a). open intervals (a, b) ;
- (b). closed intervals $[a, b]$;
- (c). half-open intervals $(a, b]$ or $[a, b)$;
- (d). open rays (a, ∞) or $(-\infty, a)$;
- (e). closed rays $[a, \infty)$ or $(-\infty, a]$.

It is enough in each case to restrict the endpoints to rational numbers whenever the resulting family still separates all real endpoints by countable unions or intersections.

Proof. Every displayed set is Borel, so the sigma-algebra generated by any listed family is contained in $\mathcal{B}_{\mathbb{R}}$. Conversely, open intervals can be obtained from any of the other families by countable unions, intersections, and complements. Every open subset of $\mathbb{R}$ is a countable union of open intervals with rational endpoints, so each listed family generates all Borel sets. $\square$

Definition 2.1.6 (Product sigma-algebra). Let $(X_\alpha, \mathcal{M}_\alpha)_{\alpha \in A}$ be measurable spaces and let $X = \prod_{\alpha \in A} X_\alpha$. The product sigma-algebra

$$\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$$

is generated by the coordinate cylinders $\pi_\alpha^{-1}(E)$, where $E \in \mathcal{M}_\alpha$ and $\pi_\alpha : X \rightarrow X_\alpha$ is the coordinate projection.

Proposition 2.1.7. If A is countable, then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by all measurable rectangles $\prod_{\alpha \in A} E_\alpha$ with $E_\alpha \in \mathcal{M}_\alpha$.

Proof. Every cylinder is a rectangle in which all but one coordinate set is the whole factor. Conversely, when A is countable,

$$\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha),$$

a countable intersection of cylinders. $\square$

Proposition 2.1.8 (Generators in the factors). Suppose $\mathcal{M}_\alpha = \sigma(\mathcal{E}_\alpha)$ for every α . Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by the cylinders $\pi_\alpha^{-1}(E)$ with $E \in \mathcal{E}_\alpha$.

Proof. Let $\mathcal{F}$ be the sigma-algebra generated by those cylinders. For fixed α , define

$$\mathcal{C}_\alpha = \{B \subseteq X_\alpha : \pi_\alpha^{-1}(B) \in \mathcal{F}\}.$$

Then $\mathcal{C}_\alpha$ is a sigma-algebra containing $\mathcal{E}_\alpha$, hence it contains $\mathcal{M}_\alpha$. Thus every measurable cylinder belongs to $\mathcal{F}$, which proves the claim. $\square$

Proposition 2.1.9 (Borel sigma-algebras of finite products). Let $X_1, \dots, X_n$ be metric spaces and let $X = \prod_{j=1}^n X_j$ have the product topology. Then

$$\bigotimes_{j=1}^n \mathcal{B}_{X_j} \subseteq \mathcal{B}_X.$$

If all X_j are separable, equality holds.

Proof. Each coordinate projection is continuous, so every Borel cylinder is Borel in X . This proves the first inclusion.

If each X_j is separable, choose a countable base $\mathcal{U}_j$ for X_j . The finite products $U_1 \times \dots \times U_n$, with $U_j \in \mathcal{U}_j$, form a countable base for X . Every open set is therefore a countable union of measurable rectangles and belongs to the product sigma-algebra. $\square$

Corollary 2.1.10. For every $n \geq 1$,

$$\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{j=1}^n \mathcal{B}_{\mathbb{R}}.$$

Definition 2.1.11 (Elementary family). A collection $\mathcal{E} \subseteq \mathcal{P}(X)$ is an *elementary family* if

- (a). $\emptyset \in \mathcal{E}$;
- (b). $E, F \in \mathcal{E}$ implies $E \cap F \in \mathcal{E}$;
- (c). for every $E \in \mathcal{E}$, the complement E^c is a finite disjoint union of members of $\mathcal{E}$.

Proposition 2.1.12. The collection $\mathcal{A}$ of finite disjoint unions of members of an elementary family $\mathcal{E}$ is an algebra.

Proof. Finite intersections of members of $\mathcal{A}$ are obtained by distributing intersections over their finite decompositions, and each resulting piece belongs to $\mathcal{E}$. Complements are obtained by expanding the elementary complement decompositions and refining the finitely many pieces into a disjoint family. Hence $\mathcal{A}$ is closed under complements and finite intersections, and therefore under finite unions. $\square$

Proposition 2.1.13 (Infinite sigma-algebras). Every infinite sigma-algebra contains a sequence of pairwise disjoint nonempty sets. Consequently, every infinite sigma-algebra has cardinality at least $2^{\aleph_0}$.

Proof. Call a nonempty measurable set an *atom* if it contains no nonempty proper measurable subset. If there are infinitely many atoms, choose a countable subfamily. If there are only finitely many atoms, their union cannot be all of X , for otherwise every measurable set would be a union of those finitely many atoms and the sigma-algebra would be finite. The complement therefore contains a nonempty atomless measurable set. Repeatedly split the remaining atomless set into two nonempty measurable pieces, retaining one piece for the next split and recording the other. This gives pairwise disjoint nonempty sets (E_n) .

The map

$$S \subseteq \mathbb{N} \mapsto \bigcup_{n \in S} E_n$$

is injective from $\mathcal{P}(\mathbb{N})$ into the sigma-algebra. $\square$

Proposition 2.1.14. An algebra $\mathcal{A}$ is a sigma-algebra if and only if it is closed under countable increasing unions.

Proof. Only the converse needs proof. Given $A_n \in \mathcal{A}$, put $B_n = \bigcup_{k=1}^n A_k$. Then $B_n \uparrow \bigcup_n A_n$, and the assumed closure gives $\bigcup_n A_n \in \mathcal{A}$. $\square$

Proposition 2.1.15 (Countable subgenerators). For every family $\mathcal{E} \subseteq \mathcal{P}(X)$,

$$\sigma(\mathcal{E}) = \bigcup \{ \sigma(\mathcal{F}) : \mathcal{F} \subseteq \mathcal{E} \text{ is countable} \}.$$

Proof. The right-hand side is closed under complements. If $A_n \in \sigma(\mathcal{F}_n)$ and each $\mathcal{F}_n$ is countable, then $\mathcal{F} = \bigcup_n \mathcal{F}_n$ is countable and all A_n belong to $\sigma(\mathcal{F})$; hence so does $\bigcup_n A_n$. Thus the right-hand side is a sigma-algebra containing $\mathcal{E}$, and equality follows. $\square$

2.2 Measures, null sets, and completion

Definition 2.2.1 (Measure). Let $(X, \mathcal{M})$ be a measurable space. A *measure* is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that $\mu(\emptyset) = 0$ and

$$\mu \left(\bigsqcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} \mu(E_n)$$

for every pairwise disjoint sequence (E_n) in $\mathcal{M}$. The triple $(X, \mathcal{M}, \mu)$ is a *measure space*.

Example. Counting measure is given by $\#(E) = \text{card}(E)$, with value ∞ for infinite sets. A Dirac measure at x_0 is $\delta_{x_0}(E) = \mathbf{1}_E(x_0)$. On an uncountable set with the countable/co-countable sigma-algebra, the formula

$$\mu(E) = \begin{cases} 0, & E \text{ countable,} \\ 1, & E^c \text{ countable} \end{cases}$$

defines a probability measure.

Theorem 2.2.2 (Basic properties). Let $(X, \mathcal{M}, \mu)$ be a measure space.

(a). If $E \subseteq F$, then $\mu(E) \leq \mu(F)$.

(b). For arbitrary $E_n \in \mathcal{M}$,

$$\mu \left(\bigcup_n E_n \right) \leq \sum_n \mu(E_n).$$

(c). If $E_n \uparrow E$, then $\mu(E_n) \uparrow \mu(E)$.

(d). If $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.

Proof. For monotonicity, write $F = E \sqcup (F \setminus E)$. For subadditivity, disjointify by setting

$$F_1 = E_1, \quad F_n = E_n \setminus \bigcup_{k < n} E_k.$$

Then $\bigcup_n F_n = \bigcup_n E_n$ and $\mu(F_n) \leq \mu(E_n)$.

For continuity from below, write E as the disjoint union of E_1 and the increments $E_n \setminus E_{n-1}$. For continuity from above, set $F_n = E_1 \setminus E_n$. Then $F_n \uparrow E_1 \setminus E$, and subtraction is legitimate because $\mu(E_1) < \infty$:

$$\mu(E_n) = \mu(E_1) - \mu(F_n) \longrightarrow \mu(E_1) - \mu(E_1 \setminus E) = \mu(E).$$

□

Corollary 2.2.3 (Liminf and limsup). For every sequence (E_n) in $\mathcal{M}$,

$$\mu(\liminf E_n) \leq \liminf_n \mu(E_n).$$

If $\mu(\bigcup_n E_n) < \infty$, then

$$\mu(\limsup E_n) \geq \limsup_n \mu(E_n).$$

Proof. The tail intersections $\bigcap_{k \geq n} E_k$ increase to $\liminf E_n$, and each has measure at most $\inf_{k \geq n} \mu(E_k)$. For the second inequality, the tail unions $\bigcup_{k \geq n} E_k$ decrease to $\limsup E_n$ and all lie in the finite-measure set $\bigcup_n E_n$, so continuity from above applies. $\square$

Proposition 2.2.4 (Finite additivity criterion). Let μ be finitely additive on an algebra $\mathcal{A}$. Then μ is a premeasure if and only if it is continuous from below: whenever $E_n \uparrow E$ with all sets in $\mathcal{A}$, one has $\mu(E_n) \uparrow \mu(E)$.

Proof. A premeasure has continuity from below by the usual increment argument. Conversely, if (E_n) is disjoint and $E = \bigcup_n E_n \in \mathcal{A}$, then $F_N = \bigcup_{n \leq N} E_n$ increases to E , and finite additivity gives

$$\mu(E) = \lim_N \mu(F_N) = \lim_N \sum_{n=1}^N \mu(E_n).$$

$\square$

Definition 2.2.5 (Null sets and completeness). A measurable set N is *null* if $\mu(N) = 0$. The measure space is *complete* if every subset of every measurable null set is measurable.

Theorem 2.2.6 (Completion). Let $(X, \mathcal{M}, \mu)$ be a measure space and define

$$\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subseteq Z \text{ for some } Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}.$$

Then $\overline{\mathcal{M}}$ is a sigma-algebra and

$$\overline{\mu}(A \cup N) = \mu(A)$$

is a well-defined complete measure on $\overline{\mathcal{M}}$ extending μ . It is the unique complete extension of μ whose measurable sets have this form.

Proof. The family is closed under countable unions because a countable union of subsets of null sets is contained in a measurable null set. If $E = A \cup N$ with $N \subseteq Z$ and $\mu(Z) = 0$, then

$$E^c = (A^c \setminus Z) \cup ((A^c \cap Z) \setminus N).$$

The first term is measurable and the second is contained in the measurable null set Z , so the complement has the required form.

If $A \cup N = B \cup M$, with N and M contained in measurable null sets, then $A \triangle B$ is null, so $\mu(A) = \mu(B)$.

Hence $\overline{\mu}$ is well defined. Disjoint representatives can be refined inside the original sigma-algebra; their overlaps are null, and countable additivity follows from countable additivity of μ . Finally, every subset of a $\overline{\mu}$ -null set is contained in an original measurable null set, proving completeness. $\square$

2.3 Sigma-finiteness, semifiniteness, and saturation

Definition 2.3.1. A measure is *finite* if $\mu(X) < \infty$, *sigma-finite* if X is a countable union of finite-measure measurable sets, and *semifinite* if every measurable E with $\mu(E) = \infty$ contains a measurable F with $0 < \mu(F) < \infty$.

Proposition 2.3.2. Every sigma-finite measure is semifinite.

Proof. Choose increasing measurable sets $X_n \uparrow X$ with $\mu(X_n) < \infty$. If $\mu(E) = \infty$, then $\mu(E \cap X_n) \uparrow \infty$. Hence some $E \cap X_n$ has positive finite measure. $\square$

Proposition 2.3.3. If μ is semifinite, $\mu(E) = \infty$, and $C > 0$, then there is a measurable $F \subseteq E$ with

$$C < \mu(F) < \infty.$$

Proof. Let

$$S = \sup\{\mu(F) : F \subseteq E, F \in \mathcal{M}, \mu(F) < \infty\}.$$

If $S < \infty$, choose F_n with $\mu(F_n) > S - 1/n$ and set $G_n = \bigcup_{k \leq n} F_k$. Each G_n has finite measure and therefore $\mu(G_n) \leq S$. Thus $G = \bigcup_n G_n$ has measure S . Since $\mu(E) = \infty$, the set $E \setminus G$ has infinite measure. Semifiniteness supplies a measurable $H \subseteq E \setminus G$ with $0 < \mu(H) < \infty$, and then $G \cup H$ is a finite-measure subset of E with measure larger than S , a contradiction. Hence $S = \infty$. $\square$

Proposition 2.3.4 (Semifinite part). For an arbitrary measure μ , define

$$\mu_{\text{sf}}(E) = \sup\{\mu(F) : F \subseteq E, F \in \mathcal{M}, \mu(F) < \infty\}.$$

Then μ_{sf} is a semifinite measure and $\mu_{\text{sf}} \leq \mu$. Moreover, $\mu_{\text{sf}} = \mu$ if and only if μ is semifinite. If

$$\nu(E) = \begin{cases} 0, & E \text{ is a countable union of finite-}\mu\text{-measure sets,} \\ \infty, & \text{otherwise,} \end{cases}$$

then ν is a measure and $\mu = \mu_{\text{sf}} + \nu$.

Proof. Let (E_n) be disjoint. For every finite N , finite-measure subsets $F_n \subseteq E_n$ give

$$\mu_{\text{sf}}\left(\bigcup_n E_n\right) \geq \sum_{n=1}^N \mu(F_n).$$

Taking suprema and then $N \rightarrow \infty$ yields the lower inequality for countable additivity. Conversely, if $F \subseteq \bigcup_n E_n$ and $\mu(F) < \infty$, then

$$\mu(F) = \sum_n \mu(F \cap E_n) \leq \sum_n \mu_{\text{sf}}(E_n).$$

Taking the supremum over F gives the reverse inequality.

Semifiniteness of μ_{sf} follows directly from the definition. If μ is semifinite, the preceding proposition shows that an infinite-measure set has finite-measure subsets of arbitrarily large measure, so $\mu_{\text{sf}}(E) = \mu(E)$ for every E .

The class of countable unions of finite- μ -measure sets is a sigma-ideal. Hence the displayed $\{0, \infty\}$ -valued function ν is countably additive: a countable union fails to be sigma-finite exactly when at least one member of a disjoint decomposition fails to be sigma-finite. If E is sigma-finite, then $\mu_{\text{sf}}(E) = \mu(E)$ and $\nu(E) = 0$; if it is not sigma-finite, then $\mu(E) = \nu(E) = \infty$. Thus $\mu = \mu_{\text{sf}} + \nu$. $\square$

Definition 2.3.5 (Local measurability and saturation). A set $E \subseteq X$ is *locally measurable* if

$$E \cap A \in \mathcal{M} \quad \text{for every } A \in \mathcal{M} \text{ with } \mu(A) < \infty.$$

Let $\mathcal{M}_{\text{loc}}$ denote the collection of locally measurable sets. The measure space is *saturated* if $\mathcal{M}_{\text{loc}} = \mathcal{M}$.

Proposition 2.3.6. The collection $\mathcal{M}_{\text{loc}}$ is a sigma-algebra containing $\mathcal{M}$. If μ is sigma-finite, then μ is saturated.

Proof. Closure of $\mathcal{M}_{\text{loc}}$ under complements and countable unions follows after intersecting with an arbitrary finite-measure measurable set. If $X_n \uparrow X$ and $\mu(X_n) < \infty$, then for $E \in \mathcal{M}_{\text{loc}}$,

$$E = \bigcup_n (E \cap X_n) \in \mathcal{M}.$$

$\square$

Proposition 2.3.7 (Saturated extension). Assume that μ is semifinite. For $E \in \mathcal{M}_{\text{loc}}$, set

$$\tilde{\mu}(E) = \sup\{\mu(A) : A \in \mathcal{M}, A \subseteq E, \mu(A) < \infty\}.$$

Then $\tilde{\mu}$ is a saturated measure on $\mathcal{M}_{\text{loc}}$ extending μ .

Proof. The proof of countable additivity is the same two-inequality argument used for the semifinite part: finite-measure measurable subsets of a disjoint union decompose into measurable intersections with the locally measurable summands. Semifiniteness shows that $\tilde{\mu}(E) = \mu(E)$ for $E \in \mathcal{M}$.

Suppose E is locally measurable with respect to $(\mathcal{M}_{\text{loc}}, \tilde{\mu})$. If $A \in \mathcal{M}$ has finite μ -measure, then $A \in \mathcal{M}_{\text{loc}}$ and $\tilde{\mu}(A) < \infty$, so $E \cap A \in \mathcal{M}_{\text{loc}}$. Intersecting once more with A shows $E \cap A \in \mathcal{M}$. Therefore $E \in \mathcal{M}_{\text{loc}}$, proving saturation. $\square$

Remark. Without semifiniteness, the extension that assigns ∞ to every locally measurable set outside $\mathcal{M}$ can differ from the supremum construction. Here is a concrete example. Let $X = X_1 \sqcup X_2$, where both sets are uncountable, and let

$$\mathcal{M} = \{A_1 \cup A_2 : A_1 \subseteq X_1, A_2 \subseteq X_2 \text{ is countable or co-countable}\}.$$

Define

$$\mu(A_1 \cup A_2) = \#A_1 + \nu_2(A_2), \quad \nu_2(A_2) = \begin{cases} 0, & A_2 \text{ countable,} \\ \infty, & A_2 \text{ co-countable.} \end{cases}$$

Every subset of X_2 is locally measurable, because every finite-measure measurable set meets X_2 in a countable set. Choose $E \subseteq X_2$ such that both E and $X_2 \setminus E$ are uncountable. Then $E \notin \mathcal{M}$. The extension assigning ∞ outside $\mathcal{M}$ gives E value ∞ , whereas the supremum over finite-measure measurable subsets of E is 0.

2.4 Outer measures and the Carathéodory construction

Definition 2.4.1 (Outer measure). An *outer measure* on X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ satisfying

- (a). $\mu^*(\emptyset) = 0$;
- (b). $A \subseteq B$ implies $\mu^*(A) \leq \mu^*(B)$;
- (c). $\mu^*(\bigcup_n A_n) \leq \sum_n \mu^*(A_n)$.

Proposition 2.4.2 (Outer measure from coverings). Let $\mathcal{E} \subseteq \mathcal{P}(X)$ contain $\emptyset$ and X , and let $p : \mathcal{E} \rightarrow [0, \infty]$ satisfy $p(\emptyset) = 0$. Define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} p(E_j) : E_j \in \mathcal{E}, A \subseteq \bigcup_j E_j \right\}.$$

Then μ^* is an outer measure.

Proof. Only countable subadditivity requires work. If $\sum_i \mu^*(A_i) = \infty$, there is nothing to prove. Otherwise, given $\varepsilon > 0$, choose a cover $(E_{ij})_j$ of A_i with cost below $\mu^*(A_i) + \varepsilon 2^{-i}$. The family of all E_{ij} covers $\bigcup_i A_i$, so

$$\mu^* \left(\bigcup_i A_i \right) \leq \sum_i \mu^*(A_i) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. □

Definition 2.4.3 (Carathéodory measurability). A set $A \subseteq X$ is μ^* -*measurable* if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{for every } E \subseteq X.$$

Theorem 2.4.4 (Carathéodory). The collection $\mathcal{M}^*$ of μ^* -measurable sets is a sigma-algebra. The restriction $\mu^* \upharpoonright \mathcal{M}^*$ is a complete measure.

Proof. Complements are immediate. To prove closure under countable unions, first disjointify a sequence (A_n) by

$$C_1 = A_1, \quad C_n = A_n \setminus \bigcup_{k < n} A_k.$$

Finite iteration of the Carathéodory identity gives

$$\mu^*(E) \geq \sum_{n=1}^N \mu^*(E \cap C_n) + \mu^* \left(E \setminus \bigcup_{n=1}^N C_n \right).$$

Letting $N \rightarrow \infty$ and using subadditivity yields the equality required for $\bigcup_n A_n$. Setting $E = \bigcup_n C_n$ proves countable additivity on disjoint measurable sets.

If N has outer measure zero, then every subset $A \subseteq N$ has outer measure zero and satisfies the Carathéodory identity. Thus the measure is complete. □

Definition 2.4.5 (Premeasure). Let $\mathcal{A}$ be an algebra. A function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a *premeasure* if $\mu_0(\emptyset) = 0$

and

$$\mu_0 \left(\bigsqcup_n A_n \right) = \sum_n \mu_0(A_n)$$

whenever the A_n are disjoint members of $\mathcal{A}$ and their union belongs to $\mathcal{A}$.

Theorem 2.4.6 (Extension from a premeasure). Let μ_0 be a premeasure on an algebra $\mathcal{A}$. Define

$$\mu^*(E) = \inf \left\{ \sum_n \mu_0(A_n) : A_n \in \mathcal{A}, E \subseteq \bigcup_n A_n \right\}.$$

Then

- (a). $\mu^*(A) = \mu_0(A)$ for every $A \in \mathcal{A}$;
- (b). every $A \in \mathcal{A}$ is μ^* -measurable;
- (c). $\mu = \mu^* \upharpoonright \sigma(\mathcal{A})$ is a measure extending μ_0 .

If μ_0 is sigma-finite, this extension is unique on $\sigma(\mathcal{A})$.

Proof. The cover consisting only of A gives $\mu^*(A) \leq \mu_0(A)$. Conversely, if $A \subseteq \bigcup_n A_n$, disjointify the cover inside A :

$$B_n = (A \cap A_n) \setminus \bigcup_{k < n} A_k.$$

Then $A = \bigsqcup_n B_n$, so

$$\mu_0(A) = \sum_n \mu_0(B_n) \leq \sum_n \mu_0(A_n).$$

Taking the infimum gives equality.

For $A \in \mathcal{A}$, split every covering set B_n into $B_n \cap A$ and $B_n \cap A^c$. Finite additivity of μ_0 and the definition of μ^* yield

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

The reverse inequality is subadditivity, so A is measurable. Existence follows from Carathéodory's theorem.

For uniqueness, first assume $\mu_0(X) < \infty$. If ν is another extension, the class

$$\mathcal{L} = \{E \in \sigma(\mathcal{A}) : \mu(E) = \nu(E)\}$$

is a Dynkin system containing the pi-system $\mathcal{A}$, hence contains $\sigma(\mathcal{A})$. In the sigma-finite case, partition X into disjoint sets in $\mathcal{A}$ of finite premeasure and apply the finite argument to the restrictions before summing. $\square$

Proposition 2.4.7 (Measurable hulls). Let $\mathcal{A}_\sigma$ be the collection of countable unions of members of $\mathcal{A}$.

- (a). For every $E \subseteq X$ and $\varepsilon > 0$, there is $A \in \mathcal{A}_\sigma$ with $E \subseteq A$ and

$$\mu^*(A) \leq \mu^*(E) + \varepsilon.$$

- (b). If $\mu^*(E) < \infty$, there is a measurable $B \supseteq E$ with $\mu^*(B) = \mu^*(E)$. If, in addition, $E \in \mathcal{M}^*$, then $\mu^*(B \setminus E) = 0$.

(c). Under sigma-finiteness, every $E \in \mathcal{M}^*$ has a measurable hull $B \supseteq E$ satisfying $\mu^*(B \setminus E) = 0$, even when $\mu^*(E) = \infty$.

Proof. The first assertion is the definition of the covering outer measure. If $\mu^*(E) < \infty$, choose $A_n \in \mathcal{A}_\sigma$ containing E with $\mu^*(A_n) \leq \mu^*(E) + 2^{-n}$, and replace A_n by $\bigcap_{k \leq n} A_k$ to make the sequence decreasing. Then $B = \bigcap_n A_n$ is measurable, contains E , and continuity from above inside the finite-measure set A_1 gives $\mu^*(B) = \mu^*(E)$. If E is measurable, then

$$\mu^*(B) = \mu^*(E) + \mu^*(B \setminus E),$$

so the difference is null.

Under sigma-finiteness, take a disjoint measurable partition $X = \bigsqcup_n X_n$ with $\mu^*(X_n) < \infty$. Apply the finite construction to $E \cap X_n$, obtaining measurable $B_n \subseteq X_n$ with $E \cap X_n \subseteq B_n$ and $\mu^*(B_n \setminus E) = 0$. Then $B = \bigcup_n B_n$ has the required properties. $\square$

Definition 2.4.8 (Inner measure on a finite space). If $\mu^*(X) < \infty$, define

$$\mu_*(E) = \mu^*(X) - \mu^*(X \setminus E).$$

Proposition 2.4.9. When $\mu^*(X) < \infty$, a set $E \subseteq X$ is μ^* -measurable if and only if $\mu^*(E) = \mu_*(E)$.

Proof. Measurability immediately gives equality by applying the Carathéodory identity to X . Conversely, suppose equality holds and let $B \supseteq E$ be a measurable hull with $\mu^*(B) = \mu^*(E)$. Since $X \setminus B$ is measurable,

$$\mu^*(X \setminus E) = \mu^*(X \setminus B) + \mu^*(B \setminus E).$$

The assumed equality forces $\mu^*(B \setminus E) = 0$. Completeness then gives $E = B \setminus (B \setminus E) \in \mathcal{M}^*$. $\square$

Proposition 2.4.10 (Regenerating an outer measure). Let $\mu = \mu^* \upharpoonright \mathcal{M}^*$ and define

$$\mu^+(E) = \inf\{\mu(A) : A \in \mathcal{M}^*, E \subseteq A\}.$$

Then $\mu^*(E) \leq \mu^+(E)$ for every E . Equality holds precisely when E has a measurable hull of the same outer measure.

Proof. Every measurable cover $A \supseteq E$ satisfies $\mu^*(E) \leq \mu(A)$, so $\mu^*(E) \leq \mu^+(E)$. If E has a measurable hull B with $\mu(B) = \mu^*(E)$, then $\mu^+(E) \leq \mu(B)$ and equality follows. Conversely, if $\mu^+(E) = \mu^*(E) < \infty$, choose measurable $A_n \supseteq E$ with $\mu(A_n) \leq \mu^*(E) + 2^{-n}$, make the sequence decreasing, and put $B = \bigcap_n A_n$; continuity from above gives $\mu(B) = \mu^*(E)$. If the common value is infinite, every measurable superset of E already has measure ∞ , so any such superset, for example X , is a hull of the same outer measure. $\square$

Example. Let $X = \{a, b\}$ and define

$$\mu^*(\emptyset) = 0, \quad \mu^*(\{a\}) = 2, \quad \mu^*(\{b\}) = 3, \quad \mu^*(X) = 4.$$

The only Carathéodory-measurable sets are $\emptyset$ and X . Hence the regenerated outer measure is 4 on every nonempty set and differs from μ^* on the singletons.

Proposition 2.4.11 (Saturation of the induced measure). The complete measure $\mu^* \upharpoonright \mathcal{M}^*$ obtained from a premeasure is saturated.

Proof. Let E be locally measurable. If $\mu^*(E) < \infty$, choose a finite measurable hull $B \supseteq E$. Then $E = E \cap B$ is measurable by local measurability. If $\mu^*(E) = \infty$ and E were not measurable, the Carathéodory identity would fail for some test set A . The strict failure forces $\mu^*(A) < \infty$. Replacing A by a measurable hull of the same finite outer measure preserves the strict failure, but local measurability makes both intersections with that hull measurable, a contradiction. $\square$

Proposition 2.4.12 (Completion and the Carathéodory domain). If the premeasure μ_0 is sigma-finite, the Carathéodory measure space is the completion of the measure on $\sigma(\mathcal{A})$. Without sigma-finiteness, every finite-measure Carathéodory-measurable set belongs to that completion, and the full Carathéodory space is the saturation of the completion.

Proof. Under sigma-finiteness, every Carathéodory-measurable set has a measurable hull in $\sigma(\mathcal{A})$ whose difference is null, so it belongs to the completion. The same argument works for every finite-measure set without sigma-finiteness. Every Carathéodory-measurable set is locally measurable with respect to the completion, and saturation then recovers the remaining sets. $\square$

Proposition 2.4.13 (Measure on traces). Let $(X, \mathcal{M}, \mu)$ be finite, let μ^* be the outer measure induced by μ , and let $E \subseteq X$ satisfy $\mu^*(E) = \mu(X)$. Put

$$\mathcal{M}_E = \{A \cap E : A \in \mathcal{M}\}, \quad \nu(A \cap E) = \mu(A).$$

Then ν is a well-defined measure on E .

Proof. For every $A \in \mathcal{M}$,

$$\mu(X) = \mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu(A) + \mu(A^c) = \mu(X).$$

Therefore $\mu^*(E \cap A) = \mu(A)$. If $A \cap E = B \cap E$, then $E \cap (A \triangle B) = \emptyset$, so $\mu(A \triangle B) = 0$ and $\mu(A) = \mu(B)$. This proves well-definedness. For disjoint traces, measurable representatives can be disjointified without changing their intersections with E , after which countable additivity follows from that of μ . $\square$

2.5 Lebesgue–Stieltjes and Lebesgue measure

Definition 2.5.1 (Half-open intervals). An h -interval is a set of the form $(a, b]$, where $-\infty \leq a < b \leq \infty$, with the evident conventions at infinite endpoints. Finite disjoint unions of h -intervals form an algebra $\mathcal{A}$ on $\mathbb{R}$.

Theorem 2.5.2 (Lebesgue–Stieltjes premeasure). Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous. Set

$$\mu_0((a, b]) = F(b) - F(a)$$

for finite endpoints and use the monotone limits of F at $\pm\infty$ for unbounded intervals. Extend μ_0 additively to finite disjoint unions of h -intervals. Then μ_0 is a premeasure on $\mathcal{A}$.

Proof. A common refinement by all endpoints shows that the value assigned to a finite union is independent of its representation; the sums telescope on each refined interval.

Suppose first that a bounded interval $(a, b]$ is the disjoint union of intervals $(a_j, b_j]$. Finite partial unions give

$$\mu_0((a, b]) \geq \sum_{j=1}^N \mu_0((a_j, b_j]).$$

For the reverse inequality, fix $\varepsilon > 0$. Right continuity permits a slight rightward enlargement of each covering interval with total added F -increment below ε , and a slight truncation near a with error below ε . The enlarged open intervals cover a compact interval, so a finite subcover exists. Ordering this finite cover and using telescoping yields

$$\mu_0((a, b]) \leq \sum_j \mu_0((a_j, b_j]) + 2\varepsilon.$$

Let $\varepsilon \downarrow 0$. Finite unions and infinite endpoints follow by refinement and monotone limits. $\square$

Theorem 2.5.3 (Lebesgue–Stieltjes measure). There is a unique Borel measure μ_F on $\mathbb{R}$, finite on bounded sets, such that

$$\mu_F((a, b]) = F(b) - F(a).$$

Conversely, every Borel measure on $\mathbb{R}$ that is finite on bounded sets is of this form for an increasing right-continuous function F , unique up to an additive constant.

Proof. Existence and uniqueness follow from the extension theorem and sigma-finiteness of the interval algebra. Conversely, given μ , define

$$F(x) = \begin{cases} \mu((0, x]), & x \geq 0, \\ -\mu((x, 0]), & x < 0. \end{cases}$$

Continuity from above and below gives right continuity, and the interval identity determines μ . $\square$

Theorem 2.5.4 (Regularity on $\mathbb{R}$). Let μ_F be a Lebesgue–Stieltjes measure and let E be measurable in its completion. Then

$$\mu_F(E) = \inf\{\mu_F(U) : E \subseteq U, U \text{ open}\}$$

after the usual interpretation when the value is infinite. If E has finite measure, then

$$\mu_F(E) = \sup\{\mu_F(K) : K \subseteq E, K \text{ compact}\}.$$

Every measurable set differs from a G_δ set and from an F_σ set by a null set.

Proof. Open-interval covers and the defining outer measure give outer regularity. For a bounded finite-measure E , choose a compact interval I containing E and an open set $U \supseteq I \setminus E$ with

$$\mu_F(U) \leq \mu_F(I \setminus E) + \varepsilon.$$

Then $K = I \setminus U$ is compact, $K \subseteq E$, and

$$\mu_F(K) \geq \mu_F(E) - \varepsilon.$$

An increasing bounded exhaustion handles unbounded sets.

For the descriptive-set statements, choose open supersets with summable excess, make them nested, and intersect them to obtain a G_δ hull with null difference. Apply the same construction to the complement to obtain the F_σ representation. A disjoint finite-measure exhaustion handles the sigma-finite case without relying on an invalid assertion that an arbitrary countable union of G_δ sets is again G_δ . $\square$

Definition 2.5.5 (Lebesgue measure). Taking $F(x) = x$ gives Lebesgue measure m on the Borel sets of $\mathbb{R}$. Its completion is also denoted by m , and its domain is the sigma-algebra of Lebesgue-measurable sets.

Theorem 2.5.6 (Translation and dilation). If E is Lebesgue measurable, $t \in \mathbb{R}$, and $r \neq 0$, then $E + t$ and rE are Lebesgue measurable and

$$m(E + t) = m(E), \quad m(rE) = |r|m(E).$$

Proof. First work on Borel sets. The measures $E \mapsto m(E + t)$ and $E \mapsto |r|^{-1}m(rE)$ agree with m on half-open intervals, so uniqueness of the sigma-finite extension gives equality on all Borel sets. Borel null sets are therefore sent to null sets, and the identities pass to the completion. $\square$

Proposition 2.5.7 (The Cantor set). The middle-thirds Cantor set $C \subseteq [0, 1]$ is compact, perfect, totally disconnected, and has Lebesgue measure zero. The Cantor function is continuous and nondecreasing, is constant on each component of C^c , and maps $[0, 1]$ onto $[0, 1]$.

Proof. At stage n , the construction leaves 2^n intervals of length 3^{-n} , whose total length is $(2/3)^n$. Compactness follows from nested closedness, and the remaining topological properties follow from the ternary construction. $\square$

2.6 Consequences and corrected exercises

Theorem 2.6.1 (Positive-measure sets contain nonmeasurable subsets). Every Lebesgue-measurable set $E \subseteq \mathbb{R}$ with $m(E) > 0$ contains a nonmeasurable subset.

Proof. Replace E by a bounded positive-measure subset. On E , declare $x \sim y$ when $x - y \in \mathbb{Q}$, and choose one representative from each equivalence class, obtaining $V \subseteq E$. Distinct rational translates of V are disjoint. If V were measurable with positive measure, infinitely many translates $V + q$ with q in a bounded rational interval would be pairwise disjoint and contained in one bounded interval, which is impossible. Hence $m(V) = 0$. But E is contained in the union of countably many rational translates of V , which would then be null, a contradiction. Thus V is nonmeasurable. $\square$

Proposition 2.6.2 (High density in an interval). If E is Lebesgue measurable and $m(E) > 0$, then for every $\alpha < 1$ there is an open interval I such that

$$m(E \cap I) > \alpha m(I).$$

Proof. For $\alpha \leq 0$, choose an interval I with $m(E \cap I) > 0$; the conclusion is immediate. Thus assume $0 < \alpha < 1$ and replace E by a bounded positive-measure subset. If the conclusion failed, every interval would satisfy $m(E \cap I) \leq \alpha m(I)$. Choose an open-interval cover (I_j) of E with

$$\sum_j m(I_j) < \frac{m(E)}{\alpha}.$$

Then

$$m(E) \leq \sum_j m(E \cap I_j) \leq \alpha \sum_j m(I_j) < m(E),$$

a contradiction. □

Theorem 2.6.3 (Steinhaus). If $E \subseteq \mathbb{R}$ is Lebesgue measurable and $m(E) > 0$, then the difference set

$$E - E = \{x - y : x, y \in E\}$$

contains an open interval centered at 0.

Proof. Choose a compact set $K \subseteq E$ with $m(K) > 0$, and then an open set $U \supseteq K$ with $m(U) < 2m(K)$. Compactness gives $\delta > 0$ such that $K + h \subseteq U$ whenever $|h| < \delta$. If K and $K + h$ were disjoint, translation invariance would give

$$m(U) \geq m(K \cup (K + h)) = 2m(K),$$

a contradiction. Hence $K \cap (K + h) \neq \emptyset$, so $h \in K - K \subseteq E - E$. □

Proposition 2.6.4. There exists a Borel set $A \subseteq [0, 1]$ such that for every nonempty interval $I \subseteq [0, 1]$,

$$0 < m(A \cap I) < m(I).$$

Proof. Enumerate the open intervals with rational endpoints contained in $[0, 1]$ as (I_n) . Recursively choose disjoint compact nowhere-dense sets $K_n, L_n \subseteq I_n$, each of positive measure, and disjoint from all previously chosen sets. This is possible because a finite union of compact nowhere-dense sets has open dense complement in I_n , inside which one can place two small positive-measure fat Cantor sets.

Set $A = \bigcup_n K_n$. Then A is F_σ , hence Borel. Every nonempty interval contains some I_n ; the set K_n gives positive measure inside A , while L_n remains disjoint from every K_j and gives positive measure inside A^c . □

Chapter 3 Integration

3.1 Measurable functions

Definition 3.1.1 (Measurable map). Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A map $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable if

$$f^{-1}(B) \in \mathcal{M} \quad (B \in \mathcal{N}).$$

When the sigma-algebras are understood, we simply say that f is measurable.

Proposition 3.1.2 (Checking a generator). If $\mathcal{N} = \sigma(\mathcal{E})$, then $f : X \rightarrow Y$ is measurable if and only if $f^{-1}(E) \in \mathcal{M}$ for every $E \in \mathcal{E}$.

Proof. The class

$$\mathcal{C} = \{B \subseteq Y : f^{-1}(B) \in \mathcal{M}\}$$

is a sigma-algebra on Y . Thus $\mathcal{E} \subseteq \mathcal{C}$ implies $\mathcal{N} \subseteq \mathcal{C}$. □

Corollary 3.1.3. Every continuous map between topological spaces is Borel measurable.

Proposition 3.1.4 (Real-valued criterion). For $f : X \rightarrow \mathbb{R}$, the following are equivalent:

- (a). f is measurable;
- (b). $\{f > a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$;
- (c). $\{f \geq a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$;
- (d). $\{f < a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$;
- (e). $\{f \leq a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$.

It is enough to test rational a .

Proof. Each listed family of rays generates $\mathcal{B}_{\mathbb{R}}$, and strict and non-strict inequalities are related by countable unions and intersections, for example

$$\{f > a\} = \bigcup_{n=1}^{\infty} \{f \geq a + 1/n\}.$$

□

Definition 3.1.5 (Measurability on a subset). If $E \subseteq X$, the trace sigma-algebra is $\mathcal{M}_E = \{A \cap E : A \in \mathcal{M}\}$. A function $f : E \rightarrow Y$ is measurable when it is $(\mathcal{M}_E, \mathcal{N})$ -measurable.

Proposition 3.1.6 (Coordinate maps). Let $Y = \prod_{\alpha \in A} Y_\alpha$ with the product sigma-algebra. A map $f : X \rightarrow Y$ is measurable if and only if every coordinate $\pi_\alpha \circ f$ is measurable.

Proof. The forward implication follows because the coordinate projections are measurable. Conversely, the inverse image under f of a generating cylinder $\pi_\alpha^{-1}(E)$ is $(\pi_\alpha \circ f)^{-1}(E)$. $\square$

Corollary 3.1.7. A complex-valued function is measurable if and only if its real and imaginary parts are measurable.

Proposition 3.1.8 (Algebraic operations). If $f, g : X \rightarrow \mathbb{C}$ are measurable, then $f + g$, fg , and $|f|$ are measurable. If f and g are real-valued, then $\max(f, g)$ and $\min(f, g)$ are also measurable.

Proof. The map $x \mapsto (f(x), g(x))$ is measurable into $\mathbb{C}^2$. Addition and multiplication are continuous on $\mathbb{C}^2$, and absolute value is continuous. For real-valued functions,

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \quad \min(f, g) = \frac{f + g - |f - g|}{2}.$$

$\square$

Proposition 3.1.9 (Limits). If (f_n) is a sequence of extended-real-valued measurable functions, then

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_n f_n, \quad \liminf_n f_n$$

are measurable. If $f_n(x)$ converges for every x in a measurable set E , the pointwise limit on E is measurable.

Proof. For example,

$$\left\{ \sup_n f_n > a \right\} = \bigcup_n \{f_n > a\},$$

and

$$\limsup_n f_n = \inf_k \sup_{n \geq k} f_n, \quad \liminf_n f_n = \sup_k \inf_{n \geq k} f_n.$$

On the convergence set, the two limits agree. $\square$

Definition 3.1.10 (Positive and negative parts). For a real-valued function f , define

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0).$$

Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Definition 3.1.11 (Simple function). A measurable function $\phi : X \rightarrow [0, \infty)$ is *simple* if it has finite range. Its standard representation is

$$\phi = \sum_{j=1}^n a_j 1_{E_j},$$

where the a_j are distinct nonnegative numbers and the E_j are pairwise disjoint measurable sets.

Theorem 3.1.12 (Dyadic approximation). If $f : X \rightarrow [0, \infty]$ is measurable, there is an increasing sequence of nonnegative simple functions ϕ_n such that $\phi_n \uparrow f$. One convenient choice is

$$\phi_n = \sum_{k=0}^{n2^n-1} k2^{-n} \mathbf{1}_{\{k2^{-n} \leq f < (k+1)2^{-n}\}} + n \mathbf{1}_{\{f \geq n\}}.$$

Moreover, $0 \leq f - \phi_n < 2^{-n}$ on $\{f < n\}$.

Proof. Each level set is measurable. The formula rounds f downward to the nearest dyadic multiple of 2^{-n} below height n and truncates higher values at n . Refinement of the dyadic grid and the increasing truncation level give $\phi_n \leq \phi_{n+1} \leq f$, and the stated error estimate implies pointwise convergence. $\square$

Theorem 3.1.13 (Completeness and almost-everywhere modification). The following statements are equivalent for a measure space $(X, \mathcal{M}, \mu)$:

- (a). μ is complete;
- (b). whenever f is measurable and $g = f$ almost everywhere, then g is measurable;
- (c). whenever f_n are measurable and $f_n \rightarrow f$ almost everywhere, then f is measurable.

Proof. Assume completeness and let N be a measurable null set outside which $f = g$. For every Borel set B ,

$$g^{-1}(B) = (f^{-1}(B) \cap N^c) \cup (g^{-1}(B) \cap N),$$

and the second term is measurable by completeness. The limit assertion follows by replacing the functions on one null exceptional set and using the measurable limsup and liminf.

Conversely, if completeness fails, choose a nonmeasurable $A \subseteq N$ with N measurable and null. Then $\mathbf{1}_A = 0$ almost everywhere but is not measurable, contradicting either assertion. $\square$

Proposition 3.1.14 (Representative on the original sigma-algebra). Let $\overline{\mathcal{M}}$ be the completion of $\mathcal{M}$. Every real- or complex-valued $\overline{\mathcal{M}}$ -measurable function f has an $\mathcal{M}$ -measurable representative g such that $f = g$ almost everywhere.

Proof. For a nonnegative function, replace the measurable sets in the standard representations of simple approximants by $\mathcal{M}$ -measurable sets differing from them only on null sets. A countable union of the exceptional null sets is still null. Outside that union, the resulting $\mathcal{M}$ -measurable simple functions converge to f ; define g as their limit there and set $g = 0$ on the measurable null set. Apply this construction to positive and negative parts, and then to real and imaginary parts, for the general case. $\square$

Remark (Arithmetic on $\overline{\mathbb{R}}$). When extended-real-valued functions are used, products are interpreted with the convention $0(\pm\infty) = 0$. Addition is not intrinsically defined at $(\infty, -\infty)$ and $(-\infty, \infty)$; if a convention is imposed, the resulting operation must be treated as a piecewise Borel map. Proofs should never conceal an expression of the form $\infty - \infty$.

Proposition 3.1.15 (The convergence set). For real-valued measurable functions f_n , the set on which (f_n)

converges to a finite limit is measurable and equals

$$\{\limsup f_n = \liminf f_n\} \cap \{\limsup f_n < \infty\} \cap \{\liminf f_n > -\infty\}.$$

For complex-valued functions, the same conclusion follows from the measurable Cauchy criterion.

Proposition 3.1.16 (Reconstruction from level sets). Suppose $(E_a)_{a \in \mathbb{R}}$ is an increasing family of measurable sets such that

$$E_a = \bigcap_{b>a} E_b, \quad \bigcup_{a \in \mathbb{R}} E_a = X, \quad \bigcap_{a \in \mathbb{R}} E_a = \emptyset.$$

Then there is a measurable $f : X \rightarrow \mathbb{R}$ satisfying

$$\{f \leq a\} = E_a \quad (a \in \mathbb{R}).$$

Proof. For each x , the set $A_x = \{a : x \in E_a\}$ is nonempty, proper, and upward closed. Define

$$f(x) = \inf A_x = \sup\{a : x \notin E_a\}.$$

The right-continuity condition on the family gives $x \in E_a$ exactly when $f(x) \leq a$. Measurability also follows from

$$\{f < a\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q < a}} E_q.$$

□

Proposition 3.1.17 (A Lebesgue-measurable non-Borel set). There exists a Lebesgue-measurable subset of $[0, 1]$ that is not Borel.

Proof. Let c be the Cantor function and define $g(x) = x + c(x)$. The map $g : [0, 1] \rightarrow [0, 2]$ is strictly increasing and continuous, hence a homeomorphism. On each complementary interval of the Cantor set C , the function c is constant, so g preserves the length of that interval. Since the complementary intervals have total length 1, one obtains $m(g(C)) = 1$.

Choose a nonmeasurable set $A \subseteq g(C)$ and put $B = g^{-1}(A) \subseteq C$. Because C is null, B is Lebesgue measurable. If B were Borel, then $A = g(B)$ would be Borel under the homeomorphism, a contradiction.

□

3.2 The integral of nonnegative functions

Definition 3.2.1 (Integral of a simple function). For a nonnegative simple function $\phi = \sum_{j=1}^n a_j \mathbf{1}_{E_j}$ in standard form, define

$$\int_X \phi \, d\mu = \sum_{j=1}^n a_j \mu(E_j),$$

with the convention $0 \cdot \infty = 0$.

Proposition 3.2.2. For nonnegative simple functions ϕ, ψ and $c \geq 0$:

- (a). $\int c\phi = c \int \phi$;
- (b). $\int(\phi + \psi) = \int \phi + \int \psi$;
- (c). $\phi \leq \psi$ implies $\int \phi \leq \int \psi$;
- (d). $E \mapsto \int_E \phi \, d\mu$ is a measure.

Proof. Refine the standard representations by all intersections of their level sets. The assertions then reduce to finite sums and countable additivity of μ . □

Definition 3.2.3 (Nonnegative integral). For a measurable $f : X \rightarrow [0, \infty]$, define

$$\int_X f \, d\mu = \sup \left\{ \int_X \phi \, d\mu : \phi \text{ simple and } 0 \leq \phi \leq f \right\}.$$

Theorem 3.2.4 (Monotone convergence theorem). If $0 \leq f_n \uparrow f$ almost everywhere, then

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

Proof. After setting all functions equal to zero on one measurable null set, assume pointwise increase everywhere. Monotonicity gives $\lim_n \int f_n \leq \int f$. Let $0 \leq \phi \leq f$ be simple and fix $0 < c < 1$. Put

$$E_n = \{f_n \geq c\phi\}.$$

Then $E_n \uparrow X$ on the support of ϕ , and

$$\int f_n \geq c \int_{E_n} \phi.$$

Continuity from below for the measure $A \mapsto \int_A \phi \, d\mu$ gives

$$\lim_n \int f_n \geq c \int \phi.$$

Let $c \uparrow 1$ and take the supremum over ϕ . □

Corollary 3.2.5 (Linearity on nonnegative functions). For nonnegative measurable f, g and $a, b \geq 0$,

$$\int (af + bg) \, d\mu = a \int f \, d\mu + b \int g \, d\mu.$$

Proof. Approximate f and g from below by increasing simple functions and apply monotone convergence. □

Proposition 3.2.6. For a nonnegative measurable function f ,

$$\int f \, d\mu = 0 \iff f = 0 \text{ almost everywhere.}$$

Proof. If $f = 0$ almost everywhere, every simple $\phi \leq f$ vanishes almost everywhere. Conversely,

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \{f \geq 1/n\}.$$

If one level set had positive measure, then $\int f \geq n^{-1} \mu\{f \geq 1/n\} > 0$. □

Corollary 3.2.7. If $0 \leq f_n \uparrow f$ almost everywhere, then $\int f_n \uparrow \int f$ even when the monotonicity and convergence fail on a null set.

Theorem 3.2.8 (Fatou's lemma). For nonnegative measurable functions f_n ,

$$\int \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int f_n \, d\mu.$$

Proof. For each k , the functions $g_k = \inf_{n \geq k} f_n$ increase to $\liminf f_n$. Since $g_k \leq f_n$ for every $n \geq k$,

$$\int g_k \leq \inf_{n \geq k} \int f_n.$$

Apply monotone convergence and let $k \rightarrow \infty$. □

Proposition 3.2.9 (Finite integral). If $f \geq 0$ and $\int f \, d\mu < \infty$, then $f < \infty$ almost everywhere and the set $\{f > 0\}$ is sigma-finite.

Proof. If $E = \{f = \infty\}$ had positive measure, then $a\mathbf{1}_E \leq f$ for every $a > 0$, forcing $\int f = \infty$. Moreover,

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \{f \geq 1/n\}, \quad \mu\{f \geq 1/n\} \leq n \int f.$$

□

Proposition 3.2.10 (A measure with density). Let $f \geq 0$ be measurable and define

$$\lambda(E) = \int_E f \, d\mu.$$

Then λ is a measure. For every nonnegative measurable g ,

$$\int g \, d\lambda = \int gf \, d\mu.$$

Proof. Countable additivity follows from monotone convergence applied to disjoint indicator sums. The integral identity is immediate for indicators, then for simple functions, and finally for arbitrary nonnegative g by monotone convergence. □

Proposition 3.2.11 (Convergence on every measurable subset). Suppose $f_n, f \geq 0$, $f_n \rightarrow f$ pointwise, and

$$\int f_n \, d\mu \rightarrow \int f \, d\mu < \infty.$$

Then for every $E \in \mathcal{M}$,

$$\int_E f_n \, d\mu \rightarrow \int_E f \, d\mu.$$

Proof. Fatou's lemma applied on E and E^c gives lower bounds for both pieces. Since the sums of the two pieces converge to the finite total integral, neither piece can have a limsup larger than its limiting integral. Thus both converge. □

Proposition 3.2.12 (Finite-measure support approximation). If $f \geq 0$ and $\int f < \infty$, then for every $\varepsilon > 0$ there is a measurable set E with $\mu(E) < \infty$ and

$$\int_E f > \int f - \varepsilon.$$

Proof. Choose a simple $\phi \leq f$ with $\int \phi > \int f - \varepsilon$. In its standard representation, every positive coefficient is attached to a finite-measure set, because $\int \phi < \infty$. The union of those finitely many sets is a suitable E . $\square$

3.3 Integrable functions and convergence theorems

Definition 3.3.1 (Signed and complex integrals). For a measurable real-valued f , define

$$\int f = \int f^+ - \int f^-$$

whenever at least one term is finite. The function is *integrable* if $\int |f| < \infty$. A complex-valued function is integrable if its real and imaginary parts are integrable, and then

$$\int f = \int \Re f + i \int \Im f.$$

The space of almost-everywhere equivalence classes of integrable functions is denoted $L^1(\mu)$, with norm

$$\|f\|_1 = \int |f| d\mu.$$

Proposition 3.3.2. The space $L^1(\mu)$ is a complex vector space, the integral is linear, and

$$\left| \int f d\mu \right| \leq \int |f| d\mu.$$

Moreover, $\|f - g\|_1 = 0$ if and only if $f = g$ almost everywhere.

Proof. The real case follows by applying the nonnegative theory to positive and negative parts. For complex f , choose $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $\alpha \int f = |\int f|$. Then

$$\left| \int f \right| = \Re \int \alpha f = \int \Re(\alpha f) \leq \int |f|.$$

The last assertion follows from the zero-integral criterion for nonnegative functions. $\square$

Theorem 3.3.3 (Dominated convergence theorem). Let $f_n \rightarrow f$ almost everywhere and suppose $|f_n| \leq g$ almost everywhere for an integrable g . Then f is integrable and

$$\int f_n d\mu \longrightarrow \int f d\mu, \quad \|f_n - f\|_1 \longrightarrow 0.$$

Proof. The bound passes to the limit, so $|f| \leq g$. For real-valued functions, Fatou's lemma applied to $g + f_n$ and $g - f_n$ yields

$$\int f \leq \liminf_n \int f_n \leq \limsup_n \int f_n \leq \int f.$$

Apply the same argument to $|f_n - f| \leq 2g$ to obtain L^1 convergence. For complex functions, apply the real theorem separately to real and imaginary parts; both are dominated by g . $\square$

Theorem 3.3.4 (Termwise integration of an absolutely summable series). If $(f_n) \subseteq L^1(\mu)$ and

$$\sum_{n=1}^{\infty} \int |f_n| d\mu < \infty,$$

then $\sum_n f_n(x)$ converges absolutely for almost every x , its sum belongs to $L^1(\mu)$, and

$$\int \sum_{n=1}^{\infty} f_n d\mu = \sum_{n=1}^{\infty} \int f_n d\mu.$$

Proof. Monotone convergence gives

$$\int \sum_n |f_n| = \sum_n \int |f_n| < \infty,$$

so the series of absolute values is finite almost everywhere. The partial sums are dominated by $\sum_n |f_n|$, and dominated convergence completes the proof. $\square$

Theorem 3.3.5 (Approximation in $L^1(\mathbb{R})$). For every $f \in L^1(\mathbb{R})$ and $\varepsilon > 0$, there exist

- (a). an integrable simple function ϕ with finite-measure support;
- (b). a simple function whose level sets are finite unions of bounded intervals;
- (c). a continuous compactly supported function g ;

within ε of f in L^1 .

Proof. Truncate f in value and in space, then use dyadic simple approximation to obtain the first statement. By regularity, every finite-measure level set can be approximated in symmetric difference by a finite union of bounded intervals, which gives the second statement. Indicators of bounded intervals can be approximated in L^1 by continuous piecewise-linear functions; linearity gives the final approximation. $\square$

Theorem 3.3.6 (Continuity under the integral sign). Let $f : X \times [a, b] \rightarrow \mathbb{C}$. Assume $f(\cdot, t)$ is measurable for every t , $f(x, \cdot)$ is continuous for almost every x , and

$$|f(x, t)| \leq g(x)$$

for an integrable g . Then

$$F(t) = \int_X f(x, t) d\mu(x)$$

is continuous on $[a, b]$.

Proof. If $t_n \rightarrow t$, then $f(x, t_n) \rightarrow f(x, t)$ almost everywhere and the sequence is dominated by g . Apply dominated convergence. $\square$

Theorem 3.3.7 (Differentiation under the integral sign). Assume $f(\cdot, t)$ is measurable for every $t \in [a, b]$,

$f(x, \cdot)$ is differentiable on (a, b) for almost every x , and

$$\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$$

for all $t \in (a, b)$ and an integrable g . If $f(\cdot, t_0)$ is integrable for one $t_0 \in (a, b)$, then $f(\cdot, t)$ is integrable for every $t \in (a, b)$, and $F : (a, b) \rightarrow \mathbb{C}$ defined by $F(t) = \int f(x, t) d\mu(x)$ is differentiable with

$$F'(t) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x).$$

Proof. The derivative bound and the mean value theorem show that every section $f(\cdot, t)$ is integrable. For $h \neq 0$, the difference quotient is bounded by g and converges almost everywhere to $\partial_t f(x, t)$. Dominated convergence gives the derivative formula. $\square$

Theorem 3.3.8 (Riemann and Lebesgue integration). If $f : [a, b] \rightarrow \mathbb{R}$ is bounded and Riemann integrable, then f is Lebesgue measurable and integrable, and the two integrals agree. More generally, a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has Lebesgue measure zero.

Proof. Choose nested partitions with mesh tending to zero. The lower step functions increase, the upper step functions decrease, and their integrals converge to the lower and upper Riemann integrals. Their pointwise limits agree at every continuity point. Thus the two limiting step functions agree almost everywhere exactly when the discontinuity set is null, and in that case their common almost-everywhere value is f . The Lebesgue integral is then squeezed between the lower and upper sums. $\square$

Proposition 3.3.9 (Uniform convergence). If $\mu(X) < \infty$, $f_n \in L^1(\mu)$, and $f_n \rightarrow f$ uniformly, then $f \in L^1(\mu)$ and

$$\|f_n - f\|_1 \leq \mu(X) \|f_n - f\|_\infty \rightarrow 0.$$

On an infinite-measure space this can fail.

Example. On $\mathbb{R}$,

$$f_n = \frac{1}{n} \mathbf{1}_{[0, n]}$$

converges uniformly to 0, but $\int f_n = 1$. Also,

$$f_n = f \mathbf{1}_{[-n, n]}, \quad f(x) = \frac{1}{1 + |x|},$$

converges uniformly to the nonintegrable function f .

Theorem 3.3.10 (Generalized dominated convergence). Suppose $f_n \rightarrow f$ almost everywhere, $|f_n| \leq g_n$, the functions g_n are nonnegative and integrable, $g_n \rightarrow g$ almost everywhere, and

$$\int g_n d\mu \rightarrow \int g d\mu < \infty.$$

Then $f \in L^1$ and $\int f_n \rightarrow \int f$.

Proof. The limit satisfies $|f| \leq g$. For real-valued f_n , Fatou's lemma applied to the nonnegative functions $g_n + f_n$ and $g_n - f_n$ gives the two inequalities needed to squeeze the integrals of f_n to $\int f$. For complex-valued functions, apply the real-valued result separately to the real and imaginary parts, both of which are dominated in absolute value by g_n . $\square$

Theorem 3.3.11 (Scheffé's lemma). Let $f_n, f \geq 0$. If $f_n \rightarrow f$ almost everywhere and $\int f_n \rightarrow \int f < \infty$, then

$$\|f_n - f\|_1 \rightarrow 0.$$

The same conclusion holds if convergence is only in measure.

Proof. Since $f_n \wedge f \rightarrow f$ almost everywhere and $0 \leq f_n \wedge f \leq f$, dominated convergence gives

$$\int (f_n \wedge f) \rightarrow \int f.$$

Now use

$$\|f_n - f\|_1 = \int f_n + \int f - 2 \int (f_n \wedge f).$$

For convergence in measure, every subsequence has a further subsequence converging almost everywhere; the preceding argument then forces every subsequence of the L^1 distances to have a further subsequence tending to zero. $\square$

Proposition 3.3.12 (Simple-function sandwiches on a completion). Assume $\mu(X) < \infty$, let $\overline{\mathcal{M}}$ be the completion of $\mathcal{M}$, and let $f : X \rightarrow \mathbb{R}$ be bounded. Then f is $\overline{\mathcal{M}}$ -measurable if and only if there are $\mathcal{M}$ -measurable simple functions ϕ_n, ψ_n such that

$$\phi_n \leq f \leq \psi_n \quad \text{almost everywhere,} \quad \int (\psi_n - \phi_n) d\mu < 2^{-n}.$$

In that case,

$$\int \phi_n d\mu \rightarrow \int f d\overline{\mu} \leftarrow \int \psi_n d\mu.$$

Proof. Replace f by an $\mathcal{M}$ -measurable representative g . Choose lower and upper simple approximations with pointwise gap below $2^{-n}/(1 + \mu(X))$. They sandwich f outside the null set on which f and g differ, and

$$\int (\psi_n - \phi_n) d\mu < 2^{-n}.$$

Conversely, take the union of the measurable null exceptional sets in the sandwiches and modify the simple functions there so that $\phi_n \leq \psi_n$ everywhere. Since

$$\sum_n \int (\psi_n - \phi_n) d\mu < \infty,$$

monotone convergence applied to the series of nonnegative gaps shows that $\sum_n (\psi_n - \phi_n) < \infty$ almost everywhere. Hence the gaps tend to zero, and the inequalities force $\phi_n \rightarrow f$ and $\psi_n \rightarrow f$ almost everywhere. The limit is an $\mathcal{M}$ -measurable representative of f , proving completed measurability. The integral convergence follows from the same sandwich estimate. $\square$

Example (Dense singularities). Let

$$h(x) = x^{-1/2} \mathbf{1}_{(0,1)}(x),$$

let (r_n) enumerate $\mathbb{Q}$, and set

$$g(x) = \sum_{n=1}^{\infty} 2^{-n} h(x - r_n).$$

Then $g \in L^1(\mathbb{R})$ and

$$\int g = 2.$$

The function is essentially unbounded on every interval and has no continuous representative obtained by changing it on a null set. Moreover, g^2 is not integrable on any interval.

Proof. Termwise integration gives $\int g = \sum_n 2^{-n} \int h = 2$. Every interval contains some r_n , and near that point the n th summand exceeds any prescribed bound on a set of positive measure. Hence g is essentially unbounded on every interval, which is incompatible with equality almost everywhere to a continuous function. Finally, on a one-sided neighborhood of r_n ,

$$g(x)^2 \geq 4^{-n} (x - r_n)^{-1},$$

whose integral diverges. □

Theorem 3.3.13 (Continuity of an indefinite integral). If $f \in L^1(\mathbb{R})$ and

$$F(x) = \int_{(-\infty, x]} f(t) dt,$$

then F is continuous.

Proof. The Lebesgue integral is absolutely continuous: for every $\varepsilon > 0$ there is $\delta > 0$ such that $m(A) < \delta$ implies $\int_A |f| < \varepsilon$. Therefore

$$|F(x+h) - F(x)| \leq \int_{[\min(x, x+h), \max(x, x+h)]} |f(t)| dt$$

is smaller than ε when $|h| < \delta$. □

Exercise 3.3.14 (An instructive series). Let $0 < a < b$ and

$$f_n(x) = ae^{-nax} - be^{-nbx}, \quad x \geq 0.$$

Then

$$\sum_{n=1}^{\infty} \int_0^{\infty} |f_n(x)| dx = \infty, \quad \sum_{n=1}^{\infty} \int_0^{\infty} f_n(x) dx = 0,$$

but $\sum_n f_n \in L^1(0, \infty)$ and

$$\int_0^{\infty} \sum_{n=1}^{\infty} f_n(x) dx = \log \frac{b}{a}.$$

Proof. The sign change of f_n occurs at $x = c/n$, where

$$c = \frac{\log(b/a)}{b-a}.$$

A direct computation gives

$$\int_0^\infty |f_n(x)| dx = \frac{2(e^{-ac} - e^{-bc})}{n},$$

so the absolute integrals form a divergent harmonic series. Each signed integral is zero. On the other hand,

$$\sum_{n=1}^\infty f_n(x) = \frac{a}{e^{ax} - 1} - \frac{b}{e^{bx} - 1},$$

which is bounded near 0 and integrable at infinity. Its antiderivative is

$$\log(1 - e^{-ax}) - \log(1 - e^{-bx}),$$

and evaluation at 0 and ∞ gives $\log(b/a)$. □

3.4 Modes of convergence

Definition 3.4.1 (Convergence in measure). A sequence of measurable functions f_n converges in measure to f if for every $\varepsilon > 0$,

$$\mu\{|f_n - f| > \varepsilon\} \longrightarrow 0.$$

It is *Cauchy in measure* if for every $\varepsilon > 0$,

$$\mu\{|f_n - f_m| > \varepsilon\} \longrightarrow 0 \quad (n, m \rightarrow \infty).$$

Proposition 3.4.2. Convergence in L^1 implies convergence in measure.

Proof. For every $\varepsilon > 0$, Markov's inequality gives

$$\mu\{|f_n - f| > \varepsilon\} \leq \frac{1}{\varepsilon} \int |f_n - f| d\mu.$$

□

Theorem 3.4.3 (Cauchy in measure). If (f_n) is Cauchy in measure, there is a measurable f such that $f_n \rightarrow f$ in measure. Moreover, there is a subsequence converging to f almost everywhere. Limits in measure are unique almost everywhere.

Proof. Choose a subsequence $g_j = f_{n_j}$ such that

$$\mu\{|g_{j+1} - g_j| > 2^{-j}\} < 2^{-j}.$$

Let $A_j = \{|g_{j+1} - g_j| > 2^{-j}\}$. Since $\sum_j \mu(A_j) < \infty$,

$$\mu(\limsup_j A_j) = 0.$$

Outside this null set, only finitely many bad increments occur, and the tail is dominated by the convergent series $\sum_j 2^{-j}$. Thus $g_j(x)$ is Cauchy almost everywhere; let f be its limit there and define it to be zero on the exceptional set.

More quantitatively,

$$\mu\{|g_j - f| > 2^{-j+1}\} \leq \sum_{k \geq j} \mu(A_k) \leq 2^{-j+1},$$

so $g_j \rightarrow f$ in measure. Since the original sequence is Cauchy in measure, for fixed j and sufficiently large n ,

$$\mu\{|f_n - g_j| > \varepsilon/2\}$$

is small. Combining this with $g_j \rightarrow f$ in measure proves $f_n \rightarrow f$ in measure.

If $f_n \rightarrow f$ and $f_n \rightarrow g$ in measure, then

$$\mu\{|f - g| > \varepsilon\} \leq \mu\{|f - f_n| > \varepsilon/2\} + \mu\{|f_n - g| > \varepsilon/2\},$$

so $f = g$ almost everywhere. □

Corollary 3.4.4 (Subsequence principle). If $f_n \rightarrow f$ in measure, there is a subsequence $f_{n_j} \rightarrow f$ almost everywhere. More generally, a numerical property $a_n \rightarrow 0$ follows once every subsequence has a further subsequence along which the property holds.

Proof. Choose n_j so that

$$\mu\{|f_{n_j} - f| > 2^{-j}\} < 2^{-j}$$

and apply the same summable-exceptional-set argument. □

Theorem 3.4.5 (Egoroff). Suppose $\mu(X) < \infty$ and $f_n \rightarrow f$ almost everywhere. For every $\varepsilon > 0$, there is a measurable $A \subseteq X$ with $\mu(A) < \varepsilon$ such that $f_n \rightarrow f$ uniformly on $X \setminus A$.

Proof. After changing the functions on a null set, assume convergence everywhere. For $k, n \geq 1$, put

$$E_{n,k} = \bigcup_{m \geq n} \{|f_m - f| \geq 1/k\}.$$

For fixed k , the sets $E_{n,k}$ decrease to $\emptyset$. Since $\mu(X) < \infty$, continuity from above gives $\mu(E_{n,k}) \rightarrow 0$. Choose n_k so that $\mu(E_{n_k,k}) < \varepsilon 2^{-k}$ and let $A = \bigcup_k E_{n_k,k}$. If $x \notin A$ and $m \geq n_k$, then $|f_m(x) - f(x)| < 1/k$, which is uniform convergence. □

Definition 3.4.6 (Almost uniform convergence). The sequence f_n converges *almost uniformly* to f if for every $\varepsilon > 0$ there is a measurable A with $\mu(A) < \varepsilon$ such that $f_n \rightarrow f$ uniformly on $X \setminus A$.

Proposition 3.4.7. Almost uniform convergence implies almost-everywhere convergence and convergence in measure.

Proof. Choose exceptional sets A_k with $\mu(A_k) < 2^{-k}$. Outside $\limsup A_k$, which is null, each point is excluded from all sufficiently large A_k and hence sees pointwise convergence. For convergence in measure, fix $\delta > 0$, choose A with arbitrarily small measure, and use uniform convergence on A^c . □

Theorem 3.4.8 (Fatou under convergence in measure). If $f_n \geq 0$ and $f_n \rightarrow f$ in measure, then

$$\int f \, d\mu \leq \liminf_n \int f_n \, d\mu.$$

Proof. Choose a subsequence along which the integrals converge to the liminf. That subsequence has a further subsequence converging almost everywhere to f . Apply Fatou's lemma to the subsubsequence. $\square$

Theorem 3.4.9 (Dominated convergence in measure). If $f_n \rightarrow f$ in measure and $|f_n| \leq g$ for an integrable g , then

$$\|f_n - f\|_1 \rightarrow 0 \quad \text{and} \quad \int f_n \rightarrow \int f.$$

Proof. Every subsequence has a further subsequence converging almost everywhere to f . Dominated convergence gives L^1 convergence along that subsubsequence. Therefore no subsequence of $\|f_n - f\|_1$ can stay bounded away from zero. $\square$

Proposition 3.4.10 (Limits of indicator functions). If $\mu(E_n) < \infty$ and $\mathbf{1}_{E_n} \rightarrow f$ in L^1 , then $f = \mathbf{1}_E$ almost everywhere for some measurable set E .

Proof. Choose a subsequence converging almost everywhere. At every point where a sequence of zeros and ones converges, its limit is zero or one. Thus f is almost everywhere $\{0, 1\}$ -valued. The set $E = \{f > 1/2\}$ is measurable and has the required property. $\square$

Proposition 3.4.11 (Composition and convergence). Let $\Phi : \mathbb{C} \rightarrow \mathbb{C}$ be continuous.

- (a). If $f_n \rightarrow f$ almost everywhere, then $\Phi(f_n) \rightarrow \Phi(f)$ almost everywhere.
- (b). If Φ is uniformly continuous, it preserves uniform, almost uniform, and in-measure convergence.

The uniform-continuity hypothesis in the second statement cannot in general be omitted.

Proof. The first assertion is pointwise continuity. For the second, choose δ from uniform continuity. Then

$$\{|\Phi(f_n) - \Phi(f)| > \varepsilon\} \subseteq \{|f_n - f| > \delta\},$$

which proves the in-measure assertion; the uniform and almost uniform cases are immediate. $\square$

Theorem 3.4.12 (Dominated Egoroff theorem). Suppose $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ for some $g \in L^1(\mu)$. Then $f_n \rightarrow f$ almost uniformly, even when $\mu(X) = \infty$.

Proof. Partition $\{g > 0\}$ into the finite-measure set $\{g \geq 1\}$ and the measurable sets

$$B_k = \left\{ \frac{1}{k+1} \leq g < \frac{1}{k} \right\}, \quad k \geq 1.$$

Given $\varepsilon > 0$, apply Egoroff on each member of this countable partition and remove a set A_k of measure below a summable allocation of ε . Let A be the union of those exceptional sets.

For a prescribed tolerance $\eta > 0$, choose K so large that $2/K < \eta$. On the tail $\bigcup_{k \geq K} B_k$, the domination gives $|f_n - f| \leq 2g < \eta$. On the finitely many remaining pieces after removal of their exceptional sets, choose a common index from uniform convergence. On $\{g = 0\}$ all the functions and the limit vanish.

This proves uniform convergence on $X \setminus A$. $\square$

Theorem 3.4.13 (Lusin). Let $f : [a, b] \rightarrow \mathbb{C}$ be Lebesgue measurable and finite almost everywhere. For every $\varepsilon > 0$, there is a compact set $K \subseteq [a, b]$ such that

$$m([a, b] \setminus K) < \varepsilon$$

and $f \upharpoonright K$ is continuous.

Proof. Choose M so that $m\{|f| > M\} < \varepsilon/3$, and let f_M be the bounded truncation of f . The function f_M belongs to $L^1([a, b])$, so choose continuous functions g_n converging to it in L^1 . A subsequence converges almost everywhere; Egoroff gives a measurable set of measure below $\varepsilon/3$ outside which the convergence is uniform. Remove also the set $\{|f| > M\}$, where f and f_M may differ. By inner regularity, choose a compact subset of the remaining measurable set losing less than another $\varepsilon/3$ of measure. On that compact set, $f = f_M$ is the uniform limit of continuous functions. $\square$

Proposition 3.4.14 (Uniformity in a parameter). Let $\mu(X) < \infty$ and let $f : X \times [0, 1] \rightarrow \mathbb{C}$ satisfy: $f(\cdot, y)$ is measurable for every y , and $f(x, \cdot)$ is continuous for every x . For $0 < \eta, \delta < 1$, define

$$E_{\eta, \delta} = \{x : |f(x, y) - f(x, 0)| \leq \eta \text{ for all } 0 \leq y < \delta\}.$$

Then $E_{\eta, \delta}$ is measurable. Moreover, for every $\varepsilon > 0$, there is a measurable A with $\mu(A) < \varepsilon$ such that

$$f(x, y) \rightarrow f(x, 0)$$

uniformly for $x \in X \setminus A$ as $y \downarrow 0$.

Proof. Continuity in y gives

$$E_{\eta, \delta} = X \setminus \bigcup_{q \in \mathbb{Q} \cap [0, \delta)} \{x : |f(x, q) - f(x, 0)| > \eta\},$$

so the set is measurable. For fixed k , the sets $E_{1/k, 1/n}$ increase to X as $n \rightarrow \infty$. Choose n_k such that the complement has measure below $\varepsilon 2^{-k}$, and remove the union of these complements. The resulting estimates are uniform in x . $\square$

3.5 Product measures and iterated integration

Definition 3.5.1 (Product measure). Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. On measurable rectangles define

$$\pi(A \times B) = \mu(A)\nu(B),$$

with $0 \cdot \infty = 0$, and extend π additively to the algebra of finite disjoint unions of rectangles. The measure obtained from the Carathéodory construction is denoted $\mu \times \nu$. If both measures are sigma-finite, it is the unique measure on $\mathcal{M} \otimes \mathcal{N}$ satisfying

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B).$$

Definition 3.5.2 (Sections). For $E \subseteq X \times Y$, define

$$E_x = \{y \in Y : (x, y) \in E\}, \quad E^y = \{x \in X : (x, y) \in E\}.$$

For a function $f : X \times Y \rightarrow \overline{\mathbb{R}}$, define $f_x(y) = f(x, y)$ and $f^y(x) = f(x, y)$.

Proposition 3.5.3 (Measurable sections). If $E \in \mathcal{M} \otimes \mathcal{N}$, then $E_x \in \mathcal{N}$ for every x and $E^y \in \mathcal{M}$ for every y . If f is product-measurable, then all its sections are measurable.

Proof. The sets whose sections are measurable form a sigma-algebra containing all rectangles. For functions, apply the set result to inverse images of Borel sets. $\square$

Definition 3.5.4 (Monotone class). A collection $\mathcal{C} \subseteq \mathcal{P}(X)$ is a *monotone class* if it is closed under increasing countable unions and decreasing countable intersections.

Theorem 3.5.5 (Monotone class lemma). If $\mathcal{A}$ is an algebra, then the monotone class generated by $\mathcal{A}$ equals $\sigma(\mathcal{A})$.

Proof. Let $\mathcal{C}$ be the monotone class generated by $\mathcal{A}$. For fixed $E \in \mathcal{C}$, define

$$\mathcal{C}(E) = \{F \in \mathcal{C} : E \setminus F, F \setminus E, E \cap F \in \mathcal{C}\}.$$

This is a monotone class. If $E \in \mathcal{A}$, then $\mathcal{A} \subseteq \mathcal{C}(E)$, so $\mathcal{C}(E) = \mathcal{C}$. It follows in turn that for every $F \in \mathcal{C}$, one has $\mathcal{A} \subseteq \mathcal{C}(F)$ and hence $\mathcal{C}(F) = \mathcal{C}$. Thus $\mathcal{C}$ is closed under relative complements and finite intersections; since $X \in \mathcal{A}$, it is an algebra. Closure under increasing unions now implies closure under arbitrary countable unions, so $\mathcal{C}$ is a sigma-algebra. $\square$

Theorem 3.5.6 (Section formula). Suppose μ and ν are sigma-finite and $E \in \mathcal{M} \otimes \mathcal{N}$. Then the functions

$$x \mapsto \nu(E_x), \quad y \mapsto \mu(E^y)$$

are measurable and

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y).$$

Proof. First assume both measures are finite. The formula is immediate for rectangles. Let $\mathcal{C}$ be the class of sets for which the first section function is measurable and the first equality holds. Increasing unions are handled by monotone convergence. Decreasing intersections are handled by continuity from above, using finiteness. Thus $\mathcal{C}$ is a monotone class containing the rectangle algebra, so the monotone class lemma gives $\mathcal{C} = \mathcal{M} \otimes \mathcal{N}$. The second equality is symmetric.

For sigma-finite measures, choose disjoint finite-measure partitions $X = \bigsqcup_n X_n$ and $Y = \bigsqcup_m Y_m$, apply the finite result on each $X_n \times Y_m$, and sum by monotone convergence. $\square$

Theorem 3.5.7 (Fubini–Tonelli). Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be sigma-finite.

(a). If $f : X \times Y \rightarrow [0, \infty]$ is measurable, then the section integrals are measurable and

$$\int_{X \times Y} f \, d(\mu \times \nu) = \int_X \left(\int_Y f(x, y) \, d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) \, d\mu(x) \right) d\nu(y).$$

(b). If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for almost every x , $f^y \in L^1(\mu)$ for almost every y , both iterated integrals are integrable, and the same equality holds.

Proof. For an indicator function, the assertion is the section formula. It follows for nonnegative simple functions by linearity and for arbitrary nonnegative functions by monotone convergence. This proves Tonelli's theorem.

If $f \in L^1$, apply Tonelli to $|f|$. The resulting finite iterated integral shows that almost every section is integrable. Apply the nonnegative result to f^+ and f^- , or separately to real and imaginary parts in the complex case. $\square$

Theorem 3.5.8 (Completed product spaces). Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be complete sigma-finite measure spaces, and let λ be the completion of $\mu \times \nu$. If f is λ -measurable and either nonnegative or integrable, then its sections are measurable for almost every base point and the Fubini–Tonelli identities remain valid.

Proof. A product-null measurable set has null sections for almost every x and almost every y by the section formula. Choose an $(\mathcal{M} \otimes \mathcal{N})$ -measurable representative g equal to f outside a λ -null set. Fubini–Tonelli applies to g , and completeness of the factor measures makes the corresponding sections of f measurable almost everywhere. The integrals agree because $f = g$ almost everywhere. $\square$

Exercise 3.5.9 (A diagonal when sigma-finiteness fails). Let $X = Y = [0, 1]$, let μ be Lebesgue measure, let ν be counting measure, and let

$$D = \{(x, x) : x \in [0, 1]\}.$$

The diagonal is a Borel subset of $[0, 1]^2$, hence product-measurable. Then

$$\int_X \nu(D_x) \, d\mu(x) = 1, \quad \int_Y \mu(D^y) \, d\nu(y) = 0, \quad (\mu \times \nu)(D) = \infty.$$

Proof. Every vertical section is a singleton of counting measure 1, whereas every horizontal section is a singleton of Lebesgue measure zero.

To compute the product outer measure, suppose D had a rectangle cover $(A_j \times B_j)$ of finite total cost. Whenever B_j is infinite, finiteness of the cost forces $\mu(A_j) = 0$. Such rectangles cover diagonal coordinates only in a null set. Rectangles with finite B_j cover only countably many diagonal coordinates altogether. A null set together with a countable set cannot cover $[0, 1]$, a contradiction. Thus every rectangle cover has infinite cost. $\square$

Exercise 3.5.10 (Unequal iterated sums). On $\mathbb{N}^2$ with counting measure, define

$$f(m, n) = \begin{cases} 1, & m = n, \\ -1, & m = n + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\sum_{m=1}^{\infty} \left(\sum_{n=1}^{\infty} f(m, n) \right) = 1, \quad \sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} f(m, n) \right) = 0.$$

Proof. For fixed m , the row sum is 1 when $m = 1$ and 0 otherwise. For fixed n , the two nonzero column entries cancel. Since $\sum_{m,n} |f(m, n)| = \infty$, Fubini's theorem does not apply. $\square$

Proposition 3.5.11 (Subgraphs). Let $f : X \rightarrow [0, \infty]$ be measurable and define

$$G_f = \{(x, t) \in X \times [0, \infty) : 0 \leq t < f(x)\}.$$

Then G_f is measurable and

$$(\mu \times m)(G_f) = \int_X f \, d\mu.$$

The same conclusion holds with $0 \leq t \leq f(x)$.

Proof. The extended-real map $(x, t) \mapsto f(x) - t$ is measurable, so G_f is its inverse image of $(0, \infty]$. Its vertical section at x has length $f(x)$, and Tonelli gives the formula. $\square$

Proposition 3.5.12 (Products of integrable functions). Let the measure spaces be arbitrary. If $f \in L^1(\mu)$ and $g \in L^1(\nu)$, then

$$h(x, y) = f(x)g(y)$$

belongs to $L^1(\mu \times \nu)$ and

$$\int_{X \times Y} h \, d(\mu \times \nu) = \left(\int_X f \, d\mu \right) \left(\int_Y g \, d\nu \right).$$

Proof. The function h is measurable because $(x, y) \mapsto (f(x), g(y))$ is measurable and multiplication is continuous. Approximate $|f|$ and $|g|$ from below by simple functions. The rectangle formula gives

$$\int |f(x)g(y)| \, d(\mu \times \nu) = \int |f| \, d\mu \int |g| \, d\nu < \infty.$$

Approximating f and g by complex simple functions and passing in L^1 gives the signed formula. $\square$

Proposition 3.5.13 (A countable counting-measure factor). Let Y be countable with counting measure. For every measurable $E \subseteq X \times Y$,

$$(\mu \times \#)(E) = \sum_{y \in Y} \mu(E^y).$$

For every nonnegative measurable f ,

$$\int_{X \times Y} f \, d(\mu \times \#) = \sum_{y \in Y} \int_X f(x, y) \, d\mu(x).$$

Proof. The disjoint decomposition

$$E = \bigsqcup_{y \in Y} E^y \times \{y\}$$

gives the set formula. Apply it to simple functions and then use monotone convergence. $\square$

3.6 Lebesgue measure in Euclidean space and changes of variables

Definition 3.6.1. Lebesgue measure m_n on $\mathbb{R}^n$ is the completion of the n -fold product of one-dimensional Lebesgue measure. Its measurable sets are called Lebesgue-measurable subsets of $\mathbb{R}^n$.

Theorem 3.6.2 (Regularity and approximation in $\mathbb{R}^n$). For every Lebesgue-measurable $E \subseteq \mathbb{R}^n$,

$$m_n(E) = \inf\{m_n(U) : E \subseteq U, U \text{ open}\}.$$

If $m_n(E) < \infty$, then

$$m_n(E) = \sup\{m_n(K) : K \subseteq E, K \text{ compact}\}.$$

Every measurable set differs from a G_δ set and from an F_σ set by a null set. Moreover, every $f \in L^1(\mathbb{R}^n)$ can be approximated in L^1 by simple functions supported on finite unions of rectangles, and then by continuous compactly supported functions.

Proof. Open sets are countable unions of rectangles with rational endpoints. The product outer measure therefore gives outer approximation by such rectangles. For a finite-measure set, work inside a large bounded rectangle and apply outer regularity to the complement to obtain compact inner approximation. A monotone exhaustion handles infinite sets. The G_δ and F_σ statements follow from nested open approximations and their complements.

For functions, first approximate by finite-support simple functions. Approximate each finite-measure level set by a finite union of rectangles in symmetric difference. Finally, approximate rectangle indicators by continuous functions that are 1 on a slightly smaller rectangle and taper linearly to 0 in a thin boundary layer. $\square$

Theorem 3.6.3 (Translation invariance). For $a \in \mathbb{R}^n$, let $\tau_a(x) = x + a$. If E is Lebesgue measurable and f is nonnegative or integrable, then

$$m_n(E + a) = m_n(E), \quad \int_{\mathbb{R}^n} f(x + a) \, dx = \int_{\mathbb{R}^n} f(x) \, dx.$$

Proof. On Borel sets, $E \mapsto m_n(E + a)$ is a sigma-finite measure agreeing with m_n on rectangles, so uniqueness gives equality. Translations therefore preserve Borel null sets and hence the completion. The integral formula follows first for indicators and simple functions, then by monotone convergence and decomposition into positive and negative parts. $\square$

Theorem 3.6.4 (Invertible linear changes of variables). Let $T \in GL(n, \mathbb{R})$. If E is Lebesgue measurable, then $T(E)$ is Lebesgue measurable and

$$m_n(T(E)) = |\det T| m_n(E).$$

If f is nonnegative or integrable, then

$$\int_{\mathbb{R}^n} f(Tx) dx = \frac{1}{|\det T|} \int_{\mathbb{R}^n} f(y) dy.$$

Proof. First work with Borel sets. Every invertible matrix is a product of elementary matrices: coordinate permutations, nonzero diagonal scalings, and shears. A permutation preserves measure. A diagonal scaling follows from the one-dimensional formula and Fubini. A shear has determinant 1 and preserves measure because one coordinate is translated by a function of the remaining coordinates; Fubini reduces the calculation to translation invariance on each section. Composing the elementary formulas gives the determinant factor.

The image of a Borel null set is null by the formula. Hence an arbitrary Lebesgue-measurable set, written as a Borel set modified by a null set, is sent to a Lebesgue-measurable set. The integral formula follows from the set formula by the standard indicator-simple-monotone approximation argument. $\square$

Theorem 3.6.5 (Nonlinear change of variables). Let $\Omega \subseteq \mathbb{R}^n$ be open and let $G : \Omega \rightarrow G(\Omega)$ be a C^1 diffeomorphism. If f is nonnegative measurable or integrable on $G(\Omega)$, then

$$\int_{G(\Omega)} f(y) dy = \int_{\Omega} f(G(x)) |\det DG(x)| dx.$$

In particular, for measurable $E \subseteq \Omega$,

$$m_n(G(E)) = \int_E |\det DG(x)| dx.$$

Proof. We outline the standard local-linearization proof. On a compact set $K \Subset \Omega$, the derivative DG is uniformly continuous and $|\det DG|$ is bounded away from zero. For every $\varepsilon > 0$, sufficiently small cubes $Q \subset K$ satisfy

$$G(x) = G(x_Q) + DG(x_Q)(x - x_Q) + R_Q(x), \quad |R_Q(x) - R_Q(z)| \leq \varepsilon|x - z|.$$

The linear change-of-variables theorem and elementary inner and outer estimates then give

$$(1 - \eta_\varepsilon) |\det DG(x_Q)| m_n(Q) \leq m_n(G(Q)) \leq (1 + \eta_\varepsilon) |\det DG(x_Q)| m_n(Q),$$

where $\eta_\varepsilon \rightarrow 0$ as $\varepsilon \rightarrow 0$.

Partition K into small cubes, discard a boundary set of arbitrarily small measure, and sum these estimates. This proves the set formula first for compact sets and then, by regularity, for all measurable sets contained in K . Exhaust Ω by compact sets. The function formula follows for indicators, then for simple functions, nonnegative functions by monotone convergence, and integrable functions by applying the result to positive and negative parts. $\square$

Exercise 3.6.6 (Failure of Fubini without absolute integrability). On $(0, 1)^2$, define

$$f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

Then the two iterated integrals exist but are unequal:

$$\int_0^1 \left(\int_0^1 f(x, y) dx \right) dy = -\frac{\pi}{4}, \quad \int_0^1 \left(\int_0^1 f(x, y) dy \right) dx = \frac{\pi}{4}.$$

Consequently, f is not integrable on $(0, 1)^2$.

Proof. Observe that

$$\frac{x^2 - y^2}{(x^2 + y^2)^2} = -\frac{\partial}{\partial x} \frac{x}{x^2 + y^2} = \frac{\partial}{\partial y} \frac{y}{x^2 + y^2}.$$

Integrating the first identity in x and the second in y gives the displayed values. If f were integrable, Fubini's theorem would force equality. $\square$

Proposition 3.6.7 (A triangular integral operator). Let $f \in L^1(0, a)$ and define

$$g(x) = \int_x^a \frac{f(t)}{t} dt, \quad 0 < x < a.$$

Then $g \in L^1(0, a)$ and

$$\int_0^a g(x) dx = \int_0^a f(t) dt.$$

Proof. First verify absolute integrability on the triangle:

$$\int_0^a \int_x^a \frac{|f(t)|}{t} dt dx = \int_0^a \frac{|f(t)|}{t} \int_0^t dx dt = \int_0^a |f(t)| dt < \infty.$$

Fubini's theorem then permits the same interchange without absolute values and gives the formula. $\square$

Definition 3.6.8 (Fractional integral). For $\alpha > 0$ and a suitable function f on $[0, \infty)$, define

$$(I_\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x - t)^{\alpha-1} f(t) dt.$$

Proposition 3.6.9 (Semigroup property). Whenever the relevant integrals are absolutely convergent,

$$I_\alpha(I_\beta f) = I_{\alpha+\beta} f.$$

If $n \in \mathbb{N}$ and f is continuous, then $I_n f$ is n times differentiable and

$$\frac{d^n}{dx^n} I_n f = f.$$

Proof. Fubini gives

$$I_\alpha(I_\beta f)(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x f(u) \int_u^x (x - t)^{\alpha-1} (t - u)^{\beta-1} dt du.$$

With $t = u + s(x - u)$, the inner integral equals

$$(x - u)^{\alpha+\beta-1} B(\alpha, \beta) = (x - u)^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

This proves the semigroup identity. The differentiation formula follows by iterating the fundamental theorem of calculus. $\square$

3.7 Polar coordinates

Definition 3.7.1 (Surface measure). Let

$$S^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$$

and let

$$\Phi : (0, \infty) \times S^{n-1} \longrightarrow \mathbb{R}^n \setminus \{0\}, \quad \Phi(r, \omega) = r\omega.$$

For a Borel set $E \subseteq S^{n-1}$, define

$$\sigma(E) = n m_n\{r\omega : 0 < r \leq 1, \omega \in E\}.$$

Theorem 3.7.2 (Polar-coordinate formula). The set function σ is a finite Borel measure on S^{n-1} . If f is nonnegative measurable or integrable on $\mathbb{R}^n$, then

$$\int_{\mathbb{R}^n} f(x) dx = \int_{S^{n-1}} \int_0^\infty f(r\omega) r^{n-1} dr d\sigma(\omega).$$

Proof. The definition is countably additive because disjoint Borel subsets of the sphere generate disjoint cones. Let ρ be the measure on $(0, \infty)$ with density r^{n-1} with respect to Lebesgue measure. Pull Lebesgue measure back through the Borel isomorphism Φ . On a rectangle $(a, b] \times E$,

$$\begin{aligned} m_n\{r\omega : a < r \leq b, \omega \in E\} &= m_n(bC_E) - m_n(aC_E) \\ &= \frac{b^n - a^n}{n} \sigma(E) \\ &= \rho((a, b]) \sigma(E), \end{aligned}$$

where $C_E = \{r\omega : 0 < r \leq 1, \omega \in E\}$. Uniqueness of the sigma-finite product extension gives the equality of measures on $(0, \infty) \times S^{n-1}$. The integral formula follows by transporting indicators, then simple and general functions. $\square$

Corollary 3.7.3 (Radial functions). If $f(x) = g(|x|)$ is nonnegative or integrable, then

$$\int_{\mathbb{R}^n} f(x) dx = \sigma(S^{n-1}) \int_0^\infty g(r) r^{n-1} dr.$$

More generally, if $f(x) = g(|x|)h(x/|x|)$ is measurable and integrable (or nonnegative), then the integral factors into the corresponding radial and spherical integrals.

Proposition 3.7.4 (Area of the sphere and volume of the ball). One has

$$\sigma(S^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(n/2)}$$

and, for the unit ball B^n ,

$$m_n(B^n) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

Proof. Fubini and the one-dimensional Gaussian integral give

$$\int_{\mathbb{R}^n} e^{-|x|^2} dx = \pi^{n/2}.$$

Polar coordinates give

$$\pi^{n/2} = \sigma(S^{n-1}) \int_0^\infty e^{-r^2} r^{n-1} dr = \frac{1}{2} \sigma(S^{n-1}) \Gamma(n/2).$$

The ball formula follows from

$$m_n(B^n) = \int_{S^{n-1}} \int_0^1 r^{n-1} dr d\sigma = \frac{1}{n} \sigma(S^{n-1}).$$

□

Proposition 3.7.5 (Absolute monomials on the sphere). Let $\alpha_j > -1$ and put $\beta_j = (\alpha_j + 1)/2$. Then

$$\int_{S^{n-1}} \prod_{j=1}^n |\omega_j|^{\alpha_j} d\sigma(\omega) = \frac{2 \prod_{j=1}^n \Gamma(\beta_j)}{\Gamma(\beta_1 + \cdots + \beta_n)}.$$

Proof. Evaluate

$$\int_{\mathbb{R}^n} e^{-|x|^2} \prod_{j=1}^n |x_j|^{\alpha_j} dx$$

in two ways. Fubini gives $\prod_j \Gamma(\beta_j)$. Polar coordinates give the desired spherical integral times

$$\int_0^\infty e^{-r^2} r^{n-1 + \sum_j \alpha_j} dr = \frac{1}{2} \Gamma\left(\sum_j \beta_j\right).$$

Equating the two expressions yields the formula.

□

3.8 Absolute continuity, uniform integrability, and tightness

Lemma 3.8.1 (Finite partition for Lebesgue measure). If $E \subseteq \mathbb{R}$ is measurable with $m(E) < \infty$ and $\delta > 0$, then E is the disjoint union of finitely many measurable sets, each of measure below δ .

Proof. Choose N so large that $m(E \setminus [-N, N]) < \delta$. Partition $[-N, N]$ into finitely many half-open intervals of length below δ , intersect them with E , and add the remaining set $E \setminus [-N, N]$. □

Theorem 3.8.2 (Absolute continuity of the integral). If $f \in L^1(E)$, then for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$A \subseteq E, \quad m(A) < \delta \quad \implies \quad \int_A |f| < \varepsilon.$$

Conversely, if $m(E) < \infty$, f is measurable, and there are numbers $\delta > 0$ and $C < \infty$ such that $m(A) < \delta$ implies $\int_A |f| \leq C$, then $f \in L^1(E)$.

Proof. Choose $M > 0$ so that

$$\int_{\{|f|>M\}} |f| < \varepsilon/2$$

and take $\delta = \varepsilon/(2M)$. Then

$$\int_A |f| \leq Mm(A) + \int_{\{|f|>M\}} |f| < \varepsilon.$$

For the converse, partition E into finitely many sets of measure below δ and sum the corresponding bounds. $\square$

Definition 3.8.3 (Uniform integrability). A family $\mathcal{F} \subseteq L^1(E)$ is *uniformly integrable* on E if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$m(A) < \delta \implies \sup_{f \in \mathcal{F}} \int_A |f| < \varepsilon.$$

Proposition 3.8.4. Every finite family of integrable functions is uniformly integrable.

Proof. Apply absolute continuity of the integral to each member and take the minimum of the finitely many resulting δ values. $\square$

Theorem 3.8.5 (Vitali convergence on a finite-measure set). Let $m(E) < \infty$. Suppose $f_n \rightarrow f$ almost everywhere on E and (f_n) is uniformly integrable. Then $f \in L^1(E)$ and

$$\|f_n - f\|_{L^1(E)} \longrightarrow 0.$$

In particular, $\int_E f_n \rightarrow \int_E f$.

Proof. First choose δ so that every set of measure below δ carries less than 1 of the L^1 mass of every f_n . Partition E into finitely many such sets. Fatou's lemma on each piece shows that f is integrable.

Now fix $\varepsilon > 0$ and choose δ so that

$$\sup_n \int_A |f_n| < \varepsilon/4$$

whenever $m(A) < \delta$. Fatou's lemma also gives $\int_A |f| \leq \varepsilon/4$. By Egoroff's theorem, choose $A \subseteq E$ with $m(A) < \delta$ such that $f_n \rightarrow f$ uniformly on $E \setminus A$. If $m(E \setminus A) > 0$, take n large enough that

$$\sup_{E \setminus A} |f_n - f| < \frac{\varepsilon}{2m(E \setminus A)};$$

the zero-measure case is immediate. Then

$$\int_E |f_n - f| \leq \int_A |f_n| + \int_A |f| + \int_{E \setminus A} |f_n - f| < \varepsilon.$$

$\square$

Corollary 3.8.6. Let $m(E) < \infty$, let $h_n \geq 0$ be integrable, and suppose $h_n \rightarrow 0$ almost everywhere. Then

$$\int_E h_n \longrightarrow 0$$

if and only if (h_n) is uniformly integrable.

Proof. Uniform integrability implies convergence by Vitali's theorem. Conversely, if the integrals tend to zero, the tail is uniformly controlled because $\int_A h_n \leq \int_E h_n$, and the finitely many initial functions form a uniformly integrable family. $\square$

Example (Why finite measure matters). On $\mathbb{R}$, the functions

$$f_n = \mathbf{1}_{[n, n+1]}$$

are uniformly integrable and converge pointwise to 0, but $\int f_n = 1$ for every n . Their mass escapes to infinity.

Example (Signed cancellation does not imply uniform integrability). On $[-1, 1]$, let

$$h_n = n(\mathbf{1}_{[0, 1/n]} - \mathbf{1}_{[-1/n, 0]}).$$

Then $\int h_n = 0$ for every n , but the family is not uniformly integrable because

$$m([0, 1/n]) \rightarrow 0, \quad \int_0^{1/n} |h_n| = 1.$$

Proposition 3.8.7 (Three useful tests and counterexamples). On $[0, 1]$:

- (a). The unit ball $\{f : \int |f| \leq 1\}$ of L^1 is not uniformly integrable.
- (b). The family $\{f : |f| \leq 1\}$ is uniformly integrable.
- (c). Even among continuous functions, a uniform bound on signed integrals over intervals does not imply uniform integrability.

Proof. For the first assertion, use $f_n = n\mathbf{1}_{[0, 1/n]}$. The second follows from $\int_A |f| \leq m(A)$. For the third, let

$$f_n(x) = n^2 \sin(2\pi n^2 x).$$

Every interval integral has absolute value at most $1/\pi$, but with $A_n = [0, 1/n]$,

$$m(A_n) \rightarrow 0, \quad \int_{A_n} |f_n(x)| dx = \frac{2n}{\pi},$$

so uniform integrability fails. $\square$

Definition 3.8.8 (Tightness). A family $\mathcal{F} \subseteq L^1(E)$ is *tight* on E if for every $\varepsilon > 0$ there is a measurable $E_0 \subseteq E$ with $m(E_0) < \infty$ such that

$$\sup_{f \in \mathcal{F}} \int_{E \setminus E_0} |f| < \varepsilon.$$

Theorem 3.8.9 (Vitali convergence on a general space). Let $f_n \rightarrow f$ almost everywhere on a measurable set E . If (f_n) is uniformly integrable and tight on E , then $f \in L^1(E)$ and

$$\|f_n - f\|_{L^1(E)} \rightarrow 0.$$

Proof. Given $\varepsilon > 0$, choose a finite-measure $E_0 \subseteq E$ such that

$$\sup_n \int_{E \setminus E_0} |f_n| < \varepsilon.$$

Fatou's lemma gives the same bound for f . On E_0 , the restrictions remain uniformly integrable, so the finite-measure Vitali theorem gives

$$\int_{E_0} |f_n - f| \rightarrow 0.$$

Therefore

$$\int_E |f_n - f| \leq \int_{E_0} |f_n - f| + \int_{E \setminus E_0} |f_n| + \int_{E \setminus E_0} |f|$$

is eventually smaller than a constant multiple of ε . □

Corollary 3.8.10. Let $h_n \geq 0$ be integrable and suppose $h_n \rightarrow 0$ almost everywhere on E . Then

$$\int_E h_n \rightarrow 0$$

if and only if (h_n) is uniformly integrable and tight.

Proof. The reverse implication is the general Vitali theorem. Conversely, if the integrals tend to zero, the tail is uniformly integrable and tight simply because its total mass is small. The finitely many initial functions are uniformly integrable, and each can be made small outside a suitable finite-measure set; taking the union of those finitely many sets proves tightness of the whole sequence. □

Chapter 4 Differentiation on the Real Line

Throughout this section, m denotes Lebesgue measure on $\mathbb{R}$. Intervals are assumed nondegenerate unless stated otherwise.

4.5 Maximal functions and the differentiation theorem

Proposition 4.5.1 (Markov's inequality). Let $(X, \mathcal{M}, \mu)$ be a measure space and let $h \in L^1(\mu)$. For every $c > 0$,

$$\mu(\{x : |h(x)| \geq c\}) \leq \frac{\|h\|_1}{c}.$$

Proof. If $E_c = \{|h| \geq c\}$, then

$$\|h\|_1 \geq \int_{E_c} |h| \, d\mu \geq c\mu(E_c).$$

□

Lemma 4.5.2 (Three-times covering lemma). Let $I_1, \dots, I_N$ be bounded open intervals. There is a pairwise disjoint subcollection $I_{j_1}, \dots, I_{j_k}$ such that

$$\bigcup_{i=1}^N I_i \subseteq \bigcup_{\ell=1}^k 3I_{j_\ell},$$

where $3I$ is the interval with the same center as I and three times its length.

Proof. Choose an interval of maximal length, remove every interval meeting it, and repeat. The selected intervals are disjoint. Every discarded interval I meets a selected interval J with $|I| \leq |J|$. The distance between their centers is at most $(|I| + |J|)/2 \leq |J|$, so $I \subseteq 3J$. □

Definition 4.5.3 (Hardy–Littlewood maximal function). For $h \in L^1_{\text{loc}}(\mathbb{R})$, define

$$Mh(x) = \sup_{r>0} \frac{1}{2r} \int_{x-r}^{x+r} |h(y)| \, dy.$$

Theorem 4.5.4 (Weak $(1,1)$ maximal inequality on $\mathbb{R}$). If $h \in L^1(\mathbb{R})$ and $c > 0$, then

$$m(\{x : Mh(x) > c\}) \leq \frac{3}{c} \|h\|_1.$$

Proof. Let $E_c = \{Mh > c\}$. For each $x \in E_c$, choose an interval I_x centered at x such that

$$\frac{1}{|I_x|} \int_{I_x} |h| > c.$$

If $K \subseteq E_c$ is compact, finitely many I_x cover K . Apply the three-times covering lemma to obtain pairwise disjoint $I_1, \dots, I_N$ such that $K \subseteq \bigcup_j 3I_j$. Then

$$m(K) \leq 3 \sum_{j=1}^N |I_j| < \frac{3}{c} \sum_{j=1}^N \int_{I_j} |h| \leq \frac{3}{c} \|h\|_1.$$

The set E_c is open, because one interval witnessing $Mh(x) > c$ also witnesses the inequality at all nearby centers after a small enlargement. Inner regularity of Lebesgue measure now gives the estimate for E_c . $\square$

Theorem 4.5.5 (Lebesgue differentiation theorem, oscillation form). If $f \in L^1(\mathbb{R})$, then for almost every $x \in \mathbb{R}$,

$$\lim_{r \downarrow 0} \frac{1}{2r} \int_{x-r}^{x+r} |f(y) - f(x)| \, dy = 0.$$

Proof. For a locally integrable function u , set

$$Du(x) = \limsup_{r \downarrow 0} \frac{1}{2r} \int_{x-r}^{x+r} |u(y) - u(x)| \, dy.$$

If $g \in C_c(\mathbb{R})$, uniform continuity gives $Dg = 0$ everywhere. For arbitrary $f \in L^1$ and $g \in C_c$,

$$Df(x) \leq M(f - g)(x) + |f(x) - g(x)| + Dg(x).$$

Consequently, for $\alpha > 0$,

$$\{Df > 2\alpha\} \subseteq \{M(f - g) > \alpha\} \cup \{|f - g| > \alpha\}.$$

The maximal inequality and Markov's inequality imply

$$m(\{Df > 2\alpha\}) \leq \frac{4}{\alpha} \|f - g\|_1.$$

Because $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$, the right-hand side can be made arbitrarily small. Thus $m(\{Df > 2\alpha\}) = 0$ for every rational $\alpha > 0$, and hence $Df = 0$ almost everywhere. $\square$

Corollary 4.5.6 (Lebesgue differentiation theorem, average form). If $f \in L^1_{\text{loc}}(\mathbb{R})$, then

$$\lim_{r \downarrow 0} \frac{1}{2r} \int_{x-r}^{x+r} f(y) \, dy = f(x)$$

for almost every $x \in \mathbb{R}$.

Proof. Localize f to a bounded interval and use

$$\left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) \, dy - f(x) \right| \leq \frac{1}{2r} \int_{x-r}^{x+r} |f(y) - f(x)| \, dy.$$

$\square$

Corollary 4.5.7 (Differentiation of indefinite integrals). Let $f \in L^1(\mathbb{R})$ and define

$$F(x) = \int_{-\infty}^x f(t) \, dt.$$

Then $F'(x) = f(x)$ for almost every x .

Proof. For $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

The one-sided version of the differentiation theorem follows from the symmetric version by comparing $[x, x+h]$ with $[x-h, x+h]$. The same argument applies for $h < 0$. $\square$

Definition 4.5.8 (Density). For a measurable set $E \subseteq \mathbb{R}$, the density of E at x is the limit, when it exists,

$$\lim_{r \downarrow 0} \frac{m(E \cap (x-r, x+r))}{2r}.$$

Theorem 4.5.9 (Lebesgue density theorem). If $E \subseteq \mathbb{R}$ is measurable, then the density of E is 1 at almost every point of E and 0 at almost every point of E^c .

Proof. Apply the differentiation theorem locally to $\mathbf{1}_E$. The limiting average is $\mathbf{1}_E(x)$ almost everywhere. $\square$

Exercise 4.5.10. There is no measurable set $E \subseteq [0, 1]$ satisfying

$$m(E \cap [0, x]) = \frac{x}{2} \quad (0 \leq x \leq 1).$$

Proof. The function $G(x) = m(E \cap [0, x])$ is absolutely continuous and satisfies $G' = \mathbf{1}_E$ almost everywhere.

If $G(x) = x/2$, then $G' = 1/2$ almost everywhere, contradicting $\mathbf{1}_E \in \{0, 1\}$ almost everywhere. $\square$

4.6 Vitali coverings and monotone functions

Proposition 4.6.1 (Prescribed countable discontinuity set). If $C = \{c_1, c_2, \dots\} \subset (a, b)$ is countable, then

$$f(x) = \sum_{n=1}^{\infty} 2^{-n} \mathbf{1}_{[c_n, b)}(x)$$

is increasing and has discontinuity set exactly C .

Proof. The series converges uniformly, so f is well defined and increasing. At c_n the n th summand has a jump of size 2^{-n} , hence f is discontinuous. If $x \notin C$, choose a finite partial sum whose tail has norm below ε , and use continuity of that finite sum at x . $\square$

Definition 4.6.2 (Vitali cover). A family $\mathcal{V}$ of closed bounded intervals is a *Vitali cover* of $E \subseteq \mathbb{R}$ if for every $x \in E$ and every $\varepsilon > 0$ there is $I \in \mathcal{V}$ with $x \in I$ and $|I| < \varepsilon$.

Theorem 4.6.3 (Vitali covering theorem). Let $E \subseteq \mathbb{R}$ have finite outer measure, and let $\mathcal{V}$ be a Vitali cover of E . Then there is a finite or countable pairwise disjoint family $(I_k) \subseteq \mathcal{V}$ such that

$$m^* \left(E \setminus \bigcup_k I_k \right) = 0.$$

Proof. Choose an open set $O \supseteq E$ with $m(O) < \infty$, and discard intervals not contained in O . If finitely many disjoint intervals already cover E , there is nothing to prove. Otherwise choose $I_1 \in \mathcal{V}$, and, after

$I_1, \dots, I_n$ have been chosen, let s_n be the supremum of the lengths of intervals in $\mathcal{V}$ disjoint from the selected ones. Choose I_{n+1} disjoint from the preceding intervals with $|I_{n+1}| > s_n/2$.

The intervals are disjoint and lie in O , so $\sum_k |I_k| < \infty$ and hence $|I_k| \rightarrow 0$. We claim

$$E \setminus \bigcup_{k=1}^n I_k \subseteq \bigcup_{k>n} 5I_k.$$

Indeed, if x lies in the left-hand side, take $I \in \mathcal{V}$ containing x and disjoint from $I_1, \dots, I_n$. It must meet some later I_N ; otherwise it would remain eligible forever, contradicting $|I_k| \rightarrow 0$. Taking the first such N , we have $|I| \leq s_{N-1} < 2|I_N|$, and the intersection implies $I \subseteq 5I_N$. Therefore

$$m^* \left(E \setminus \bigcup_k I_k \right) \leq 5 \sum_{k>n} |I_k| \xrightarrow{n \rightarrow \infty} 0.$$

□

Definition 4.6.4 (Dini derivatives). For a real-valued function f and an interior point x of its domain, define

$$\overline{D}f(x) = \limsup_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad \underline{D}f(x) = \liminf_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

The one-sided Dini derivatives are defined analogously.

Lemma 4.6.5 (A level-set estimate for an increasing function). Let f be increasing on $[a, b]$. For $\alpha > 0$,

$$m^* (\{x \in (a, b) : \overline{D}f(x) > \alpha\}) \leq \frac{f(b) - f(a)}{\alpha}.$$

In particular,

$$m^* (\{x : \overline{D}f(x) = \infty\}) = 0.$$

Proof. Let

$$E_\alpha = \{x \in (a, b) : \overline{D}f(x) > \alpha\},$$

and fix $0 < \beta < \alpha$. For every $x \in E_\alpha$, there are arbitrarily short intervals $[c, d] \subset (a, b)$ containing x for which

$$f(d) - f(c) > \beta(d - c).$$

Indeed, this follows directly from the definition of the upper Dini derivative, using $[x, x+h]$ when $h > 0$ and $[x+h, x]$ when $h < 0$. These intervals form a Vitali cover of E_α . Select a pairwise disjoint subfamily $([c_k, d_k])$ covering E_α up to a null set. Since f is increasing and the intervals are disjoint,

$$\begin{aligned} m^*(E_\alpha) &\leq \sum_k (d_k - c_k) \\ &< \frac{1}{\beta} \sum_k (f(d_k) - f(c_k)) \\ &\leq \frac{f(b) - f(a)}{\beta}. \end{aligned}$$

Letting $\beta \uparrow \alpha$ proves the first assertion. Finally,

$$\{\overline{D}f = \infty\} \subseteq \{\overline{D}f > n\}$$

for every n , and the preceding estimate tends to zero as $n \rightarrow \infty$.

□

Theorem 4.6.6 (Lebesgue's theorem for monotone functions). Every monotone function on an interval is differentiable almost everywhere.

Proof. It is enough to consider an increasing function f on a bounded interval (a, b) . For rational numbers $0 < \beta < \alpha$, define

$$E_{\alpha, \beta} = \{x \in (a, b) : \overline{D}f(x) > \alpha > \beta > \underline{D}f(x)\}.$$

We show $m^*(E_{\alpha, \beta}) = 0$. Fix $\varepsilon > 0$ and choose an open set O with

$$E_{\alpha, \beta} \subseteq O \subseteq (a, b), \quad m(O) \leq m^*(E_{\alpha, \beta}) + \varepsilon.$$

The intervals $[c, d] \subseteq O$ for which

$$f(d) - f(c) < \beta(d - c)$$

form a Vitali cover of $E_{\alpha, \beta}$. Select disjoint intervals $[c_k, d_k]$ covering all but a set of outer measure below ε .

On each selected interval, apply the preceding lemma to the restriction of f and the set $E_{\alpha, \beta} \cap [c_k, d_k]$:

$$m^*(E_{\alpha, \beta} \cap [c_k, d_k]) \leq \frac{f(d_k) - f(c_k)}{\alpha}.$$

Hence

$$\begin{aligned} m^*(E_{\alpha, \beta}) &\leq \varepsilon + \frac{1}{\alpha} \sum_k (f(d_k) - f(c_k)) \\ &< \varepsilon + \frac{\beta}{\alpha} \sum_k (d_k - c_k) \\ &\leq \varepsilon + \frac{\beta}{\alpha} (m^*(E_{\alpha, \beta}) + \varepsilon). \end{aligned}$$

Since $\beta/\alpha < 1$ and the outer measure is finite, letting $\varepsilon \downarrow 0$ gives $m^*(E_{\alpha, \beta}) = 0$.

At every point where the upper and lower Dini derivatives differ, or one is infinite, the point belongs to one of countably many sets of the form just considered or to an infinite-derivative level set. Thus the exceptional set is null. $\square$

Corollary 4.6.7. If f is increasing on $[a, b]$, then $f' \in L^1(a, b)$ and

$$\int_a^b f'(x) \, dx \leq f(b) - f(a).$$

Proof. Extend f to $[a, b+1]$ by $f(x) = f(b)$ for $x > b$. For $h > 0$, set

$$D_h f(x) = \frac{f(x+h) - f(x)}{h} \geq 0.$$

At almost every $x \in (a, b)$, $D_h f(x) \rightarrow f'(x)$. Fatou's lemma and monotonicity give

$$\begin{aligned} \int_a^b f' &\leq \liminf_{h \downarrow 0} \int_a^b D_h f(x) \, dx \\ &= \liminf_{h \downarrow 0} \frac{1}{h} \left(\int_b^{b+h} f - \int_a^{a+h} f \right) \leq f(b) - f(a). \end{aligned}$$

$\square$

4.7 Functions of bounded variation

Definition 4.7.1 (Variation). Let $f : [a, b] \rightarrow \mathbb{R}$ and let $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ be a partition.

Define

$$V(f, P) = \sum_{j=1}^n |f(x_j) - f(x_{j-1})|, \quad \text{TV}(f; [a, b]) = \sup_P V(f, P).$$

The function f is of *bounded variation* on $[a, b]$ if $\text{TV}(f; [a, b]) < \infty$.

Proposition 4.7.2 (Variation function). If $f \in \text{BV}([a, b])$, define

$$V_f(x) = \text{TV}(f; [a, x]), \quad a \leq x \leq b.$$

Then V_f is increasing, and both $V_f + f$ and $V_f - f$ are increasing.

Proof. Variation is additive over adjacent intervals:

$$\text{TV}(f; [a, v]) = \text{TV}(f; [a, u]) + \text{TV}(f; [u, v]) \quad (a \leq u \leq v \leq b).$$

Also $|f(v) - f(u)| \leq \text{TV}(f; [u, v])$. Therefore

$$V_f(v) - V_f(u) \geq |f(v) - f(u)|,$$

which is equivalent to the monotonicity of $V_f \pm f$. □

Theorem 4.7.3 (Jordan decomposition for functions). A real-valued function f on $[a, b]$ is of bounded variation if and only if it is the difference of two increasing functions.

Proof. If $f = g - h$ with g, h increasing, then for every partition P ,

$$V(f, P) \leq g(b) - g(a) + h(b) - h(a).$$

Conversely, if $f \in \text{BV}([a, b])$, then

$$g(x) = \frac{V_f(x) + f(x) - f(a)}{2}, \quad h(x) = \frac{V_f(x) - f(x) + f(a)}{2}$$

are increasing and $f = f(a) + g - h$. □

Corollary 4.7.4. Every function of bounded variation is differentiable almost everywhere. Its derivative belongs to $L^1(a, b)$ and

$$\int_a^b |f'(x)| dx \leq \text{TV}(f; [a, b]).$$

Proof. Write $f = g - h$ with g, h increasing. Both are differentiable almost everywhere, so $f' = g' - h'$ almost everywhere. Moreover,

$$\int_a^b |f'| \leq \int_a^b (g' + h') \leq g(b) - g(a) + h(b) - h(a).$$

Using the canonical Jordan decomposition above gives the stated bound. □

Proposition 4.7.5 (Algebra properties). If $f, g \in \text{BV}([a, b])$ and $\alpha, \beta \in \mathbb{C}$, then $\alpha f + \beta g \in \text{BV}([a, b])$. Since BV functions on a compact interval are bounded, $fg \in \text{BV}([a, b])$ as well, and

$$\text{TV}(fg) \leq \|f\|_\infty \text{TV}(g) + \|g\|_\infty \text{TV}(f).$$

Proof. The linear assertion follows directly from the triangle inequality. For a partition,

$$|f_j g_j - f_{j-1} g_{j-1}| \leq |f_j| |g_j - g_{j-1}| + |g_{j-1}| |f_j - f_{j-1}|.$$

Summing and taking the supremum gives the product estimate. $\square$

Proposition 4.7.6 (Lower semicontinuity of total variation). If $f_n : [a, b] \rightarrow \mathbb{R}$ converges pointwise to f , then

$$\text{TV}(f; [a, b]) \leq \liminf_{n \rightarrow \infty} \text{TV}(f_n; [a, b]).$$

Proof. For every fixed partition P ,

$$V(f, P) = \lim_{n \rightarrow \infty} V(f_n, P) \leq \liminf_{n \rightarrow \infty} \text{TV}(f_n; [a, b]).$$

Take the supremum over P . $\square$

Example (Oscillatory examples). Define $f(0) = g(0) = 0$ and, for $x \neq 0$,

$$f(x) = x^2 \cos(1/x^2), \quad g(x) = x^2 \cos(1/x).$$

Then f is not of bounded variation near 0. Indeed, at

$$x_n = (n\pi)^{-1/2}$$

the values alternate in sign and have magnitude $1/(n\pi)$, so suitable partition sums dominate a constant multiple of the harmonic series. By contrast, g is of bounded variation on $[-1, 1]$ because

$$g'(x) = 2x \cos(1/x) + \sin(1/x) \in L^1(-1, 1).$$

More generally, for $\alpha, \beta > 0$,

$$h(x) = x^\alpha \sin(x^{-\beta}), \quad h(0) = 0,$$

belongs to $\text{BV}([0, 1])$ exactly when $\alpha > \beta$.

Proof. The preceding argument proves the first assertion. When $\alpha > \beta$, the derivative

$$h'(x) = \alpha x^{\alpha-1} \sin(x^{-\beta}) - \beta x^{\alpha-\beta-1} \cos(x^{-\beta})$$

is integrable near 0. If $\alpha \leq \beta$, choose successive points where the sine equals 1 and -1 . The corresponding variation dominates a constant multiple of

$$\sum_{n=1}^{\infty} n^{-\alpha/\beta},$$

which diverges. $\square$

Proposition 4.7.7. Assume $f \in C([a, b]) \cap C^1((a, b))$. Then

$$f \in BV([a, b]) \iff f' \in L^1(a, b).$$

Proof. The forward implication is the preceding corollary. Conversely, on every compact subinterval $[u, v] \subset (a, b)$,

$$f(v) - f(u) = \int_u^v f'(t) dt.$$

By continuity at the endpoints this identity extends to $a \leq u < v \leq b$. Hence for every partition P ,

$$V(f, P) \leq \int_a^b |f'|,$$

so f has bounded variation. □

Remark. A continuous function need not have bounded variation on any nondegenerate interval. Any continuous nowhere differentiable function provides an example, because every BV function is differentiable almost everywhere.

4.8 Absolute continuity

Definition 4.8.1 (Absolute continuity). A function $f : [a, b] \rightarrow \mathbb{R}$ is *absolutely continuous* if for every $\varepsilon > 0$ there is $\delta > 0$ such that for every finite pairwise disjoint collection of intervals $(a_k, b_k) \subset [a, b]$,

$$\sum_k (b_k - a_k) < \delta \implies \sum_k |f(b_k) - f(a_k)| < \varepsilon.$$

Proposition 4.8.2. Every Lipschitz function on $[a, b]$ is absolutely continuous.

Proof. If $|f(v) - f(u)| \leq L|v - u|$, take $\delta = \varepsilon/L$. □

Theorem 4.8.3 (Jordan decomposition inside AC). If $f \in AC([a, b])$, then $f \in BV([a, b])$. Moreover, f is the difference of two increasing absolutely continuous functions.

Proof. Choose δ for $\varepsilon = 1$. Every partition can be grouped into at most $1 + (b - a)/\delta$ subcollections whose total interval lengths are below δ . Hence the variation of f is uniformly bounded.

Let V_f be the variation function. For pairwise disjoint intervals (c_k, d_k) of sufficiently small total length, refine partitions inside each interval and combine them. The absolute continuity of f gives

$$\sum_k (V_f(d_k) - V_f(c_k)) < \varepsilon.$$

Thus V_f is absolutely continuous. The canonical functions

$$g = \frac{V_f + f - f(a)}{2}, \quad h = \frac{V_f - f + f(a)}{2}$$

are increasing and absolutely continuous, and $f = f(a) + g - h$. □

Remark (Strict inclusions). On a compact interval,

$$C^1 \subsetneq \text{Lip} \subsetneq \text{AC} \subsetneq \text{BV}.$$

For example, $|x|$ is Lipschitz but not C^1 , $\sqrt{x}$ is absolutely continuous on $[0, 1]$ but not Lipschitz, and the Cantor function is continuous and of bounded variation but not absolutely continuous. A step function shows that BV functions need not be continuous.

Definition 4.8.4 (Divided differences and local averages). Extend $f \in C([a, b])$ constantly outside $[a, b]$. For $h > 0$, define

$$D_h f(x) = \frac{f(x+h) - f(x)}{h}, \quad A_h f(x) = \frac{1}{h} \int_x^{x+h} f(t) dt.$$

Then, for $a \leq u < v \leq b$,

$$\int_u^v D_h f(x) dx = A_h f(v) - A_h f(u).$$

Theorem 4.8.5 (Absolute continuity and uniform integrability). Let $f \in C([a, b])$. Then f is absolutely continuous if and only if the family

$$\{D_h f : 0 < h \leq 1\}$$

is uniformly integrable on $[a, b]$.

Proof. Assume first that f is absolutely continuous. By the Jordan decomposition inside AC, write

$$f = f(a) + g - k,$$

where g and k are increasing and absolutely continuous. Extend them constantly to $[a, b+1]$. It is enough to treat g , because

$$|D_h f| \leq D_h g + D_h k.$$

Given $\varepsilon > 0$, choose $\delta > 0$ from the absolute continuity of g . Let $E \subseteq [a, b]$ be measurable with $m(E) < \delta$. Ignoring the endpoints, which form a null set, choose a relatively open set

$$E \cap (a, b) \subseteq O = \bigsqcup_j (a_j, b_j) \subseteq (a, b), \quad m(O) < \delta.$$

Since $D_h g \geq 0$, Tonelli's theorem and the identity for local averages give, for $0 < h \leq 1$,

$$\begin{aligned} \int_E D_h g &\leq \sum_j \int_{a_j}^{b_j} D_h g(x) dx \\ &= \frac{1}{h} \int_0^h \sum_j (g(b_j + t) - g(a_j + t)) dt \\ &\leq \varepsilon. \end{aligned}$$

For each fixed t , the translated intervals remain pairwise disjoint and have total length below δ ; the estimate for a countable family follows from the finite one by monotone convergence. Thus the divided differences of g , and similarly those of k , are uniformly integrable.

Conversely, suppose the divided differences are uniformly integrable. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$m(E) < \delta \implies \sup_{0 < h \leq 1} \int_E |D_h f| < \varepsilon.$$

For pairwise disjoint intervals (c_j, d_j) of total length below δ ,

$$\begin{aligned} \sum_j |A_h f(d_j) - A_h f(c_j)| &= \sum_j \left| \int_{c_j}^{d_j} D_h f(x) \, dx \right| \\ &\leq \int_{\cup_j (c_j, d_j)} |D_h f| < \varepsilon. \end{aligned}$$

The constant extension of f is uniformly continuous on $[a, b+1]$, so $A_h f \rightarrow f$ uniformly on $[a, b]$. Letting $h \downarrow 0$ gives the defining estimate for absolute continuity. $\square$

Example (Local absolute continuity need not be global). The function

$$f(x) = \begin{cases} x \sin(1/x), & x > 0, \\ 0, & x = 0, \end{cases}$$

is absolutely continuous on $[\varepsilon, 1]$ for every $\varepsilon > 0$, but it is not of bounded variation on $[0, 1]$ and hence is not absolutely continuous there.

Proposition 4.8.6. Let $f : [0, 1] \rightarrow \mathbb{R}$ be increasing and continuous. If f is absolutely continuous on $[\varepsilon, 1]$ for every $\varepsilon > 0$, then f is absolutely continuous on $[0, 1]$.

Proof. Fix $\varepsilon > 0$. By continuity at 0, choose $\eta > 0$ so that

$$f(\eta) - f(0) < \frac{\varepsilon}{2}.$$

Absolute continuity on $[\eta, 1]$ supplies $\delta > 0$ such that every finite pairwise disjoint family of subintervals of $[\eta, 1]$ having total length below δ has total f -increment below $\varepsilon/2$.

Now take pairwise disjoint intervals in $[0, 1]$ with total length below δ , splitting at η whenever necessary. Because f is increasing, the sum of the increments contributed by the pieces in $[0, \eta]$ is at most $f(\eta) - f(0)$, while the remaining pieces contribute less than $\varepsilon/2$. Hence the total increment is below ε , proving absolute continuity on $[0, 1]$ directly from the definition. $\square$

Theorem 4.8.7 (Fundamental theorem of calculus for absolutely continuous functions). If $f \in AC([a, b])$, then f is differentiable almost everywhere, $f' \in L^1(a, b)$, and

$$f(x) = f(a) + \int_a^x f'(t) \, dt \quad (a \leq x \leq b).$$

Conversely, every function of this form is absolutely continuous.

Proof. By the Jordan decomposition inside AC, write

$$f = f(a) + g - k,$$

where g and k are increasing and absolutely continuous. The monotone differentiation theorem shows that g' and k' exist almost everywhere and belong to $L^1(a, b)$.

Fix $x \in [a, b]$. For $h > 0$, use the constant extension and set

$$D_h g(t) = \frac{g(t+h) - g(t)}{h}.$$

Then $D_h g \rightarrow g'$ almost everywhere on $[a, x]$. By the preceding theorem, the family $(D_h g)_{0 < h \leq 1}$ is uniformly integrable. The Vitali convergence theorem therefore gives

$$\int_a^x g'(t) dt = \lim_{h \downarrow 0} \int_a^x D_h g(t) dt = \lim_{h \downarrow 0} (A_h g(x) - A_h g(a)) = g(x) - g(a).$$

The same argument applies to k . Subtracting yields

$$f(x) = f(a) + \int_a^x f'(t) dt, \quad f' = g' - k' \in L^1(a, b).$$

Conversely, if $f(x) = f(a) + \int_a^x u$ with $u \in L^1$, then for every finite pairwise disjoint family of intervals,

$$\sum_j |f(b_j) - f(a_j)| \leq \int_{\cup_j (a_j, b_j)} |u|.$$

Absolute continuity of the Lebesgue integral makes the right-hand side arbitrarily small when the union has sufficiently small measure. $\square$

Corollary 4.8.8 (Monotone characterization). If f is increasing on $[a, b]$, then

$$f \in AC([a, b]) \iff \int_a^b f' = f(b) - f(a).$$

Proof. The forward implication follows from the fundamental theorem. Conversely, for $x \in [a, b]$,

$$0 \leq f(x) - f(a) - \int_a^x f' \leq f(b) - f(a) - \int_a^b f' = 0,$$

where the first inequality follows by applying the monotone derivative bound on $[a, x]$ and the second on $[x, b]$. $\square$

Lemma 4.8.9. If $u \in L^1(a, b)$ and

$$\int_c^d u(t) dt = 0$$

for every interval $[c, d] \subseteq [a, b]$, then $u = 0$ almost everywhere.

Proof. The function $U(x) = \int_a^x u$ is identically zero. By differentiation of indefinite integrals, $u = U' = 0$ almost everywhere. $\square$

Theorem 4.8.10 (Lusin's property (N)). Every absolutely continuous function maps Lebesgue-null sets to Lebesgue-null sets. It also maps Lebesgue-measurable sets to Lebesgue-measurable sets.

Proof. Let N be null and $\varepsilon > 0$. Cover N by pairwise disjoint open intervals whose total length is below the δ supplied by absolute continuity. The image of each interval is contained in an interval of length at most the oscillation of f there, and the sum of these oscillations is below ε . Hence $m^*(f(N)) = 0$.

For a measurable $E \subseteq [a, b]$, choose an F_σ set $F \subseteq E$ with $m(E \setminus F) = 0$. The image of each compact component used to build F is compact, so $f(F)$ is F_σ . The remaining image $f(E \setminus F)$ is null. Therefore $f(E)$ is measurable. $\square$

Exercise 4.8.11 (Composition can destroy absolute continuity). On $[-1, 1]$, let

$$f(x) = x^{1/3}, \quad g(x) = \begin{cases} x^2 \cos\left(\frac{\pi}{2x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f and g are absolutely continuous, but $f \circ g$ is not of bounded variation and hence is not absolutely continuous.

Proof. The function f is an indefinite integral of an L^1 function, and g is absolutely continuous because $g' \in L^1$. Along the partition containing $\pm 1/(2k)$ and $\pm 1/(2k-1)$, the variation of $f \circ g$ dominates a constant multiple of

$$\sum_{k=1}^n k^{-2/3},$$

which diverges. □

4.9 Singular functions and the Lebesgue decomposition of BV functions

Definition 4.9.1 (Singular function). A function $s \in \text{BV}([a, b])$ is called *singular* if $s' = 0$ almost everywhere. The Cantor–Lebesgue function is the standard nonconstant example.

Theorem 4.9.2 (Lebesgue decomposition of a BV function). Every $f \in \text{BV}([a, b])$ can be written

$$f = g + s,$$

where g is absolutely continuous and s is singular. Two such decompositions differ only by transferring an additive constant between g and s ; imposing $g(a) = 0$ makes the decomposition unique.

Proof. Since $f' \in L^1$, define

$$g(x) = \int_a^x f'(t) \, dt, \quad s = f - g.$$

Then g is absolutely continuous, $s \in \text{BV}$, and $s' = 0$ almost everywhere. If $f = g_1 + s_1 = g_2 + s_2$, then $g_1 - g_2 = s_2 - s_1$ is both absolutely continuous and singular. Its derivative is zero almost everywhere, so the fundamental theorem makes it constant. □

Corollary 4.9.3 (Banach–Zarecki theorem for increasing functions). Let $f : [a, b] \rightarrow \mathbb{R}$ be increasing and continuous. Then f is absolutely continuous if and only if it maps null sets to null sets.

Proof. Only the converse needs proof. Let μ_f be the Lebesgue–Stieltjes measure determined by

$$\mu_f((u, v]) = f(v) - f(u).$$

Continuity and monotonicity of f imply that the pushforward of μ_f under f is Lebesgue measure on $[f(a), f(b)]$: first check this on half-open intervals and then apply the monotone-class theorem.

Let $N \subseteq [a, b]$ be a Borel null set. By property (N), $f(N)$ is null; choose a Borel null set B containing $f(N)$. Since $N \subseteq f^{-1}(B)$,

$$0 \leq \mu_f(N) \leq \mu_f(f^{-1}(B)) = m(B) = 0.$$

Thus $\mu_f \ll m$. The Radon–Nikodym theorem gives $u \in L^1(a, b)$ with $d\mu_f = u \, dm$, and therefore

$$f(x) - f(a) = \mu_f((a, x]) = \int_a^x u(t) \, dt.$$

The fundamental theorem now shows that f is absolutely continuous. □

Proposition 4.9.4 (Variation and the derivative). If $f \in \text{BV}([a, b])$, then

$$\int_a^b |f'| \leq \text{TV}(f; [a, b]).$$

Equality holds if and only if f is absolutely continuous.

Proof. The inequality was proved above. At every point where both f' and V'_f exist, the estimate

$$V_f(y) - V_f(x) \geq |f(y) - f(x)|$$

implies

$$V'_f(x) \geq |f'(x)|.$$

Consequently,

$$\int_a^b |f'| \leq \int_a^b V'_f \leq V_f(b) - V_f(a) = \text{TV}(f; [a, b]).$$

Suppose equality holds in the stated inequality. Then both inequalities in the last display are equalities, so

$$\int_a^b V'_f = V_f(b) - V_f(a).$$

The monotone characterization of absolute continuity shows that V_f is absolutely continuous. Hence the canonical Jordan components

$$\frac{V_f + f - f(a)}{2}, \quad \frac{V_f - f + f(a)}{2}$$

are absolutely continuous, and therefore so is f .

Conversely, if f is absolutely continuous, then for every partition

$$\sum_j |f(x_j) - f(x_{j-1})| \leq \sum_j \int_{x_{j-1}}^{x_j} |f'| = \int_a^b |f'|.$$

Taking the supremum and combining with the opposite inequality gives equality. □

Exercise 4.9.5. A singular increasing function that is not constant cannot be absolutely continuous. More quantitatively, if $s(a) = 0$ and $s(b) > 0$, then for every $\varepsilon > 0$ there are pairwise disjoint intervals (a_k, b_k) with

$$\sum_k (b_k - a_k) < \varepsilon, \quad \sum_k (s(b_k) - s(a_k)) > s(b) - \varepsilon.$$

Countable nonnegative linear combinations of singular increasing functions are singular whenever the

series converges pointwise to a finite value.

Proof. Let

$$E = \{x \in (a, b) : s'(x) = 0\}, \quad m([a, b] \setminus E) = 0.$$

Fix $\varepsilon, \delta > 0$ and put $\alpha = \varepsilon/(2(b-a))$. Intervals $[c, d]$ containing a point of E and satisfying

$$s(d) - s(c) < \alpha(d - c)$$

form a Vitali cover of E . Choose finitely many pairwise disjoint such intervals whose union leaves less than δ of E uncovered. Their total s -increment is below $\varepsilon/2$. The complementary intervals have total length below δ and, by additivity of increments for an increasing function, carry more than $s(b) - s(a) - \varepsilon$ of the total increase. This proves the quantitative assertion.

Conversely, let u be increasing and have this concentration property. Since $u' \in L^1(a, b)$, choose $\delta > 0$ so that $m(A) < \delta$ implies $\int_A u' < \eta$. Apply the concentration property with an error smaller than $\min\{\delta, \eta\}$, and let A be the union of the resulting pairwise disjoint intervals. Thus $m(A) < \delta$, while the complementary intervals J carry total u -increment below η . The monotone derivative bound on each complementary interval gives

$$\int_a^b u' \leq \int_A u' + \sum_J (u(\sup J) - u(\inf J)) < 2\eta.$$

Letting $\eta \downarrow 0$ shows that $u' = 0$ almost everywhere. Finally, let $s = \sum_n s_n$ be finite-valued, with each s_n singular and increasing. Given ε and δ , choose N so that

$$\sum_{n>N} (s_n(b) - s_n(a)) < \frac{\varepsilon}{2}.$$

The finite sum $\sum_{n \leq N} s_n$ is singular and has the concentration property just proved. The same intervals then miss at most an additional $\varepsilon/2$ of the increase of the full series. Hence s has the concentration property and is singular. $\square$

Chapter 5 Signed Measures and Differentiation

5.1 Signed and complex measures

Definition 5.1.1 (Signed measure). Let $(X, \mathcal{M})$ be a measurable space. A *signed measure* is a countably additive function

$$\nu : \mathcal{M} \longrightarrow [-\infty, \infty]$$

with $\nu(\emptyset) = 0$ that does not take both values $+\infty$ and $-\infty$.

Proposition 5.1.2 (Continuity). Let ν be a signed measure.

- (a). If $E_n \uparrow E$, then $\nu(E_n) \rightarrow \nu(E)$ in the extended real line.
- (b). If $E_n \downarrow E$ and $\nu(E_1)$ is finite, then $\nu(E_n) \rightarrow \nu(E)$.

Proof. Disjointify the increasing sequence. For the decreasing case, apply the increasing result to $E_1 \setminus E_n$ and subtract from the finite number $\nu(E_1)$. $\square$

Definition 5.1.3 (Positive and negative sets). A measurable set P is *positive* for ν if $\nu(E) \geq 0$ for every measurable $E \subseteq P$. A measurable set N is *negative* if $\nu(E) \leq 0$ for every measurable $E \subseteq N$. A set that is both positive and negative is ν -null.

Lemma 5.1.4. A countable union of positive sets is positive, and a countable union of negative sets is negative.

Proof. Disjointify the family. Every measurable subset of the union decomposes into countably many subsets of the original sets, and countable additivity preserves the sign. $\square$

Lemma 5.1.5 (Positive-subset lemma). If $\nu(E) > 0$, then E contains a positive measurable subset P with $\nu(P) > 0$.

Proof. No measurable subset of E can have measure $-\infty$. Indeed, if $A \subseteq E$ had $\nu(A) = -\infty$, then the definition of a signed measure would rule out $\nu(E \setminus A) = +\infty$, and countable additivity would force $\nu(E) = -\infty$, a contradiction.

Put $E_0 = E$. If E_{k-1} is not positive, let n_k be the least positive integer for which there is a measurable set $A_k \subseteq E_{k-1}$ satisfying

$$\nu(A_k) < -\frac{1}{n_k},$$

and set $E_k = E_{k-1} \setminus A_k$. If no such integer exists, the current remainder is positive and the construction stops.

The sets A_k are pairwise disjoint, and the integers n_k are nondecreasing. They must tend to infinity: otherwise they would eventually be bounded by some N , and the union of infinitely many disjoint A_k would have measure at most $-\sum 1/N = -\infty$, which is impossible for a subset of E . Likewise, $\sum_k \nu(A_k) > -\infty$.

Let

$$P = E \setminus \bigcup_k A_k.$$

Then

$$\nu(P) = \nu(E) - \sum_k \nu(A_k) > 0.$$

If P contained a measurable A with $\nu(A) < 0$, choose n with $\nu(A) < -1/n$. For all sufficiently large k , $n_k > n$, while $A \subseteq E_{k-1}$ would still be available in the construction, contradicting the minimality of n_k .

Thus P is positive. $\square$

Theorem 5.1.6 (Hahn decomposition). For every signed measure ν there are a positive set P and a negative set N such that

$$X = P \sqcup N.$$

If $X = P' \sqcup N'$ is another such decomposition, then $P \triangle P'$ and $N \triangle N'$ are ν -null.

Proof. Assume, after replacing ν by $-\nu$ if needed, that ν never takes $+\infty$. Let

$$\alpha = \sup\{\nu(A) : A \text{ is positive}\}.$$

The number α is finite: otherwise a countable union of positive sets would have measure $+\infty$. Choose positive sets P_n with $\nu(P_n) \rightarrow \alpha$ and set $P = \bigcup_n P_n$. Then P is positive and $\nu(P) = \alpha$. Put $N = X \setminus P$.

If N were not negative, some measurable $E \subseteq N$ would satisfy $\nu(E) > 0$. By the positive-subset lemma, E would contain a positive set A with $\nu(A) > 0$. Then $P \cup A$ would be positive and have measure larger than α , a contradiction. Hence N is negative.

For uniqueness, $P \cap N'$ is both positive and negative, hence null; similarly $P' \cap N$ is null. These two sets form the symmetric differences. $\square$

Definition 5.1.7 (Mutual singularity). Positive measures λ and μ are *mutually singular*, written $\lambda \perp \mu$, if there are disjoint measurable sets A, B with $A \cup B = X$ such that λ is concentrated on A and μ is concentrated on B . For a signed or complex measure ν , the notation $\nu \perp \mu$ means $|\nu| \perp \mu$.

Theorem 5.1.8 (Jordan decomposition). For every signed measure ν there are unique mutually singular positive measures ν^+ and ν^- such that

$$\nu = \nu^+ - \nu^-.$$

If $X = P \sqcup N$ is a Hahn decomposition, then

$$\nu^+(E) = \nu(E \cap P), \quad \nu^-(E) = -\nu(E \cap N).$$

At least one of $\nu^+(X)$ and $\nu^-(X)$ is finite.

Proof. The formulas define positive measures concentrated on disjoint sets and clearly give $\nu = \nu^+ - \nu^-$. If $\nu = \lambda - \mu$ with $\lambda \perp \mu$, choose disjoint carriers A, B . Then A is positive and B negative for ν , so the Hahn uniqueness statement identifies the two parts. $\square$

Definition 5.1.9 (Total variation). The *total variation* of a signed measure is

$$|\nu| = \nu^+ + \nu^-.$$

A signed measure is finite or σ -finite when $|\nu|$ has the corresponding property. Integration is defined by

$$\int f \, d\nu = \int f \, d\nu^+ - \int f \, d\nu^-$$

whenever $f \in L^1(|\nu|)$.

Proposition 5.1.10 (Basic total-variation facts). Let ν be a signed measure.

(a). A set E is ν -null if and only if $|\nu|(E) = 0$.

(b). If $f \in L^1(|\nu|)$, then

$$\left| \int f \, d\nu \right| \leq \int |f| \, d|\nu|.$$

(c). For $E \in \mathcal{M}$,

$$|\nu|(E) = \sup \left\{ \left| \int_E f \, d\nu \right| : f \text{ measurable, } |f| \leq 1 \right\}.$$

(d). If ρ, η are positive measures and $\rho \perp \eta$, then

$$\rho + \eta \perp \lambda \iff \rho \perp \lambda \text{ and } \eta \perp \lambda.$$

Proof. The first two statements follow immediately from the Jordan decomposition. For the third, the inequality “ $\leq$ ” is the second statement. On a Hahn decomposition $X = P \sqcup N$, take $f = \mathbf{1}_P - \mathbf{1}_N$ on E to obtain equality. The last assertion follows by intersecting or uniting carriers. $\square$

Proposition 5.1.11 (Extremal formulas). For $E \in \mathcal{M}$,

$$\nu^+(E) = \sup\{\nu(F) : F \subseteq E, F \in \mathcal{M}\},$$

$$\nu^-(E) = -\inf\{\nu(F) : F \subseteq E, F \in \mathcal{M}\},$$

and

$$|\nu|(E) = \sup \left\{ \sum_{j=1}^n |\nu(E_j)| : E = \bigsqcup_{j=1}^n E_j \right\}.$$

The same supremum is obtained if countable measurable partitions are allowed.

Proof. On a Hahn decomposition, $\nu(F) \leq \nu^+(F) \leq \nu^+(E)$, with equality for $F = E \cap P$. The negative formula is analogous. For a partition,

$$\sum_j |\nu(E_j)| \leq \sum_j |\nu|(E_j) = |\nu|(E).$$

Equality is attained by the two-set partition $E \cap P, E \cap N$. □

Definition 5.1.12 (Complex measure and its variation). A *complex measure* on $(X, \mathcal{M})$ is a countably additive map $\nu : \mathcal{M} \rightarrow \mathbb{C}$. Its total variation is

$$|\nu|(E) = \sup \left\{ \sum_{j=1}^n |\nu(E_j)| : E = \bigsqcup_{j=1}^n E_j, E_j \in \mathcal{M} \right\}.$$

This formula defines a finite positive measure. Indeed, countable additivity follows by refining partitions with a disjoint measurable decomposition, and finiteness follows from

$$|\nu|(X) \leq |\Re \nu|(X) + |\Im \nu|(X) < \infty.$$

Remark. Writing $\nu = \nu_1 + i\nu_2$, the real and imaginary parts are finite signed measures. Applying Jordan decomposition to each yields a representation as a linear combination of four finite positive measures. For $f \in L^1(|\nu|)$ we define

$$\int f d\nu = \int f d(\Re \nu) + i \int f d(\Im \nu).$$

This definition agrees with integration against the polar factor obtained below.

Theorem 5.1.13 (Polar decomposition of a complex measure). For every complex measure ν there is a measurable function h such that

$$|h| = 1 \quad |\nu|\text{-almost everywhere}, \quad \nu(E) = \int_E h d|\nu|.$$

Consequently,

$$\left| \int f d\nu \right| \leq \int |f| d|\nu|$$

for every $f \in L^1(|\nu|)$.

Proof. Let

$$\mu = |\Re \nu| + |\Im \nu|.$$

The finite signed Radon–Nikodym theorem, proved below, gives a function $f \in L^1(\mu)$ such that $\nu = f\mu$.

For every finite measurable partition (E_j) of E ,

$$\sum_j |\nu(E_j)| \leq \sum_j \int_{E_j} |f| d\mu = \int_E |f| d\mu,$$

so $|\nu|(E) \leq \int_E |f| d\mu$.

For the reverse inequality, put

$$\theta = \begin{cases} \bar{f}/|f|, & f \neq 0, \\ 0, & f = 0. \end{cases}$$

Choose simple functions θ_n with $|\theta_n| \leq 1$ and

$$\int_E |\theta_n - \theta| |f| d\mu \rightarrow 0.$$

For a simple θ_n , the partition definition of variation gives

$$\left| \int_E \theta_n d\nu \right| \leq |\nu|(E).$$

Passing to the limit yields $\int_E |f| d\mu \leq |\nu|(E)$. Hence

$$|\nu| = |f|\mu.$$

Define $h = f/|f|$ on $\{f \neq 0\}$ and $h = 0$ elsewhere. Then $\nu = h|\nu|$ and $|h| = 1$ $|\nu|$ -almost everywhere. The integral inequality follows immediately. $\square$

Proposition 5.1.14 (Equivalent descriptions of complex variation). For a complex measure ν and $E \in \mathcal{M}$, the following quantities are equal:

$$\begin{aligned} & \sup \left\{ \sum_{j=1}^n |\nu(E_j)| : E = \bigsqcup_{j=1}^n E_j \right\}, \\ & \sup \left\{ \sum_{j=1}^n |\nu(E_j)| : E_j \text{ disjoint and } \bigcup_j E_j \subseteq E \right\}, \\ & \sup \left\{ \left| \int_E \phi d\nu \right| : \phi \text{ simple, } |\phi| \leq 1 \right\}. \end{aligned}$$

Their common value is $|\nu|(E)$.

Proof. The first two suprema agree after adding the unused remainder. For a simple function constant on a finite partition, the triangle inequality gives an upper bound by the partition sum. Conversely, on each partition element choose a unimodular scalar that rotates $\nu(E_j)$ to the positive real axis. Approximate the polar factor $\bar{h}$ by simple functions to obtain the full value $|\nu|(E)$. $\square$

5.2 Absolute continuity and the Radon–Nikodym theorem

Definition 5.2.1 (Absolute continuity). Let ν be a signed or complex measure and μ a positive measure. We write $\nu \ll \mu$ if

$$\mu(E) = 0 \implies \nu(E) = 0.$$

Equivalently, $|\nu| \ll \mu$.

Theorem 5.2.2 (ε - δ characterization). Let ν be a finite signed or complex measure and let μ be a positive measure. Then

$$\nu \ll \mu$$

if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\mu(E) < \delta \implies |\nu|(E) < \varepsilon.$$

Proof. The ε - δ condition plainly implies absolute continuity. Conversely, suppose $|\nu| \ll \mu$ but the condition fails. Then there are $\varepsilon > 0$ and sets E_n with

$$\mu(E_n) < 2^{-n}, \quad |\nu|(E_n) \geq \varepsilon.$$

Set $F_k = \bigcup_{n \geq k} E_n$ and $F = \bigcap_k F_k$. Then $\mu(F) = 0$, while $F_k \downarrow F$ and finiteness of $|\nu|$ gives

$$|\nu|(F) = \lim_k |\nu|(F_k) \geq \varepsilon,$$

a contradiction. □

Corollary 5.2.3 (Absolute continuity of the integral). If $f \in L^1(\mu)$, then for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\mu(E) < \delta \implies \int_E |f| d\mu < \varepsilon.$$

Proof. Apply the theorem to the finite measure $\nu(E) = \int_E |f| d\mu$. □

Lemma 5.2.4 (A comparison lemma). Let λ and μ be finite positive measures. Either $\lambda \perp \mu$, or there are $\varepsilon > 0$ and a measurable set E with $\mu(E) > 0$ such that

$$\lambda(A) \geq \varepsilon \mu(A)$$

for every measurable $A \subseteq E$.

Proof. For each n , take a Hahn decomposition $X = P_n \sqcup N_n$ for the signed measure $\lambda - \mu/n$. Put

$$P = \bigcup_{n=1}^{\infty} P_n, \quad N = \bigcap_{n=1}^{\infty} N_n.$$

If $\mu(P) = 0$, then μ is concentrated on N . Moreover, for every measurable $A \subseteq N$ and every n ,

$$\lambda(A) \leq \frac{1}{n} \mu(A),$$

because $N \subseteq N_n$. Since μ is finite, letting $n \rightarrow \infty$ gives $\lambda(A) = 0$. Thus λ is concentrated on P , and $\lambda \perp \mu$.

If $\mu(P) > 0$, then $\mu(P_n) > 0$ for some n . On P_n , the signed measure $\lambda - \mu/n$ is positive, so

$$\lambda(A) \geq \frac{1}{n} \mu(A)$$

for every measurable $A \subseteq P_n$. Take $E = P_n$ and $\varepsilon = 1/n$. □

Theorem 5.2.5 (Lebesgue–Radon–Nikodym theorem, finite case). Let ν be a finite signed measure and μ a finite positive measure. There are unique objects

$$\lambda \perp \mu, \quad f \in L^1(\mu),$$

such that

$$\nu(E) = \lambda(E) + \int_E f \, d\mu \quad (E \in \mathcal{M}).$$

If $\nu \ll \mu$, then $\lambda = 0$. The function f is unique μ -almost everywhere and is denoted $d\nu/d\mu$.

Proof. First suppose ν is positive. Let $\mathcal{F}$ be the set of measurable $g \geq 0$ satisfying

$$\int_E g \, d\mu \leq \nu(E) \quad (E \in \mathcal{M}).$$

The class is closed under pointwise maxima. Put

$$\alpha = \sup_{g \in \mathcal{F}} \int g \, d\mu.$$

Choose $g_n \in \mathcal{F}$ with integrals tending to α , replace them by the successive maxima, and let $f = \sup_n g_n$.

Monotone convergence gives $f \in \mathcal{F}$ and $\int f \, d\mu = \alpha$. Define

$$\lambda(E) = \nu(E) - \int_E f \, d\mu.$$

Then λ is a positive measure. If λ were not singular to μ , the comparison lemma would give E and $\varepsilon > 0$ with $\lambda \geq \varepsilon\mu$ on E . But then $f + \varepsilon\mathbf{1}_E \in \mathcal{F}$ and has integral larger than α , a contradiction. Thus $\lambda \perp \mu$.

Apply the positive result to ν^+ and ν^- and subtract. For uniqueness, if

$$\lambda + f\mu = \lambda' + f'\mu, \quad \lambda, \lambda' \perp \mu,$$

then $\lambda - \lambda' = (f' - f)\mu$ is both singular and absolutely continuous with respect to μ , hence zero. Therefore $\lambda = \lambda'$ and $f = f'$ almost everywhere. $\square$

Theorem 5.2.6 (Sigma-finite form). Let μ be σ -finite and let ν be a σ -finite signed measure. Then there are unique signed measures λ, ρ such that

$$\nu = \lambda + \rho, \quad \lambda \perp \mu, \quad \rho \ll \mu.$$

Moreover, there is a measurable function f , finite μ -almost everywhere, such that

$$\rho(E) = \int_E f \, d\mu$$

whenever the integral is defined. On a countable partition on which $|\nu|$ and μ are finite, f is integrable on every piece. If ν is finite, then $f \in L^1(\mu)$.

Proof. Choose a countable measurable partition (X_n) such that both $\mu(X_n)$ and $|\nu|(X_n)$ are finite. Apply the finite theorem to the restrictions on each X_n , and sum the resulting mutually disjoint pieces. The local densities patch to a finite-valued measurable function almost everywhere. Uniqueness follows by restriction to the same partition. $\square$

Theorem 5.2.7 (Extended Radon–Nikodym theorem). Let μ be a σ -finite positive measure.

(a). If ν is an arbitrary positive measure with $\nu \ll \mu$, then there is a measurable $f : X \rightarrow [0, \infty]$ such that

$$\nu(E) = \int_E f \, d\mu \quad (E \in \mathcal{M}).$$

(b). If ν is a signed measure with $|\nu| \ll \mu$, then there is a measurable extended-real function f such that

$$\nu(E) = \int_E f \, d\mu$$

whenever either side is defined.

Proof. First suppose that $\mu(X) < \infty$ and ν is positive. Let $\mathcal{S}$ be the family of measurable sets on which ν is σ -finite, and put

$$c = \sup\{\mu(S) : S \in \mathcal{S}\}.$$

Choose $S_n \in \mathcal{S}$ with $\mu(S_n) \rightarrow c$ and let $S = \bigcup_n S_n$. Then $S \in \mathcal{S}$ and $\mu(S) = c$. The sigma-finite theorem supplies a density f_0 for $\nu \upharpoonright S$ with respect to $\mu \upharpoonright S$. Define the extended-valued function

$$f(x) = \begin{cases} f_0(x), & x \in S, \\ +\infty, & x \notin S. \end{cases}$$

If $\mu(E \setminus S) = 0$, absolute continuity gives $\nu(E \setminus S) = 0$, and the desired identity follows on S . If $\mu(E \setminus S) > 0$, then $\nu(E \setminus S) = \infty$; otherwise adjoining $E \setminus S$ to S would produce a ν -sigma-finite set of μ -measure larger than c . In this case both sides are infinite.

For sigma-finite μ , apply the finite construction on a disjoint finite-measure partition and patch the densities. For a signed measure, apply the positive result to the mutually singular Jordan parts and choose their densities on disjoint Hahn carriers. Their difference is then well defined almost everywhere and gives the stated formula. $\square$

Proposition 5.2.8 (Chain rule). If $\nu \ll \mu \ll \lambda$ and the measures are σ -finite, then

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} \quad \lambda\text{-almost everywhere.}$$

Proof. For every measurable E ,

$$\nu(E) = \int_E \frac{d\nu}{d\mu} \, d\mu = \int_E \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} \, d\lambda.$$

Uniqueness of the Radon–Nikodym derivative gives the result. $\square$

Proposition 5.2.9 (Product rule). For $j = 1, 2$, let $\nu_j \ll \mu_j$ be σ -finite positive measures. Then

$$\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$$

and

$$\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x_1, x_2) = \frac{d\nu_1}{d\mu_1}(x_1) \frac{d\nu_2}{d\mu_2}(x_2)$$

for $(\mu_1 \times \mu_2)$ -almost every (x_1, x_2) .

Proof. The right-hand side integrates over measurable rectangles to

$$\nu_1(A_1)\nu_2(A_2).$$

Uniqueness of the product measure and then uniqueness of the Radon–Nikodym derivative complete the proof. $\square$

Example (Why finiteness and sigma-finiteness matter). Let $X = \mathbb{N}$, let ν be counting measure, and let

$$\mu(E) = \sum_{n \in E} 2^{-n}.$$

Then $\nu \ll \mu$, but no global ε – δ estimate can hold because $\mu(\{n\}) \rightarrow 0$ while $\nu(\{n\}) = 1$.

On $[0, 1]$, Lebesgue measure is absolutely continuous with respect to counting measure, since the only counting-null set is empty. Nevertheless, it has no density with respect to counting measure: a density would have to vanish at every singleton and hence integrate to zero. Counting measure on an uncountable set is not σ -finite.

Definition 5.2.10 (Decomposable measure). A measure μ is *decomposable* if there is a pairwise disjoint family $\mathcal{F} \subseteq \mathcal{M}$ such that

- (a). $\mu(F) < \infty$ for every $F \in \mathcal{F}$;
- (b). $X = \bigcup_{F \in \mathcal{F}} F$;
- (c). $E \subseteq X$ is measurable if and only if $E \cap F$ is measurable for every $F \in \mathcal{F}$;
- (d). for measurable E ,

$$\mu(E) = \sum_{F \in \mathcal{F}} \mu(E \cap F),$$

where the unordered nonnegative sum is used.

Proposition 5.2.11. Every σ -finite measure is decomposable. Let μ be decomposable with decomposition $\mathcal{F}$, and let ν be a signed measure satisfying $|\nu| \ll \mu$. Then there is a measurable extended-real function f such that

$$\nu(E) = \int_E f \, d\mu$$

for every μ -sigma-finite measurable set E . On every $F \in \mathcal{F}$ on which $|\nu|$ is σ -finite, the function f is finite μ -almost everywhere.

Proof. A σ -finite cover can be disjointified to give a countable decomposition, including any null pieces. For general decomposable μ , apply the extended Radon–Nikodym theorem to the restrictions of ν and μ on each $F \in \mathcal{F}$, obtaining a density f_F ; on a μ -null piece take $f_F = 0$. Define $f = f_F$ on F . The measurability criterion in the definition of decomposability makes f measurable.

Suppose first that E has finite μ -measure, and put

$$\mathcal{F}_E = \{F \in \mathcal{F} : \mu(E \cap F) > 0\}.$$

The family $\mathcal{F}_E$ is countable because the unordered sum of the positive numbers $\mu(E \cap F)$ is finite. Let

$$R = E \setminus \bigcup_{F \in \mathcal{F}_E} (E \cap F).$$

Then R is measurable and the decomposition formula gives $\mu(R) = 0$, hence $|\nu|(R) = 0$. Countable additivity and the corresponding decomposition of the integral now give

$$\nu(E) = \sum_{F \in \mathcal{F}_E} \nu(E \cap F) = \sum_{F \in \mathcal{F}_E} \int_{E \cap F} f \, d\mu = \int_E f \, d\mu.$$

Disjointifying a sigma-finite cover of E extends the identity to every μ -sigma-finite set. If $|\nu| \upharpoonright F$ is sigma-finite, the ordinary Radon–Nikodym theorem yields a finite density almost everywhere on that piece. $\square$

5.3 Complex Radon–Nikodym theory

Theorem 5.3.1. Let ν be a complex measure and let μ be a σ -finite positive measure. If $\nu \ll \mu$, then there is a unique $f \in L^1(\mu)$ such that

$$\nu(E) = \int_E f \, d\mu.$$

Moreover,

$$|\nu|(E) = \int_E |f| \, d\mu, \quad \frac{d|\nu|}{d\mu} = \left| \frac{d\nu}{d\mu} \right| \quad \mu\text{-almost everywhere.}$$

Proof. Apply the signed Radon–Nikodym theorem to the real and imaginary parts and combine the two densities, obtaining $\nu = f\mu$ with $f \in L^1(\mu)$. The phase-approximation argument in the polar-decomposition proof gives

$$|\nu|(E) = \int_E |f| \, d\mu$$

for every measurable E . This is exactly the asserted derivative identity. Uniqueness follows from uniqueness of the real and imaginary Radon–Nikodym derivatives. $\square$

Corollary 5.3.2 (Polar factor and chain rule). If ν is a complex measure, then

$$\frac{d\nu}{d|\nu|} = h, \quad |h| = 1 \quad |\nu|\text{-almost everywhere.}$$

If μ and λ are σ -finite positive measures and ν is a complex measure satisfying $\nu \ll \mu \ll \lambda$, then

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} \quad \lambda\text{-almost everywhere.}$$

Proposition 5.3.3 (Triangle inequality for variation). If ν_1 and ν_2 are complex measures, then

$$|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|.$$

Proof. Take $\mu = |\nu_1| + |\nu_2|$. Then $\nu_j = f_j \mu$ and

$$|\nu_1 + \nu_2|(E) = \int_E |f_1 + f_2| \, d\mu \leq \int_E (|f_1| + |f_2|) \, d\mu.$$

$\square$

5.4 Differentiation of Measures in Euclidean Space

Let m denote Lebesgue measure on $\mathbb{R}^n$, and let $B(x, r)$ be the open Euclidean ball centered at x of radius r .

Lemma 5.4.1 (Three-times covering lemma for balls). From every finite family of balls in $\mathbb{R}^n$ one can select pairwise disjoint balls $B_1, \dots, B_N$ such that

$$\bigcup_{B \text{ in the original family}} B \subseteq \bigcup_{j=1}^N 3B_j.$$

Consequently, if the original union has measure greater than c , then

$$\sum_{j=1}^N m(B_j) > 3^{-n}c.$$

Proof. Choose successively a ball of largest remaining radius and discard all balls meeting it. A discarded ball has radius no larger than the selected ball it meets, and is therefore contained in its triple. The measure estimate follows from disjointness and $m(3B_j) = 3^n m(B_j)$. $\square$

Definition 5.4.2 (Local averages). For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $r > 0$, define

$$A_r f(x) = \frac{1}{m(B(x, r))} \int_{B(x, r)} f(y) \, dy.$$

Lemma 5.4.3. If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then $(r, x) \mapsto A_r f(x)$ is jointly continuous on $(0, \infty) \times \mathbb{R}^n$.

Proof. On a compact set of parameters, all relevant balls lie in one fixed bounded set. Changes in the center are controlled by continuity of translations in L^1 , and changes in the radius are controlled by the measure of the symmetric difference of the balls. The normalizing volume is continuous and bounded away from zero there. $\square$

Definition 5.4.4 (Hardy–Littlewood maximal function in $\mathbb{R}^n$). For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, define

$$Mf(x) = \sup_{r>0} A_r |f|(x).$$

The level sets $\{Mf > \alpha\}$ are open by the preceding lemma.

Theorem 5.4.5 (Hardy–Littlewood maximal inequality). For $f \in L^1(\mathbb{R}^n)$ and $\alpha > 0$,

$$m(\{Mf > \alpha\}) \leq \frac{3^n}{\alpha} \|f\|_1.$$

Proof. Let K be a compact subset of $\{Mf > \alpha\}$. For each $x \in K$, choose a ball B_x centered at x with

$$\int_{B_x} |f| > \alpha m(B_x).$$

A finite subfamily covers K . Select disjoint balls by the covering lemma. Then

$$m(K) \leq 3^n \sum_j m(B_j) < \frac{3^n}{\alpha} \sum_j \int_{B_j} |f| \leq \frac{3^n}{\alpha} \|f\|_1.$$

Take the supremum over compact K . □

Theorem 5.4.6 (Lebesgue differentiation theorem in $\mathbb{R}^n$). If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then

$$\lim_{r \downarrow 0} A_r f(x) = f(x)$$

for almost every $x \in \mathbb{R}^n$.

Proof. Fix a ball $B(0, N)$. Only sufficiently small radii matter, so after multiplying f by the indicator of a slightly larger ball we may assume $f \in L^1$. Given $\varepsilon > 0$, choose $g \in C_c(\mathbb{R}^n)$ with $\|f - g\|_1 < \varepsilon$. Since $A_r g(x) \rightarrow g(x)$ everywhere,

$$\limsup_{r \downarrow 0} |A_r f(x) - f(x)| \leq M(f - g)(x) + |f(x) - g(x)|.$$

The maximal inequality and Markov's inequality show that the set where the left side exceeds a fixed positive number has measure bounded by a constant times $\|f - g\|_1$. Let $\varepsilon \downarrow 0$ and then let $N \rightarrow \infty$. □

Theorem 5.4.7 (Lebesgue points). If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then for almost every x ,

$$\lim_{r \downarrow 0} \frac{1}{m(B(x, r))} \int_{B(x, r)} |f(y) - f(x)| \, dy = 0.$$

Such points are called *Lebesgue points* of f .

Proof. Apply the differentiation theorem to $|f - c|$ for each c in a countable dense subset of $\mathbb{C}$. Outside the union of the resulting null sets, approximate $f(x)$ by such a c and use the triangle inequality. □

Definition 5.4.8 (Sets shrinking nicely). A family $(E_r)_{r>0}$ shrinks nicely to x if there is $c > 0$ such that

$$x \in E_r \subseteq B(x, r), \quad m(E_r) \geq c m(B(x, r))$$

for all sufficiently small r .

Corollary 5.4.9. If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and (E_r) shrinks nicely to x , then for almost every x ,

$$\lim_{r \downarrow 0} \frac{1}{m(E_r)} \int_{E_r} f(y) \, dy = f(x).$$

Proof. At a Lebesgue point,

$$\frac{1}{m(E_r)} \int_{E_r} |f - f(x)| \leq \frac{1}{c m(B(x, r))} \int_{B(x, r)} |f - f(x)| \rightarrow 0.$$

□

Definition 5.4.10 (Regular Borel measure). A Borel measure ν on $\mathbb{R}^n$ is *regular* if it is finite on compact sets, inner regular on open sets, and outer regular on Borel sets. A signed or complex Borel measure is regular when its total variation is regular.

Theorem 5.4.11 (Differentiation of regular measures). Let ν be a regular signed or complex Borel measure on $\mathbb{R}^n$, finite on compact sets, and write its Lebesgue decomposition with respect to m as

$$\nu = \lambda + f m, \quad \lambda \perp m.$$

For every family (E_r) shrinking nicely to x ,

$$\lim_{r \downarrow 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

for almost every x .

Proof. The absolutely continuous part follows from the preceding corollary. It remains to prove that the density of the singular part is zero. Fix $N \in \mathbb{N}$ and restrict $|\lambda|$ to the compact ball $\overline{B(0, N+1)}$; this restriction is finite and controls every sufficiently small ball centered in $B(0, N)$.

For a finite positive Borel measure η , the covering argument used for the Hardy–Littlewood theorem gives

$$m \left(\left\{ x : \sup_{r>0} \frac{\eta(B(x, r))}{m(B(x, r))} > \alpha \right\} \right) \leq \frac{3^n}{\alpha} \eta(\mathbb{R}^n).$$

Let Z be a Lebesgue-null Borel set carrying the restricted singular measure. By regularity, for every $\varepsilon > 0$ there is a compact set $K \subseteq Z$ such that

$$|\lambda|(\overline{B(0, N+1)} \setminus K) < \varepsilon.$$

If $x \notin K$, all sufficiently small balls centered at x avoid K . Therefore, outside the null set K , the set on which

$$\limsup_{r \downarrow 0} \frac{|\lambda|(B(x, r))}{m(B(x, r))} > \alpha$$

has measure at most $3^n \varepsilon / \alpha$. Letting $\varepsilon \downarrow 0$ shows that the centered density is zero almost everywhere in $B(0, N)$. Since N is arbitrary, this holds almost everywhere in $\mathbb{R}^n$.

If (E_r) shrinks nicely to x , then

$$\frac{|\lambda|(E_r)}{m(E_r)} \leq c^{-1} \frac{|\lambda|(B(x, r))}{m(B(x, r))},$$

so the same conclusion holds for the given family. Combining the singular and absolutely continuous parts proves the theorem. $\square$

Definition 5.4.12 (Uncentered maximal function). Define

$$M^* f(x) = \sup_{B \ni x} \frac{1}{m(B)} \int_B |f|,$$

where the supremum is over all balls containing x .

Proposition 5.4.13. For every locally integrable f ,

$$Mf \leq M^* f \leq 2^n Mf.$$

Proof. The first inequality is immediate. If $B(y, r)$ contains x , then $B(y, r) \subseteq B(x, 2r)$ and

$$\frac{1}{m(B(y, r))} \int_{B(y, r)} |f| \leq 2^n \frac{1}{m(B(x, 2r))} \int_{B(x, 2r)} |f|.$$

□

Exercise 5.4.14 (Sharpness at infinity). If $0 \neq f \in L^1(\mathbb{R}^n)$, then there are $C, R > 0$ such that

$$Mf(x) \geq C|x|^{-n} \quad (|x| > R).$$

In particular, the weak $(1, 1)$ estimate cannot in general be strengthened to $Mf \in L^1$.

Proof. Choose R_0 with $\int_{B(0, R_0)} |f| = a > 0$. For $|x| > 2R_0$, the ball $B(x, 2|x|)$ contains $B(0, R_0)$, so

$$Mf(x) \geq \frac{a}{m(B(x, 2|x|))} = C|x|^{-n}.$$

□

5.5 Functions of Bounded Variation and Borel Measures on the Real Line

Proposition 5.5.1 (Right-continuous modification). Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and define

$$G(x) = F(x+) = \lim_{y \downarrow x} F(y).$$

Then the discontinuity set of F is countable, G is increasing and right-continuous, and

$$F'(x) = G'(x)$$

for almost every x .

Proof. For each $k, N \in \mathbb{N}$, there are only finitely many points in $[-N, N]$ at which the jump exceeds $1/k$, because the sum of disjoint jumps is bounded by $F(N+1) - F(-N-1)$. Hence the discontinuity set D is countable. The function G is plainly increasing and right-continuous.

Both F and G are differentiable almost everywhere by Lebesgue's theorem for monotone functions. Let $x \notin D$ be a point at which both derivatives exist, and put $H = G - F$. Then $H(x) = 0$ and H vanishes on the dense set $\mathbb{R} \setminus D$. Choose $h_j \rightarrow 0$ with $x + h_j \notin D$. Along this sequence,

$$\frac{H(x + h_j) - H(x)}{h_j} = 0.$$

Since $H'(x)$ exists, it must be zero. Thus $F'(x) = G'(x)$ outside a null set. □

Definition 5.5.2 (Global variation). For $F : \mathbb{R} \rightarrow \mathbb{C}$, define

$$T_F(x) = \sup \left\{ \sum_{j=1}^n |F(x_j) - F(x_{j-1})| : -\infty < x_0 < \cdots < x_n = x \right\}.$$

We write $F \in \text{BV}(\mathbb{R})$ if $T_F(\infty) = \sup_x T_F(x) < \infty$. We write $F \in \text{NBV}(\mathbb{R})$ if, in addition, F is right-continuous and

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0.$$

Lemma 5.5.3. If $F \in \text{BV}(\mathbb{R})$ is real-valued, then $T_F + F$ and $T_F - F$ are increasing.

Proof. For $x < y$,

$$T_F(y) - T_F(x) \geq |F(y) - F(x)|.$$

□

Theorem 5.5.4 (Characterizations of $\text{BV}(\mathbb{R})$). Let $F : \mathbb{R} \rightarrow \mathbb{C}$.

- (a). $F \in \text{BV}(\mathbb{R})$ if and only if $\Re F, \Im F \in \text{BV}(\mathbb{R})$.
- (b). A real-valued F belongs to $\text{BV}(\mathbb{R})$ if and only if it is the difference of two bounded increasing functions.
- (c). Every $F \in \text{BV}(\mathbb{R})$ has finite one-sided limits at every point and finite limits at $\pm\infty$.
- (d). The discontinuity set of a BV function is at most countable.

Proof. The first assertion follows from

$$|z| \leq |\Re z| + |\Im z|, \quad |\Re z|, |\Im z| \leq |z|.$$

For real F , use

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F).$$

The remaining claims follow from the corresponding facts for bounded increasing functions. □

Proposition 5.5.5. For $F \in \text{BV}(\mathbb{R})$, the variation function T_F is right-continuous if and only if F is right-continuous.

Proof. If T_F is right-continuous, then

$$|F(x+h) - F(x)| \leq T_F(x+h) - T_F(x) \rightarrow 0.$$

Conversely, assume that F is right-continuous. Write

$$V(u, v) = \text{TV}(F; [u, v]).$$

Fix $x \in \mathbb{R}$ and $\varepsilon > 0$. Choose a partition

$$x = t_0 < t_1 < \cdots < t_N = x + 1$$

whose variation sum is larger than $V(x, x+1) - \varepsilon$. For $0 < h < t_1 - x$, the partition $x+h < t_1 < \cdots < t_N$ gives

$$V(x+h, x+1) \geq V(x, x+1) - \varepsilon - |F(x+h) - F(x)|.$$

Using additivity of variation over adjacent intervals,

$$0 \leq V(x, x+h) = V(x, x+1) - V(x+h, x+1) \leq \varepsilon + |F(x+h) - F(x)|.$$

Right continuity of F and then $\varepsilon \downarrow 0$ imply $V(x, x+h) \rightarrow 0$. Since

$$T_F(x+h) - T_F(x) = V(x, x+h),$$

the function T_F is right-continuous. □

Theorem 5.5.6 (Lebesgue–Stieltjes correspondence). If μ is a complex Borel measure on $\mathbb{R}$, then

$$F_\mu(x) = \mu((-\infty, x])$$

belongs to $\text{NBV}(\mathbb{R})$. Conversely, for every $F \in \text{NBV}(\mathbb{R})$ there is a unique complex Borel measure μ_F satisfying

$$\mu_F((a, b]) = F(b) - F(a).$$

Moreover,

$$|\mu_F| = \mu_{T_F}.$$

Proof. For a positive measure the correspondence is the Lebesgue–Stieltjes construction. Apply it to the Jordan decompositions of the real and imaginary parts. Uniqueness follows because the half-open intervals generate $\mathcal{B}(\mathbb{R})$. For a finite partition,

$$\sum_j |F(x_j) - F(x_{j-1})| = \sum_j |\mu_F((x_{j-1}, x_j])|.$$

Taking suprema and using the partition description of total variation gives $|\mu_F|((a, b]) = \mu_{T_F}((a, b])$, hence equality on all Borel sets. □

Theorem 5.5.7 (Derivative of a BV distribution function). Let $F \in \text{NBV}(\mathbb{R})$ and write

$$\mu_F = \lambda + f m, \quad \lambda \perp m.$$

Then F' exists almost everywhere, $F' \in L^1(\mathbb{R})$, and

$$F'(x) = f(x)$$

for almost every x . In particular,

$$\mu_F \perp m \iff F' = 0 \text{ almost everywhere,}$$

and

$$\mu_F \ll m \iff F(x) = \int_{-\infty}^x F'(t) dt.$$

Proof. For $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{\mu_F((x, x+h])}{h}.$$

The one-sided intervals $(x, x+h]$ are uniformly comparable with centered intervals: they lie in $(x-h, x+h)$ and have half its length. The proof of the differentiation theorem therefore applies to this one-sided family and identifies the limit with the density $f(x)$ almost everywhere. The singular part has density

zero. The two equivalences follow from uniqueness of the Lebesgue decomposition and the fundamental theorem for absolutely continuous functions. $\square$

Definition 5.5.8 (Absolute continuity of an NBV function). An $F \in \text{NBV}(\mathbb{R})$ is called *absolutely continuous* if it is absolutely continuous on every compact interval.

Proposition 5.5.9. For $F \in \text{NBV}(\mathbb{R})$,

$$F \text{ is absolutely continuous} \iff \mu_F \ll m.$$

In that case $d\mu_F = F' dm$.

Proof. If F is absolutely continuous, the interval criterion and outer regularity show that $m(E) = 0$ implies $|\mu_F|(E) = 0$. Conversely, if $\mu_F \ll m$, then $F(x) - F(a) = \int_{(a,x]} (d\mu_F/dm) dm$, so F is absolutely continuous on compact intervals. $\square$

Theorem 5.5.10 (Fundamental theorem of calculus for Lebesgue integrals). For $-\infty < a < b < \infty$, the following are equivalent for a function $F : [a, b] \rightarrow \mathbb{C}$:

- (a). F is absolutely continuous;
- (b). there exists $f \in L^1(a, b)$ such that

$$F(x) - F(a) = \int_a^x f(t) dt;$$

- (c). F is differentiable almost everywhere, $F' \in L^1(a, b)$, and

$$F(x) - F(a) = \int_a^x F'(t) dt.$$

Proof. This is the compact-interval form already proved in Chapter 2, now restated in measure language: F is absolutely continuous exactly when its Stieltjes measure has no singular part. $\square$

Theorem 5.5.11 (Lebesgue–Stieltjes integration by parts). Let $F, G \in \text{NBV}(\mathbb{R})$, and suppose at least one of them is continuous. Then for $a < b$,

$$\int_{(a,b]} F d\mu_G + \int_{(a,b]} G d\mu_F = F(b)G(b) - F(a)G(a).$$

Proof. It is enough by linearity to treat bounded increasing real-valued functions. On the triangle

$$\Omega = \{(x, y) : a < x \leq y \leq b\},$$

Fubini–Tonelli gives

$$(\mu_F \times \mu_G)(\Omega) = \int_{(a,b]} (F(y) - F(a)) d\mu_G(y),$$

and, using the complementary triangle,

$$(\mu_F \times \mu_G)(\Omega) = \mu_F((a, b])\mu_G((a, b]) - \int_{(a,b]} (G(x) - G(a)) d\mu_F(x).$$

If one function is continuous, the diagonal has zero product measure, so the two triangular decompositions match without a jump correction. Rearranging yields the formula. $\square$

Theorem 5.5.12 (Change of variables for an increasing map). Let $G : [a, b] \rightarrow [c, d]$ be continuous and increasing with $G(a) = c$ and $G(b) = d$. Then the pushforward of μ_G under G is Lebesgue measure on $[c, d]$. Thus, for every nonnegative or integrable f ,

$$\int_c^d f(y) \, dy = \int_a^b f(G(x)) \, d\mu_G(x).$$

The conclusion can fail if G is merely right-continuous.

Proof. For an interval $(u, v] \subseteq [c, d]$, the preimage under G is an interval up to endpoints, and its μ_G -measure is $v - u$ by continuity. A monotone-class argument extends equality to all Borel sets, and then to integrals. A step function G gives a counterexample without continuity: its Stieltjes measure is atomic, while Lebesgue measure is not. $\square$

Proposition 5.5.13 (Lipschitz characterization). A function $F : \mathbb{R} \rightarrow \mathbb{C}$ is Lipschitz with constant M if and only if it is absolutely continuous on every compact interval and

$$|F'(x)| \leq M$$

for almost every x .

Proof. A Lipschitz function is absolutely continuous, and difference quotients are bounded by M , so $|F'| \leq M$ almost everywhere. Conversely,

$$|F(y) - F(x)| = \left| \int_x^y F'(t) \, dt \right| \leq M|y - x|.$$

$\square$

Proposition 5.5.14 (Differentiating a monotone series). Let (F_j) be nonnegative increasing functions on $[a, b]$, and suppose

$$F(x) = \sum_{j=1}^{\infty} F_j(x) < \infty$$

for every x . Then

$$F'(x) = \sum_{j=1}^{\infty} F'_j(x)$$

for almost every x .

Proof. After normalizing at a , the associated positive measures satisfy

$$\mu_F = \sum_{j=1}^{\infty} \mu_{F_j}.$$

Taking absolutely continuous parts with respect to Lebesgue measure and using uniqueness of the Lebesgue decomposition gives

$$F' m = \sum_j F'_j m.$$

$\square$

Example (A strictly increasing singular continuous function). Enumerate the rational subintervals of $[0, 1]$ as (a_n, b_n) . Let C_n be a rescaled Cantor function that is 0 on $(-\infty, a_n]$ and 1 on $[b_n, \infty)$. Then

$$G(x) = \sum_{n=1}^{\infty} 2^{-n} C_n(x)$$

converges uniformly, is continuous, and satisfies $G' = 0$ almost everywhere. It is strictly increasing because every interval (x, y) contains some (a_n, b_n) , and the corresponding summand increases by 2^{-n} between x and y .

Example (A continuous function monotone on no interval). There is a Borel set $A \subseteq [0, 1]$ such that every nondegenerate interval $I \subseteq [0, 1]$ satisfies

$$0 < m(A \cap I) < m(I).$$

Define

$$F(x) = m(A \cap [0, x]), \quad G(x) = 2F(x) - x.$$

Then F is absolutely continuous and strictly increasing, while G is absolutely continuous and continuous but monotone on no nondegenerate interval. Indeed,

$$G' = 2\mathbf{1}_A - 1$$

almost everywhere, and both signs occur on sets of positive measure inside every interval.

Chapter 6 L^p Spaces

6.1 Definitions and basic inequalities

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $0 < p \leq \infty$.

Definition 6.1.1 (L^p norms). For a measurable function $f : X \rightarrow \mathbb{C}$, define

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p} \quad (0 < p < \infty),$$

and

$$\|f\|_\infty = \text{ess sup}_X |f| = \inf\{C \geq 0 : |f| \leq C \text{ almost everywhere}\}.$$

The space $L^p(\mu)$ consists of equivalence classes modulo equality almost everywhere for which the corresponding quantity is finite.

Proposition 6.1.2. For every $0 < p \leq \infty$, $L^p(\mu)$ is a vector space. If $1 \leq p \leq \infty$, then $\|\cdot\|_p$ is a norm. If $0 < p < 1$, then

$$\|f + g\|_p^p \leq \|f\|_p^p + \|g\|_p^p,$$

so $d(f, g) = \|f - g\|_p^p$ is a translation-invariant metric.

Proof. For $p \geq 1$, the elementary estimate

$$|f + g|^p \leq 2^{p-1}(|f|^p + |g|^p)$$

shows closure under addition; Minkowski's inequality below gives the triangle inequality. For $0 < p < 1$, concavity of t^p gives $(a + b)^p \leq a^p + b^p$. The $p = \infty$ case follows directly from the essential supremum. $\square$

Definition 6.1.3 (Conjugate exponent). For $1 \leq p \leq \infty$, the conjugate exponent p' is determined by

$$\frac{1}{p} + \frac{1}{p'} = 1,$$

with $1' = \infty$ and $\infty' = 1$.

Lemma 6.1.4 (Young's inequality). If $1 < p < \infty$ and $a, b \geq 0$, then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}.$$

Equality holds exactly when $a^p = b^{p'}$.

Proof. The convexity inequality for the exponential, or equivalently the tangent-line inequality for $t \mapsto t^p/p$, yields the result after a one-variable minimization. $\square$

Theorem 6.1.5 (Hölder's inequality). If $1 \leq p \leq \infty$ and $f \in L^p(\mu)$, $g \in L^{p'}(\mu)$, then $fg \in L^1(\mu)$ and

$$\int_X |fg| d\mu \leq \|f\|_p \|g\|_{p'}.$$

Proof. For $1 < p < \infty$, normalize so that both norms are 1, apply Young's inequality pointwise to $|f||g|$, and integrate. The endpoint cases follow directly from the definition of the essential supremum. $\square$

Corollary 6.1.6 (Finite-measure embedding). If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then

$$L^q(\mu) \subseteq L^p(\mu), \quad \|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q.$$

Proof. If $q < \infty$, apply Hölder to $|f|^p \cdot 1$ with exponents q/p and $q/(q-p)$. If $q = \infty$, use

$$\int_X |f|^p d\mu \leq \|f\|_\infty^p \mu(X).$$

$\square$

Proposition 6.1.7 (Multiplication operators). Let $1 \leq p, q, r \leq \infty$ satisfy

$$\frac{1}{q} = \frac{1}{p} + \frac{1}{r}.$$

If $h \in L^r(\mu)$, then multiplication by h defines a bounded linear map

$$M_h : L^p(\mu) \longrightarrow L^q(\mu), \quad M_h f = hf,$$

and

$$\|M_h f\|_q \leq \|h\|_r \|f\|_p.$$

Proof. When all exponents are finite, apply Hölder's inequality to $|h|^q$ and $|f|^q$ with exponents r/q and p/q . The endpoint cases follow directly from the essential-supremum definition. $\square$

Theorem 6.1.8 (Dual formula for the norm). Let $1 \leq p < \infty$. Then

$$\|f\|_p = \sup \left\{ \left| \int_X f \bar{h} d\mu \right| : h \in L^{p'}(\mu), \|h\|_{p'} \leq 1 \right\}.$$

Proof. Hölder gives the upper bound. If $1 < p < \infty$ and $f \neq 0$, take

$$h = \frac{f|f|^{p-2}}{\|f\|_p^{p-1}}.$$

Then $\|h\|_{p'} = 1$ and the integral equals $\|f\|_p$. For $p = 1$, take $h = f/|f|$ on $\{f \neq 0\}$ and $h = 0$ elsewhere. $\square$

Theorem 6.1.9 (Minkowski's inequality). If $1 \leq p \leq \infty$ and $f, g \in L^p(\mu)$, then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Proof. For $1 < p < \infty$, use the dual formula:

$$\left| \int (f + g) \bar{h} \right| \leq \|f\|_p \|h\|_{p'} + \|g\|_p \|h\|_{p'}.$$

Take the supremum over $\|h\|_{p'} \leq 1$. The cases $p = 1$ and $p = \infty$ follow from the pointwise triangle inequality. $\square$

Proposition 6.1.10 (Equality in Hölder). Let $1 < p < \infty$, and suppose $f \in L^p, g \in L^{p'}$ are not zero. Equality

$$\left| \int f \bar{g} d\mu \right| = \|f\|_p \|g\|_{p'}$$

holds if and only if there is a constant $c > 0$ such that

$$|f|^p = c|g|^{p'}$$

almost everywhere and there is a unimodular constant ζ for which

$$\zeta f \bar{g} \geq 0$$

almost everywhere. Equivalently, $f \bar{g}$ has one constant complex phase on $\{fg \neq 0\}$.

Proof. Equality in the integrated Young inequality forces equality pointwise almost everywhere after normalization, which gives proportionality of the powers. Equality in $|\int f \bar{g}| \leq \int |fg|$ gives phase alignment. $\square$

Proposition 6.1.11 (Equality in Minkowski). Let $1 < p < \infty$ and let $f, g \in L^p$ be nonzero. Equality

$$\|f + g\|_p = \|f\|_p + \|g\|_p$$

holds if and only if there are nonnegative constants a, b , not both zero, such that

$$af = bg$$

almost everywhere.

Proof. Equality in the dual proof of Minkowski forces equality in Hölder for both f and g against the same norming function, and also forces their phases to agree. The Hölder equality characterization then gives proportionality. The converse is immediate. $\square$

Proposition 6.1.12 (Equality in the L^1 triangle inequality). For $f, g \in L^1(\mu)$,

$$\|f + g\|_1 = \|f\|_1 + \|g\|_1$$

if and only if

$$f \bar{g} \geq 0$$

almost everywhere.

Proof. The equality of norms is equivalent to $|f + g| = |f| + |g|$ almost everywhere, which is exactly the condition that f and g point in the same complex direction. $\square$

Proposition 6.1.13 (Strict convexity). Let $1 < p < \infty$. If

$$\|f\|_p = \|g\|_p = 1, \quad \left\| \frac{1}{2}(f + g) \right\|_p = 1,$$

then $f = g$ almost everywhere. Thus $L^p(\mu)$ is strictly convex for $1 < p < \infty$.

Proof. The hypotheses imply equality in Minkowski's inequality. Hence $af = bg$ almost everywhere for some nonnegative constants a, b , not both zero. Equality of the two norms forces $a = b$, and therefore $f = g$ almost everywhere. $\square$

Remark. Strict convexity can fail at the endpoints. In L^1 , take two disjoint nonnegative unit vectors. In L^∞ , distinct unit vectors may have a midpoint of norm one.

Proposition 6.1.14 (A dual formula for L^∞). Assume μ is σ -finite. Then for $f \in L^\infty(\mu)$,

$$\|f\|_\infty = \sup \left\{ \left| \int_X f \bar{h} d\mu \right| : h \in L^1(\mu), \|h\|_1 \leq 1 \right\}.$$

Proof. The upper bound is Hölder. Let $M = \|f\|_\infty$. If $M = 0$, the result is immediate. Otherwise fix $0 < \varepsilon < M$. The set $E = \{|f| > M - \varepsilon\}$ has positive measure. Sigma-finiteness supplies a measurable $F \subseteq E$ with $0 < \mu(F) < \infty$. Set

$$h = \frac{f}{|f|} \frac{\mathbf{1}_F}{\mu(F)}$$

on $\{f \neq 0\}$. Then $\|h\|_1 = 1$ and the pairing exceeds $M - \varepsilon$. $\square$

6.2 Completeness and almost-everywhere subsequences

Theorem 6.2.1 (Completeness of L^p). For $1 \leq p \leq \infty$, the normed space $L^p(\mu)$ is complete.

Proof. First let $p < \infty$, and let (f_n) be Cauchy. Choose a subsequence (f_{n_k}) such that

$$\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}.$$

Set

$$g_N = |f_{n_1}| + \sum_{k=1}^N |f_{n_{k+1}} - f_{n_k}|.$$

Minkowski gives

$$\sup_N \|g_N\|_p \leq \|f_{n_1}\|_p + 1 < \infty.$$

By monotone convergence, $g = \lim_N g_N$ belongs to L^p , hence is finite almost everywhere. Therefore the telescoping series converges almost everywhere to a measurable function f , and $|f_{n_k}| \leq g$.

For fixed k , Fatou's lemma gives

$$\|f - f_{n_k}\|_p \leq \liminf_{j \rightarrow \infty} \|f_{n_j} - f_{n_k}\|_p.$$

The right side tends to zero as $k \rightarrow \infty$. Thus the subsequence converges to f in L^p , and the original Cauchy sequence does as well.

For $p = \infty$, choose the same type of subsequence with summable L^∞ differences. The resulting series converges uniformly outside one null set, producing the L^∞ limit. $\square$

Corollary 6.2.2 (Almost-everywhere subsequence). If $f_n \rightarrow f$ in $L^p(\mu)$ for $1 \leq p < \infty$, then some subsequence converges to f almost everywhere.

Proof. Choose a subsequence with

$$\sum_k \|f_{n_{k+1}} - f_{n_k}\|_p < \infty$$

and use the pointwise convergence argument in the completeness proof. $\square$

Corollary 6.2.3 (A dominating subsequence). If $f_n \rightarrow f$ in $L^p(\mu)$ for $1 \leq p < \infty$, then there are a subsequence (f_{n_k}) and a function $g \in L^p(\mu)$ such that

$$|f_{n_k}| \leq g$$

almost everywhere for every k .

Proof. Choose the subsequence so that the sum of successive L^p distances is finite and take

$$g = |f_{n_1}| + \sum_k |f_{n_{k+1}} - f_{n_k}|.$$

$\square$

Proposition 6.2.4 (The range $0 < p < 1$). For $0 < p < 1$, the metric space

$$(L^p(\mu), d(f, g) = \|f - g\|_p^p)$$

is complete, although $\|\cdot\|_p$ is not a norm unless the space is trivial.

Proof. Choose a subsequence for which the sum of the p th powers of successive distances is finite. Since

$$\left| \sum_k u_k \right|^p \leq \sum_k |u_k|^p,$$

the same telescoping argument proves almost-everywhere and metric convergence. $\square$

6.3 Duality and reflexivity

Proposition 6.3.1 (Natural map into the dual). Let $1 \leq p < \infty$ and $g \in L^{p'}(\mu)$. In the endpoint case $p = 1$, assume additionally that μ is σ -finite. Define

$$\Phi_g(f) = \int_X f \bar{g} d\mu.$$

Then $\Phi_g \in (L^p)^*$ and

$$\|\Phi_g\| = \|g\|_{p'}.$$

The map $g \mapsto \Phi_g$ is a conjugate-linear isometric injection.

Proof. Hölder gives boundedness and the upper estimate. The norm formula gives equality. If $\Phi_g = 0$, test against a norming function supported where $g \neq 0$ to obtain $g = 0$ almost everywhere. $\square$

Theorem 6.3.2 (Riesz representation for L^p). Assume μ is σ -finite and $1 \leq p < \infty$. Every bounded linear functional on $L^p(\mu)$ has the form

$$\Phi(f) = \int_X f \bar{g} d\mu$$

for a unique $g \in L^{p'}(\mu)$, and $\|\Phi\| = \|g\|_{p'}$.

Proof. Choose a measurable partition (X_n) with finite measure. On finite-measure sets define

$$\nu(E) = \Phi(\mathbf{1}_E).$$

Continuity of Φ and dominated convergence show that ν is countably additive and absolutely continuous with respect to μ . Radon–Nikodym gives a density $\bar{g}$ on each X_n , and the densities patch together.

For bounded simple f of finite-measure support, $\Phi(f) = \int f \bar{g}$. Density extends the identity to all of L^p . Finally, boundedness of Φ and the dual norm formula on finite-measure truncations show $g \in L^{p'}$ and $\|g\|_{p'} \leq \|\Phi\|$. The reverse inequality is the preceding proposition. $\square$

Corollary 6.3.3 (Dual of ℓ^p). For $1 \leq p < \infty$, the dual is conjugate-linearly and isometrically identified with

$$(\ell^p)^* \cong \ell^{p'}$$

through

$$\Phi_b(a) = \sum_{k=1}^{\infty} a_k \bar{b}_k.$$

Proof. Apply the representation theorem to counting measure on $\mathbb{N}$. $\square$

Theorem 6.3.4 (Reflexivity). If $1 < p < \infty$ and μ is σ -finite, then $L^p(\mu)$ is reflexive. In particular, ℓ^p is reflexive for $1 < p < \infty$.

Proof. The dual of L^p is $L^{p'}$, and the dual of $L^{p'}$ is L^p . Under these identifications, the canonical embedding of L^p into its bidual is the identity. $\square$

Proposition 6.3.5. The space ℓ^1 is not reflexive.

Proof. The dual of ℓ^1 is ℓ^∞ . On the subspace $c \subset \ell^\infty$ of convergent sequences, the limit map is a bounded linear functional. By Hahn–Banach it extends to a functional $L \in (\ell^\infty)^*$. If L were represented by an element $a \in \ell^1$, then $L(e_n) = a_n$. But $e_n \rightarrow 0$ pointwise and the limit functional is zero on every e_n , so $a_n = 0$ for all n , while $L(1, 1, \dots) = 1$. This is impossible. Thus the canonical image of ℓ^1 in its bidual is not onto. $\square$

6.4 Approximation, separability, and translations

Theorem 6.4.1 (Density of simple functions). If $0 < p < \infty$ and $f \in L^p(\mu)$, then for every $\varepsilon > 0$ there is an integrable simple function s such that

$$\|f - s\|_p < \varepsilon.$$

Proof. Approximate $|f|$ pointwise by dyadic simple functions bounded by $|f|$, restore the phase of f , and apply dominated convergence to $|f - s_n|^p$. $\square$

Theorem 6.4.2 (Density of compactly supported continuous functions). For $1 \leq p < \infty$, the space $C_c(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.

Proof. First approximate by simple functions of finite-measure support. By regularity, approximate the sets in those simple functions by finite unions of rectangles, and then approximate rectangle indicators by continuous cutoff functions. The resulting L^p error can be made arbitrarily small. $\square$

Remark (Failure in L^∞). Step functions built from finitely many intervals are not dense in $L^\infty([0, 1])$. Let C be a fat Cantor set with $0 < m(C) < 1$ and empty interior. Suppose a finite interval step function s satisfied

$$\|\mathbf{1}_C - s\|_\infty < \frac{1}{3}.$$

Because $m(C) > 0$, one interval of constancy I satisfies $m(I \cap C) > 0$, forcing the corresponding constant to exceed $2/3$. Since C has empty interior, $I \setminus C$ contains a nonempty open interval and therefore has positive measure; on that same set the constant would have to be below $1/3$. This contradiction proves the failure of density.

Proposition 6.4.3 (Separability). For $1 \leq p < \infty$, the spaces ℓ^p and $L^p(\mathbb{R}^n)$ are separable. The spaces ℓ^∞ and $L^\infty([a, b])$ are not separable when $a < b$.

Proof. Finitely supported sequences with rational real and imaginary parts form a countable dense subset of ℓ^p . Rational linear combinations of indicators of rational rectangles form a countable dense subset of $L^p(\mathbb{R}^n)$.

The uncountable family of 0–1 sequences is 1-separated in ℓ^∞ . Likewise, the functions $\mathbf{1}_{[a, x]}$, $x \in [a, b]$, are pairwise distance 1 in $L^\infty([a, b])$. $\square$

Corollary 6.4.4. There is no continuous surjection from $L^1([a, b])$ onto $L^\infty([a, b])$ when $a < b$.

Proof. A continuous image of a separable metric space is separable, whereas $L^\infty([a, b])$ is not. $\square$

Proposition 6.4.5 (Bounded functions and changes of exponent). Let E be measurable. If $f \in L^p(E) \cap L^\infty(E)$ and $q > p$, then

$$\|f\|_q^q \leq \|f\|_\infty^{q-p} \|f\|_p^p.$$

In particular, $f \in L^q(E)$.

Proof. Use $|f|^q = |f|^{q-p}|f|^p \leq \|f\|_\infty^{q-p} |f|^p$ almost everywhere. $\square$

Definition 6.4.6 (Translations). For $f : \mathbb{R}^n \rightarrow \mathbb{C}$ and $t \in \mathbb{R}^n$, define

$$\tau_t f(x) = f(x - t).$$

Theorem 6.4.7 (Continuity of translations). Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$. Then

$$\|\tau_t f\|_p = \|f\|_p$$

and

$$\lim_{t \rightarrow 0} \|\tau_t f - f\|_p = 0.$$

Moreover, the function $t \mapsto \|\tau_t f - f\|_p$ is bounded and uniformly continuous.

Proof. Translation invariance of Lebesgue measure gives the norm identity and the bound

$$\|\tau_t f - f\|_p \leq 2 \|f\|_p.$$

For $g \in C_c(\mathbb{R}^n)$, uniform continuity and compact support imply $\|\tau_t g - g\|_p \rightarrow 0$. Given f , choose such a g with $\|f - g\|_p$ small and use

$$\|\tau_t f - f\|_p \leq 2 \|f - g\|_p + \|\tau_t g - g\|_p.$$

Finally,

$$|\|\tau_t f - f\|_p - \|\tau_s f - f\|_p| \leq \|\tau_{t-s} f - f\|_p,$$

which proves uniform continuity. $\square$

Proposition 6.4.8 (Dilations). For $t \neq 0$ define $R_t f(x) = f(tx)$ on $\mathbb{R}^n$. Then, for $0 < p < \infty$,

$$\|R_t f\|_p = |t|^{-n/p} \|f\|_p,$$

while $\|R_t f\|_\infty = \|f\|_\infty$.

Proof. For finite p , use the change of variables $y = tx$. The essential-supremum identity is immediate because dilation preserves null sets. $\square$

Corollary 6.4.9 (L^p Lebesgue points). If $1 \leq p < \infty$ and $f \in L^p_{\text{loc}}(\mathbb{R}^n)$, then for almost every x ,

$$\lim_{r \downarrow 0} \frac{1}{m(B(x, r))} \int_{B(x, r)} |f(y) - f(x)|^p dy = 0.$$

Proof. Apply the Lebesgue differentiation theorem to $|f - c|^p$ for a countable dense set of constants c , and approximate $f(x)$ by one of them. $\square$

Theorem 6.4.10 (Riesz criterion for indefinite integrals). Let $1 < p < \infty$ and let $F : [a, b] \rightarrow \mathbb{C}$. The following are equivalent:

(a). there is $f \in L^p(a, b)$ such that

$$F(x) = F(a) + \int_a^x f(t) \, dt;$$

(b). there is $M < \infty$ such that every partition $a = x_0 < \dots < x_n = b$ satisfies

$$\sum_{k=1}^n \frac{|F(x_k) - F(x_{k-1})|^p}{|x_k - x_{k-1}|^{p-1}} \leq M.$$

The least admissible M is $\|f\|_p^p$.

Proof. If $F(x) - F(a) = \int_a^x f$, Holder's inequality on each subinterval gives

$$|F(x_k) - F(x_{k-1})|^p \leq |x_k - x_{k-1}|^{p-1} \int_{x_{k-1}}^{x_k} |f|^p.$$

Summing yields the partition bound with $M = \|f\|_p^p$.

Conversely, define on interval step functions

$$\Lambda \left(\sum_{k=1}^n c_k \mathbf{1}_{(x_{k-1}, x_k]} \right) = \sum_{k=1}^n c_k (F(x_k) - F(x_{k-1})).$$

The definition is unchanged by refinement. Discrete Holder gives

$$|\Lambda(g)| \leq \left(\sum_{k=1}^n \frac{|F(x_k) - F(x_{k-1})|^p}{|x_k - x_{k-1}|^{p-1}} \right)^{1/p} \left(\sum_{k=1}^n |c_k|^{p'} |x_k - x_{k-1}| \right)^{1/p'} \leq M^{1/p} \|g\|_{p'}.$$

Thus Λ extends to a bounded functional on $L^{p'}(a, b)$. The L^p representation theorem gives $f \in L^p(a, b)$ such that $\Lambda(g) = \int_a^b g f$. Testing $g = \mathbf{1}_{(a, x]}$ yields the desired formula for F . The norm identity in the representation theorem shows that the least M equals $\|f\|_p^p$. $\square$

Exercise 6.4.11 (A small-interval limit). For $1 \leq p < \infty$, determine all $\lambda \in \mathbb{R}$ for which

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{-\lambda} \int_0^\varepsilon f(x) \, dx = 0$$

for every $f \in L^p([0, 1])$. For $p = \infty$, answer the analogous question.

Proof. If $1 < p < \infty$, Hölder gives

$$\left| \varepsilon^{-\lambda} \int_0^\varepsilon f \right| \leq \varepsilon^{1/p' - \lambda} \left(\int_0^\varepsilon |f|^p \right)^{1/p}.$$

This tends to zero when $\lambda \leq 1/p'$; at equality the local L^p integral tends to zero. For $p = 1$, the same conclusion is $\lambda \leq 0$. Necessity follows by testing $f(x) = x^{-\alpha}$ with $\alpha < 1/p$ and letting $\alpha \uparrow 1/p$. Thus the answer for finite p is

$$\lambda \leq 1 - \frac{1}{p}.$$

For $p = \infty$, the estimate is $\|f\|_\infty \varepsilon^{1-\lambda}$, so every $\lambda < 1$ works, while $f \equiv 1$ excludes $\lambda \geq 1$. $\square$

Example (Two incomplete normed spaces). The polynomial space on $[a, b]$ with the supremum norm is not complete: the partial sums of the exponential series are uniformly Cauchy but converge to a nonpolynomial function.

If $E = [0, 1]$ and $1 \leq p_1 < p_2 < \infty$, then $L^{p_2}(E)$ equipped with the L^{p_1} norm is not complete. Choose q with $p_1 < q < p_2$ and approximate $x^{-1/q}$ by truncations in L^{p_2} . The truncations are Cauchy in L^{p_1} , but their L^{p_1} limit does not belong to L^{p_2} .

6.5 Uniform integrability and convergence

Definition 6.5.1 (Uniform integrability). A family $\mathcal{F} \subseteq L^1(E)$ is *uniformly integrable* on E if

$$\sup_{f \in \mathcal{F}} \int_E |f| < \infty$$

and, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$A \subseteq E, \quad m(A) < \delta \quad \implies \quad \sup_{f \in \mathcal{F}} \int_A |f| < \varepsilon.$$

On a general measure space, replace m by the underlying measure. For Lebesgue measure on a finite-measure set, the uniform L^1 bound follows from the small-set condition by partitioning the set into finitely many sufficiently small pieces; it is stated explicitly here so that the definition also works on spaces with atoms.

Proposition 6.5.2. Every finite family of integrable functions is uniformly integrable. If $m(E) < \infty$, every family bounded in $L^p(E)$ for some $p > 1$ is uniformly integrable in $L^1(E)$.

Proof. For a finite family, take the minimum of the finitely many δ values supplied by absolute continuity of the integral. If $\sup_{f \in \mathcal{F}} \|f\|_p \leq M$, Hölder gives

$$\int_A |f| \leq M m(A)^{1/p'}.$$

□

Theorem 6.5.3 (Vitali convergence theorem on a finite-measure set). Assume $m(E) < \infty$, $f_n \rightarrow f$ in measure on E , and (f_n) is uniformly integrable. Then $f \in L^1(E)$ and

$$\|f_n - f\|_1 \rightarrow 0.$$

In particular,

$$\int_E f_n \rightarrow \int_E f.$$

Proof. Every subsequence has a further subsequence converging almost everywhere. It therefore suffices to prove the result under almost-everywhere convergence. Fatou's lemma and uniform integrability show $f \in L^1$. Given $\varepsilon > 0$, choose δ from uniform integrability; Fatou gives the same small-set estimate for f .

Egoroff's theorem yields a measurable $A \subseteq E$ with $m(A) < \delta$ such that $f_n \rightarrow f$ uniformly on $E \setminus A$. Then

$$\int_E |f_n - f| \leq \int_A |f_n| + \int_A |f| + m(E) \sup_{E \setminus A} |f_n - f|,$$

which is eventually small. □

Corollary 6.5.4 (Scheffé–Riesz convergence criterion). Let $1 \leq p < \infty$. If $f_n \rightarrow f$ almost everywhere and

$$\int |f_n|^p \rightarrow \int |f|^p < \infty,$$

then

$$\|f_n - f\|_p \rightarrow 0.$$

Proof. Set

$$g_n = 2^{p-1}(|f_n|^p + |f|^p) - |f_n - f|^p \geq 0.$$

Then $g_n \rightarrow 2^p |f|^p$ almost everywhere. Fatou's lemma gives

$$2^p \int |f|^p \leq \liminf_n \int g_n = 2^p \int |f|^p - \limsup_n \int |f_n - f|^p.$$

Thus the last limsup is zero. □

Definition 6.5.5 (Tightness). A family $\mathcal{F} \subseteq L^1(E)$ is *tight* if for every $\varepsilon > 0$ there is a measurable $E_0 \subseteq E$ with finite measure such that

$$\sup_{f \in \mathcal{F}} \int_{E \setminus E_0} |f| < \varepsilon.$$

Theorem 6.5.6 (Vitali convergence on a general measure space). Let $f_n \rightarrow f$ in measure. If (f_n) is uniformly integrable and tight, then $f \in L^1$ and

$$\|f_n - f\|_1 \rightarrow 0.$$

Conversely, L^1 convergence implies uniform integrability and tightness.

Proof. Choose a finite-measure set E_0 making all tails small. Fatou gives the same tail estimate for f . Apply the finite-measure Vitali theorem on E_0 and then add the tails.

Conversely, if $f_n \rightarrow f$ in L^1 , then a tail of the sequence is uniformly close to the singleton $\{f\}$, which is uniformly integrable and tight; the finitely many initial terms do not affect either property. □

Theorem 6.5.7 (L^p Vitali criterion). Let $1 \leq p < \infty$ and suppose $f_n \rightarrow f$ almost everywhere. Then

$$f_n \rightarrow f \text{ in } L^p(E)$$

if and only if the family $(|f_n|^p)$ is uniformly integrable and tight in $L^1(E)$.

Proof. If $p = 1$, the implication from L^p convergence follows from

$$||f_n| - |f|| \leq |f_n - f|.$$

If $p > 1$, use

$$||u|^p - |v|^p| \leq p(|u| + |v|)^{p-1}|u - v|$$

and Hölder's inequality to obtain $|f_n|^p \rightarrow |f|^p$ in L^1 . In either case, uniform integrability and tightness follow from L^1 convergence.

Conversely, Fatou's lemma shows that $|f|^p$ is integrable and inherits the same small-set and tail bounds. Since

$$|f_n - f|^p \leq 2^{p-1}(|f_n|^p + |f|^p),$$

the family $|f_n - f|^p$ is uniformly integrable and tight. Apply the general Vitali theorem to the almost-everywhere convergent sequence $|f_n - f|^p \rightarrow 0$. $\square$

Corollary 6.5.8. Assume $m(E) < \infty$, $f_n \rightarrow f$ almost everywhere, and for some $\theta > 0$,

$$\sup_n \|f_n\|_{p+\theta} < \infty.$$

Then $f_n \rightarrow f$ in $L^p(E)$.

Proof. The family $|f_n|^p$ is bounded in $L^{1+\theta/p}(E)$ and is therefore uniformly integrable. Tightness is automatic on a finite-measure set. Apply the L^p Vitali criterion. $\square$

Example (Uniform-integrability tests). On $[0, 1]$, the family $\{f : |f| \leq 1\}$ is uniformly integrable. By contrast, the family of continuous functions satisfying $\int_0^1 |f| \leq 1$ is not uniformly integrable: continuous triangular spikes of height n and base $2/n$ give a counterexample.

Bounds on signed interval integrals do not control uniform integrability. For example,

$$f_n(x) = n^2 \sin(2\pi n^2 x)$$

has uniformly bounded integrals over every interval, but on $A_n = [0, 1/n]$,

$$\int_{A_n} |f_n| \asymp n$$

while $m(A_n) \rightarrow 0$.

Example (Escape of mass). On $\mathbb{R}$, the sequence

$$f_n = \mathbf{1}_{[n, n+1]}$$

is uniformly integrable and converges pointwise to zero, but $\int f_n = 1$ for every n . It is not tight. This is why tightness is indispensable on infinite-measure spaces.